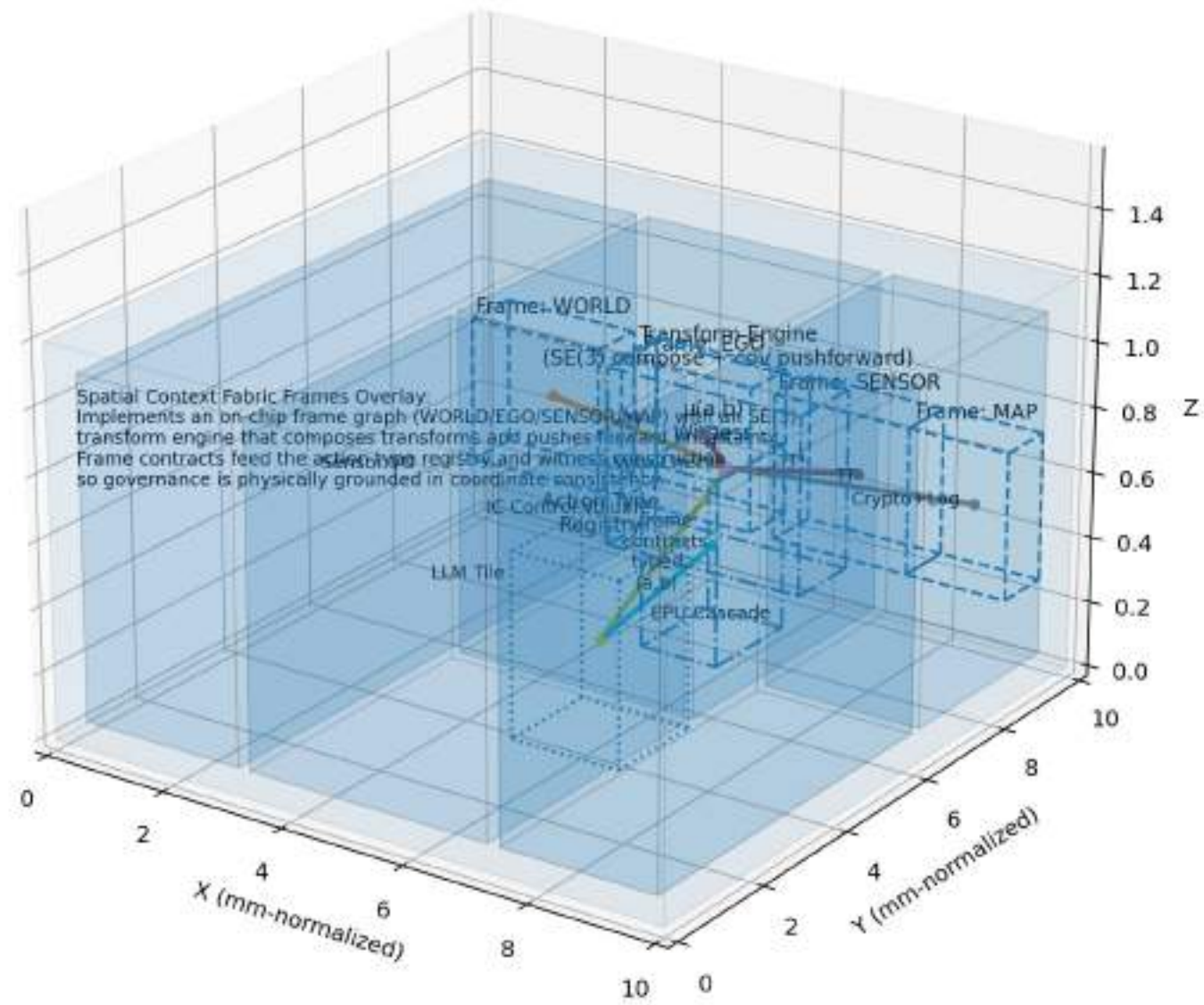
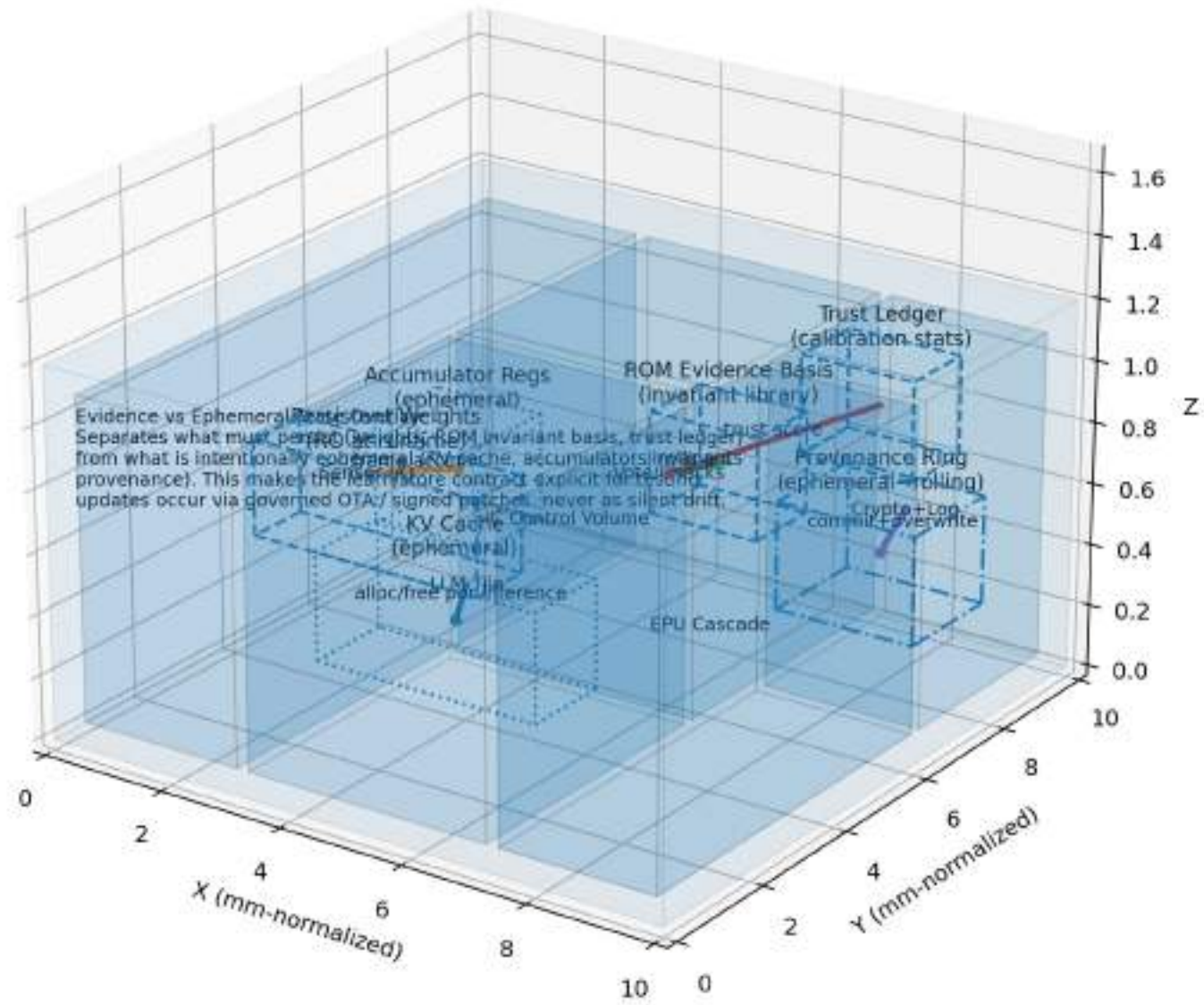


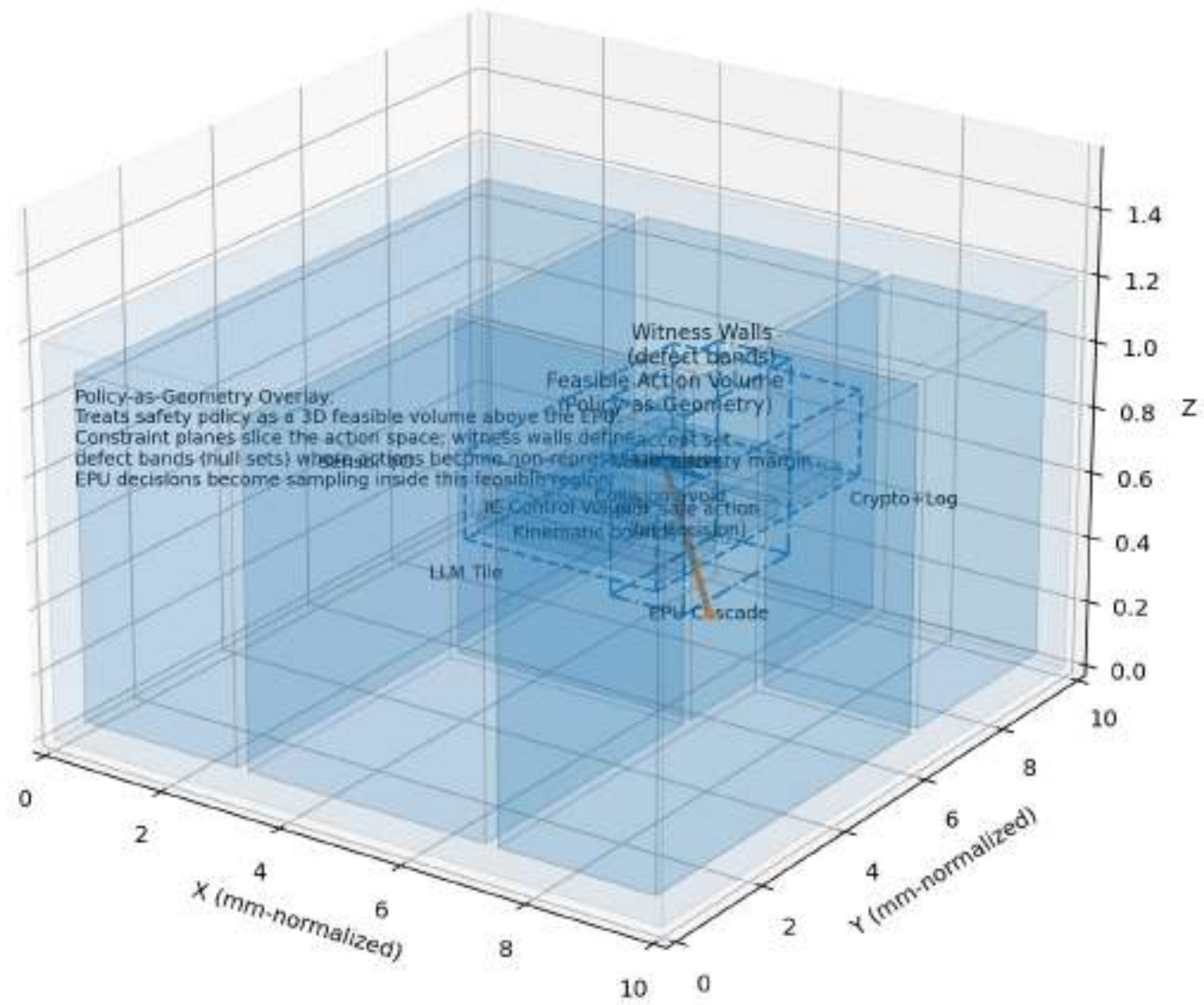
Overlay 57: Spatial Context Fabric Frames (On-chip Frame Graph)



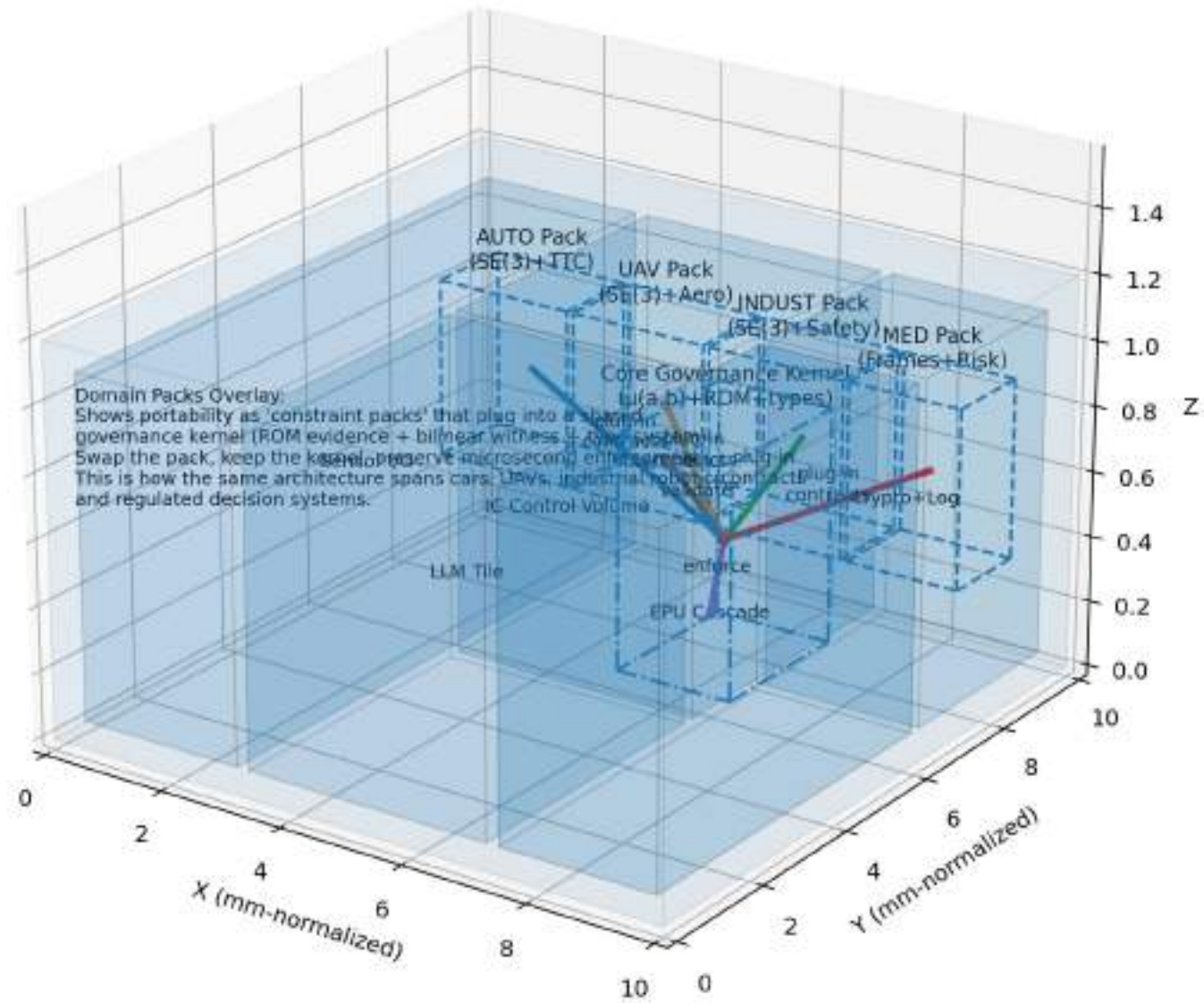
Overlay 58: Evidence Store vs Ephemeral State (Learning/Storage Contract)



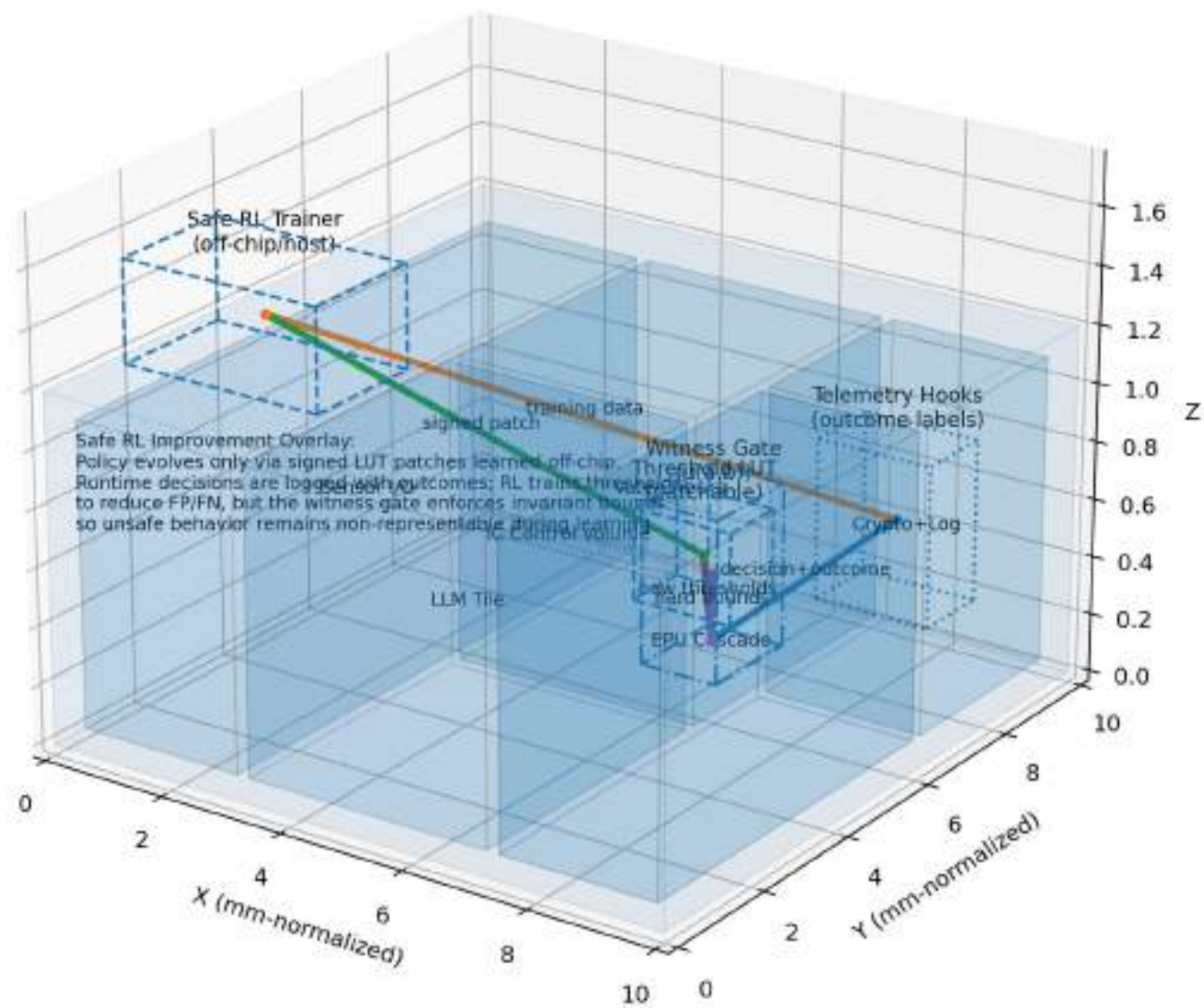
Overlay 59: Policy-as-Geometry (Feasible Action Volume)



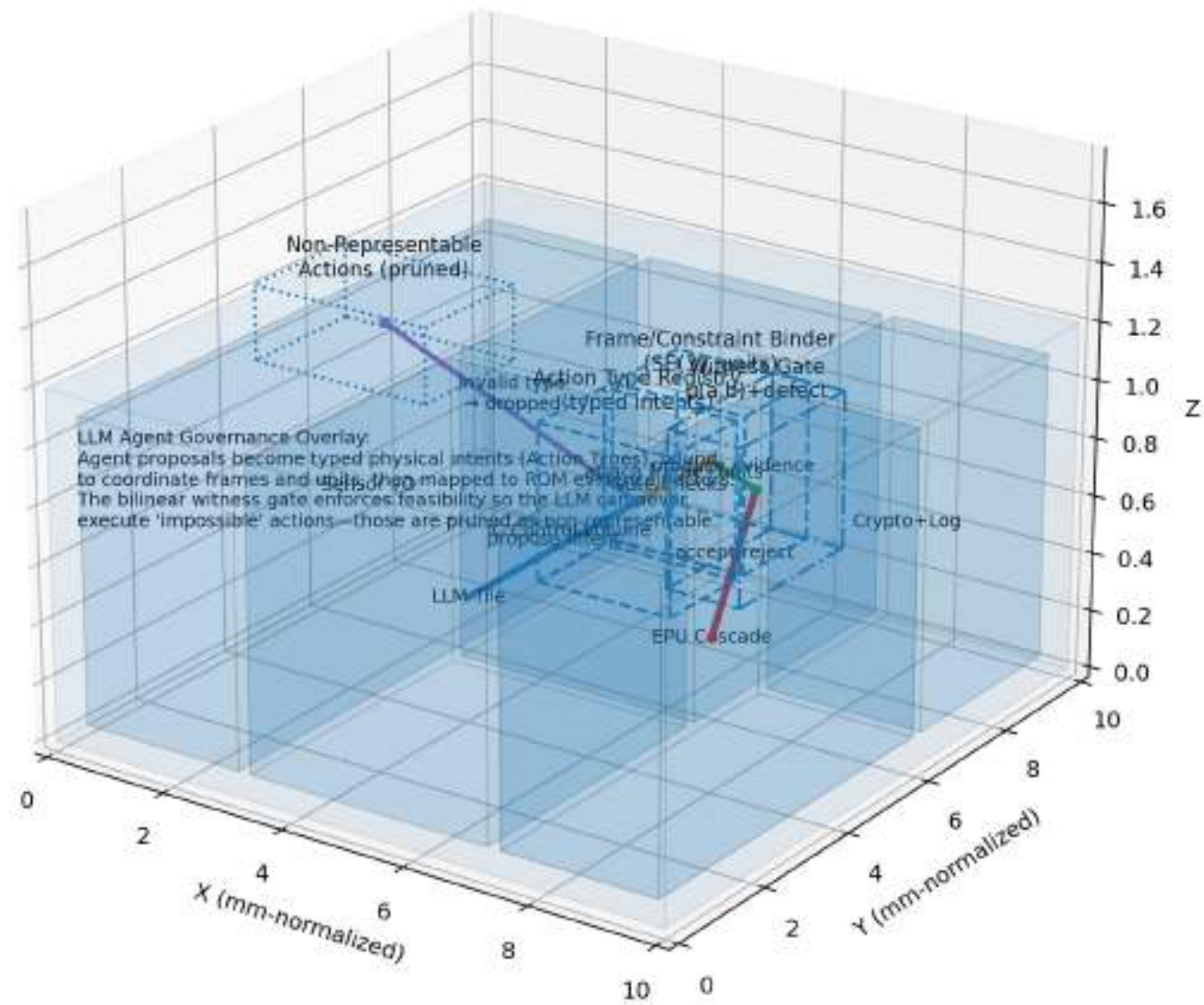
Overlay 60: Domain Packs (Portable Invariants + Coordinate Contracts)



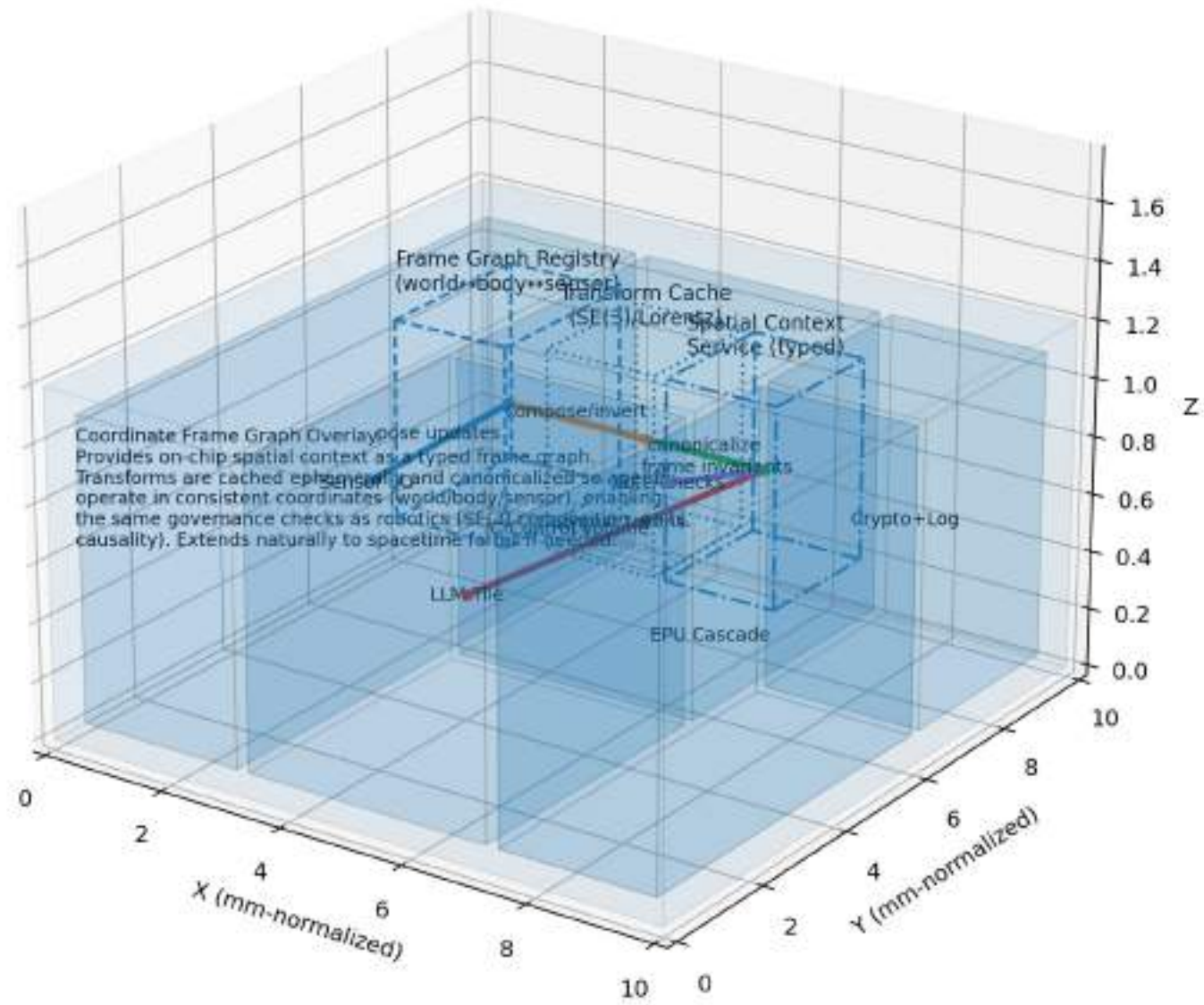
Overlay 61: Safe RL Improvement Loop (Bounded Policy Evolution)



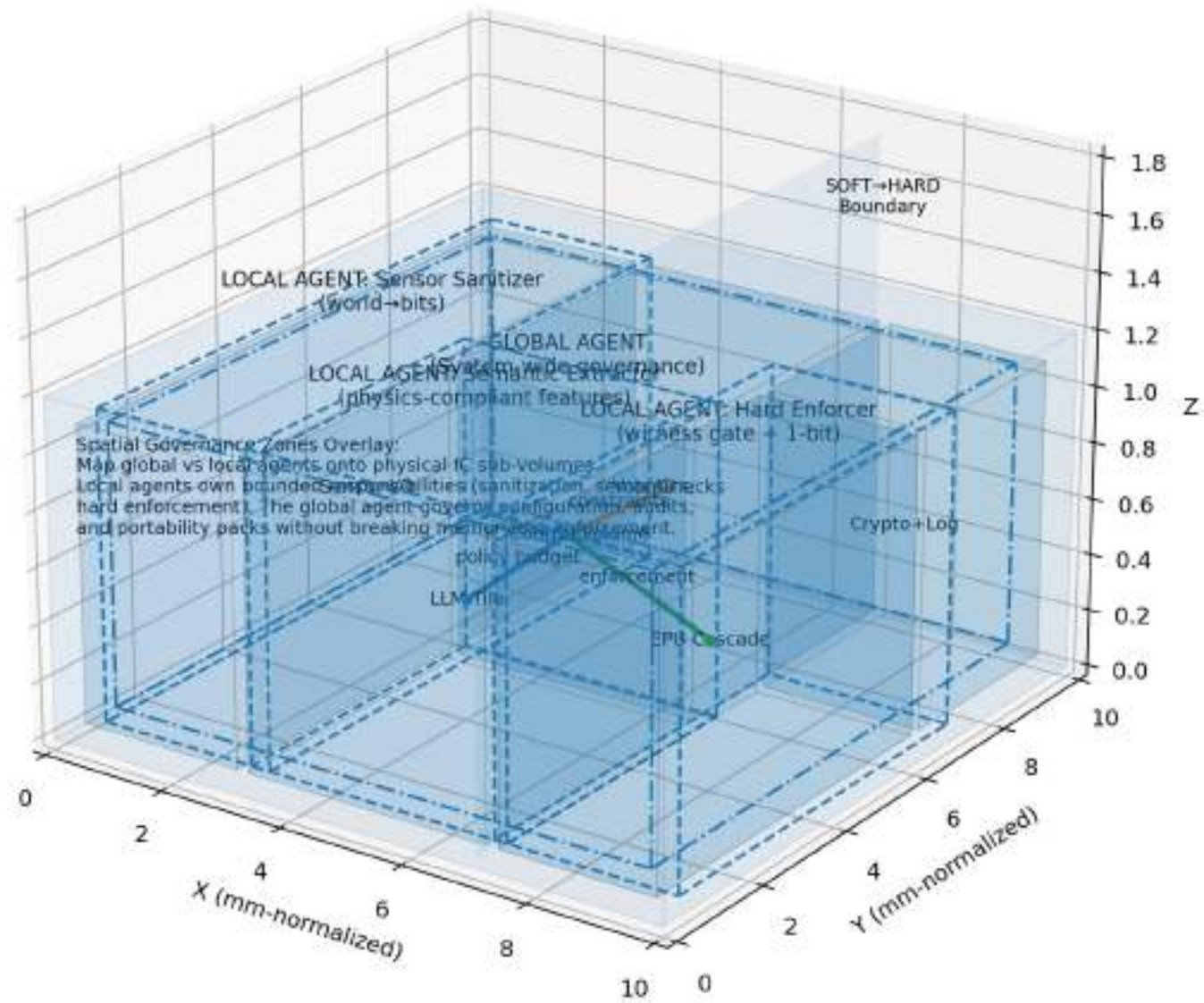
Overlay 62: LLM Agent Governance (Physical Feasibility via Typed Intents)



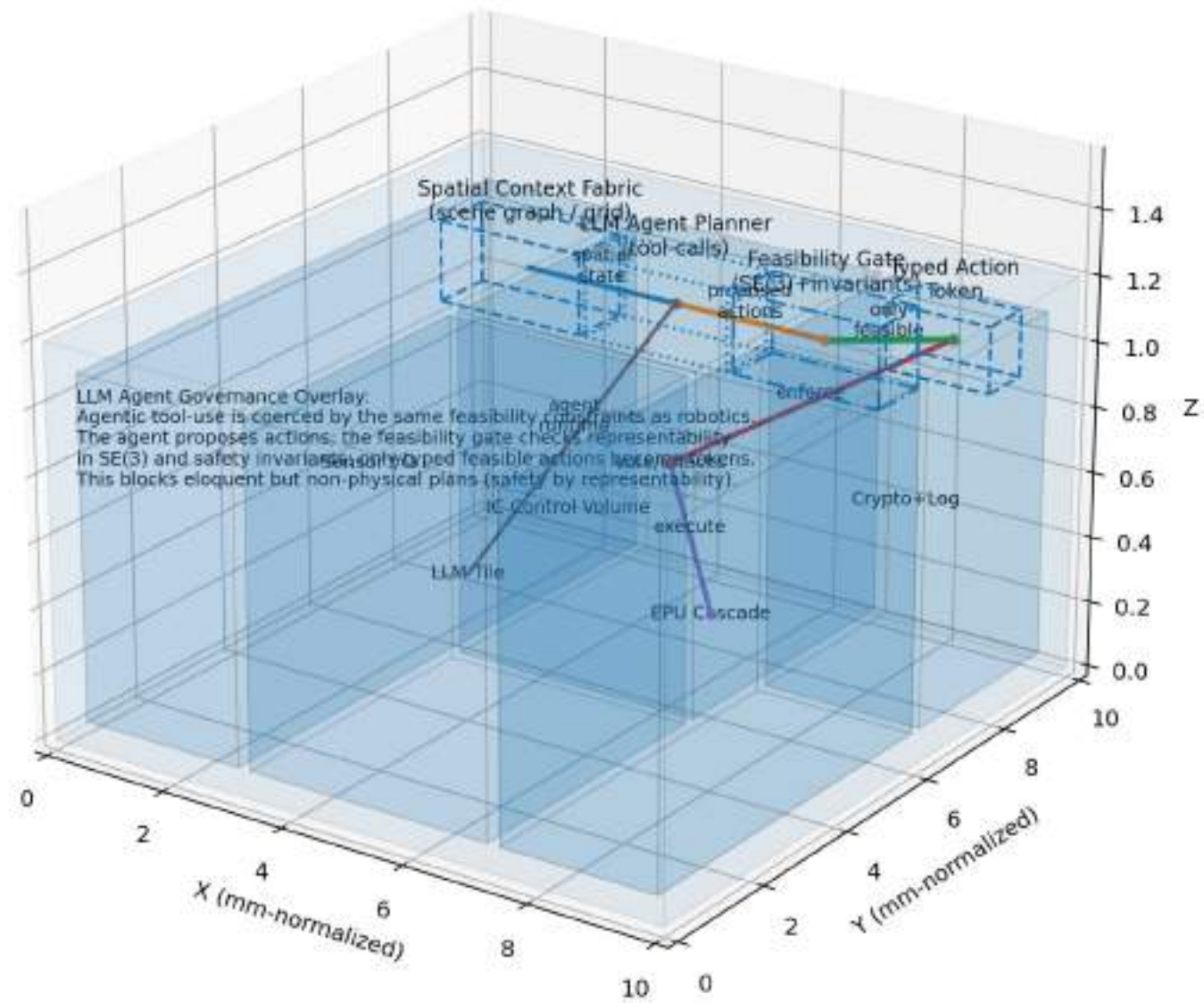
Overlay 63: Coordinate Frame Graph (On-Chip Spatial Context Fabric)



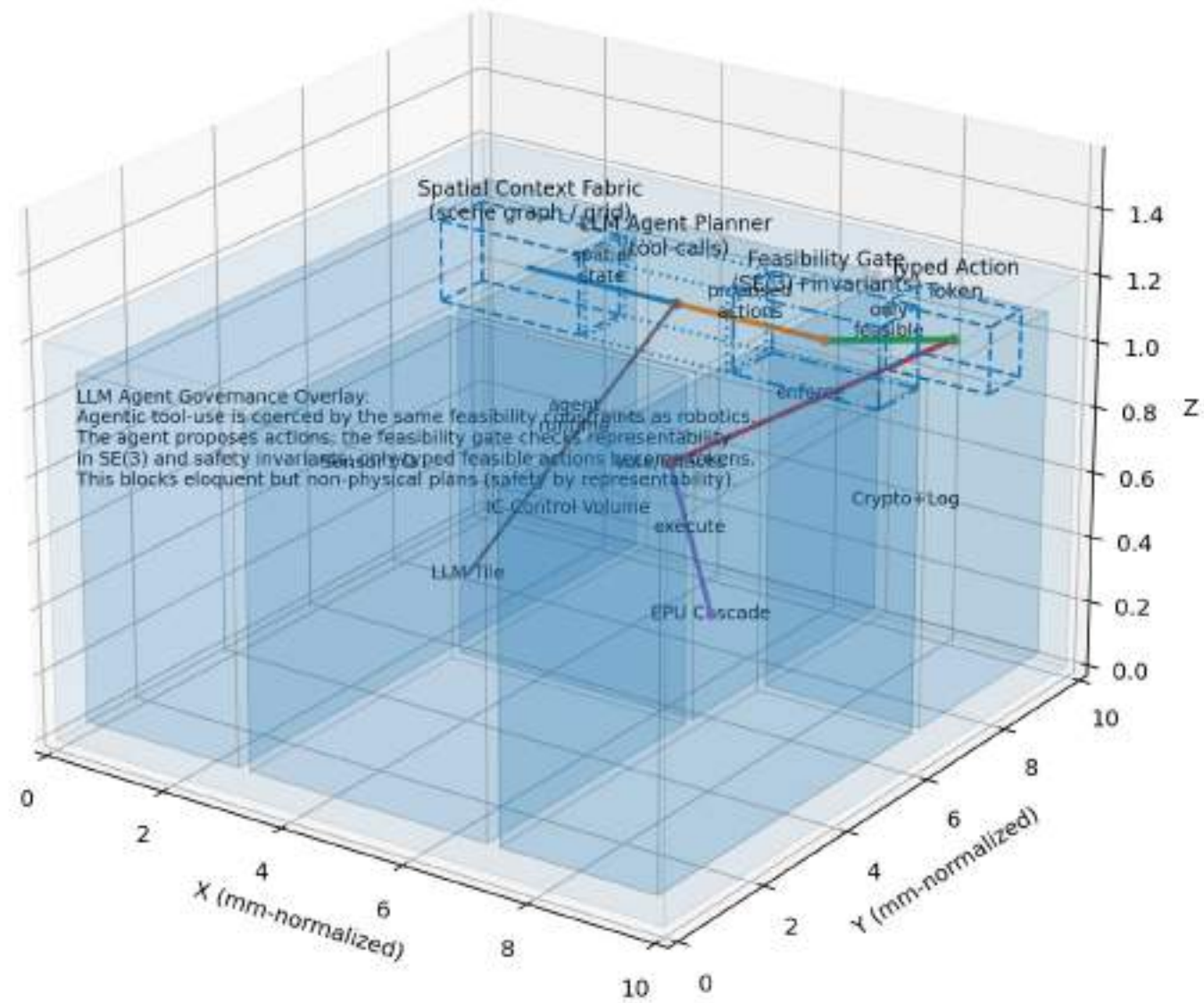
Overlay 64: Spatial Governance Zones (Global vs Local Agents)



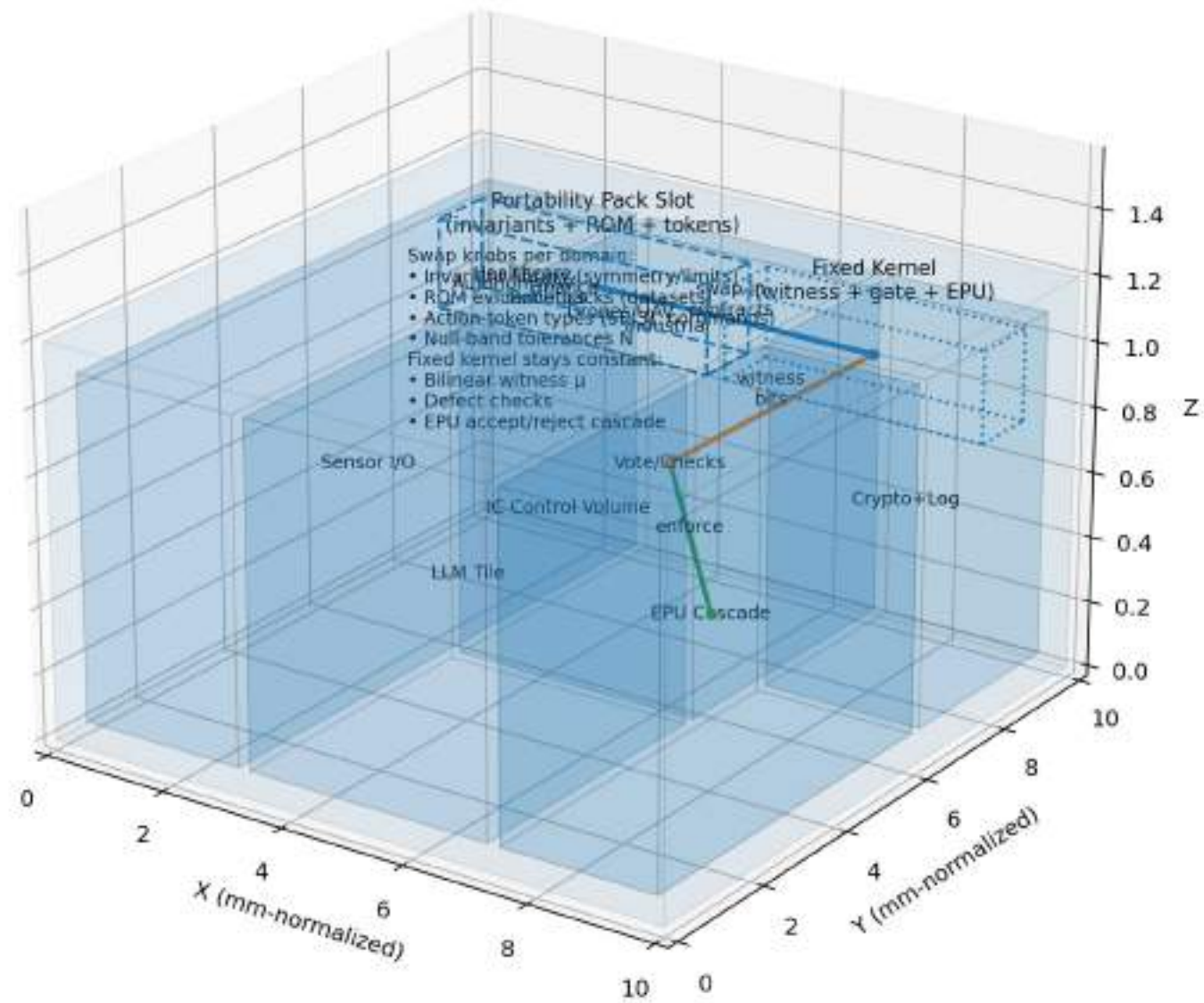
Overlay 36: LLM-Agent Governance as Physical Feasibility (Spatial Context + Tool-Use)



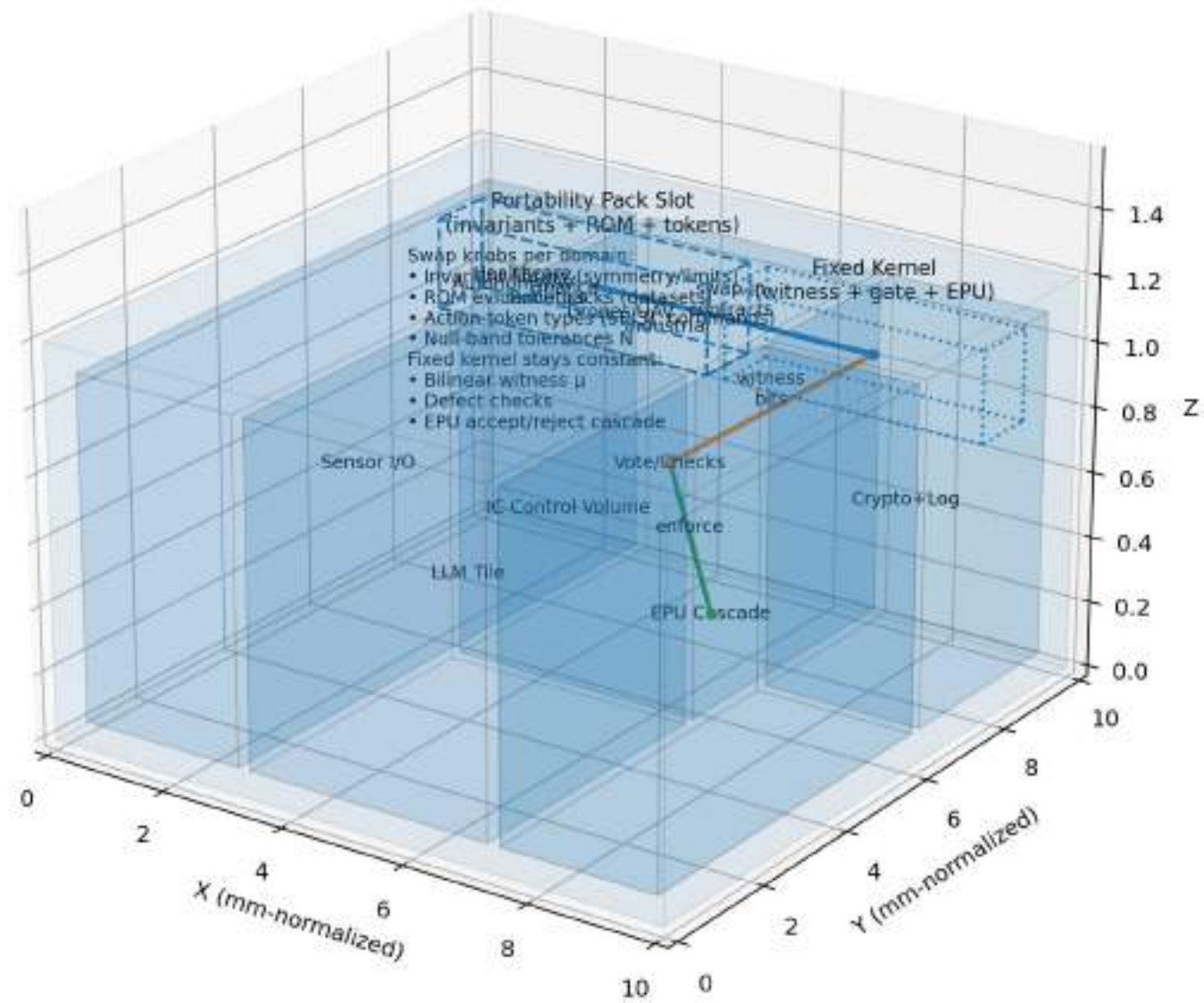
Overlay 36: LLM-Agent Governance as Physical Feasibility (Spatial Context + Tool-Use)



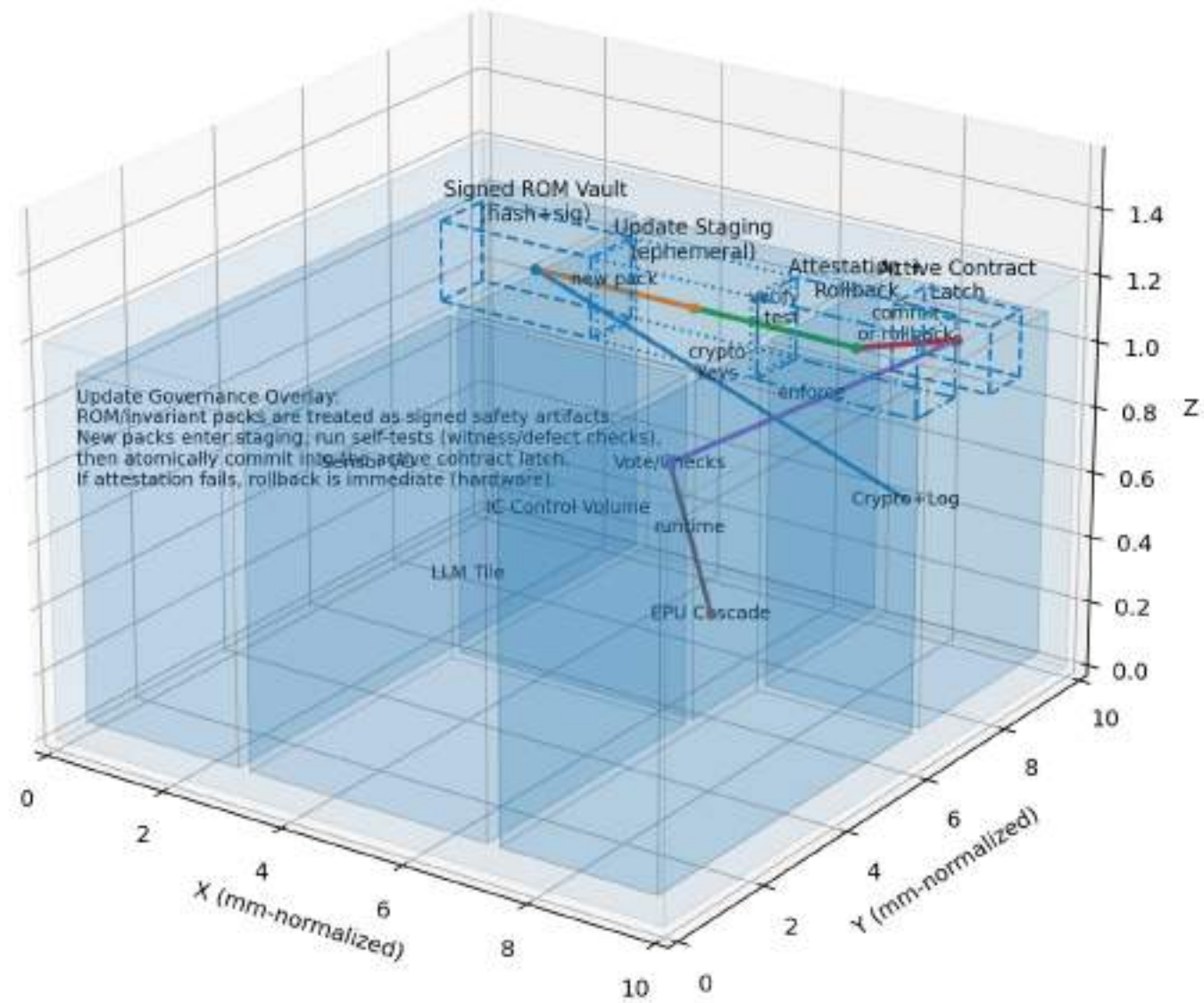
Overlay 37: Domain Portability Template (Swap-in Packs, Fixed Enforcement Kernel)



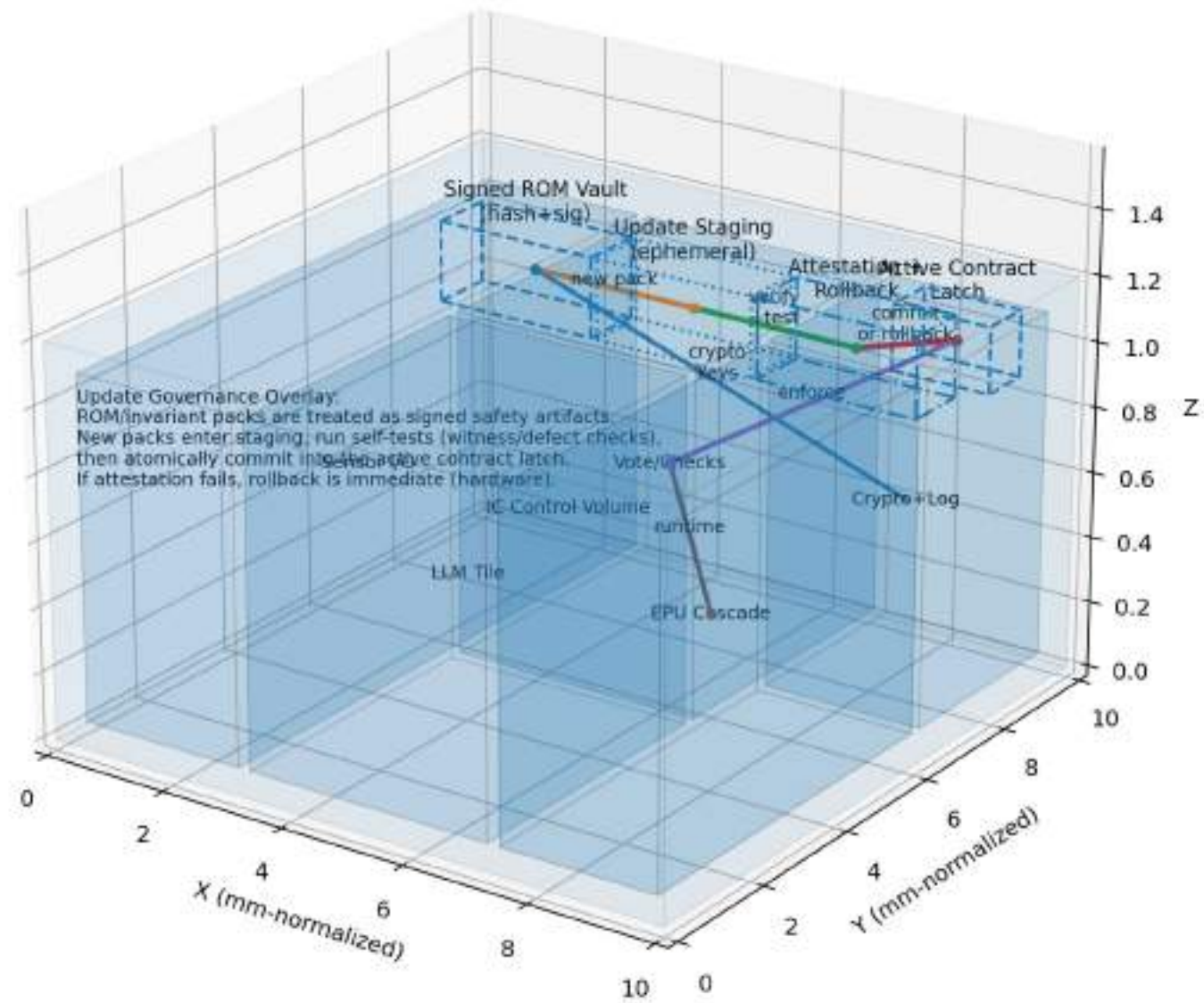
Overlay 37: Domain Portability Template (Swap-in Packs, Fixed Enforcement Kernel)



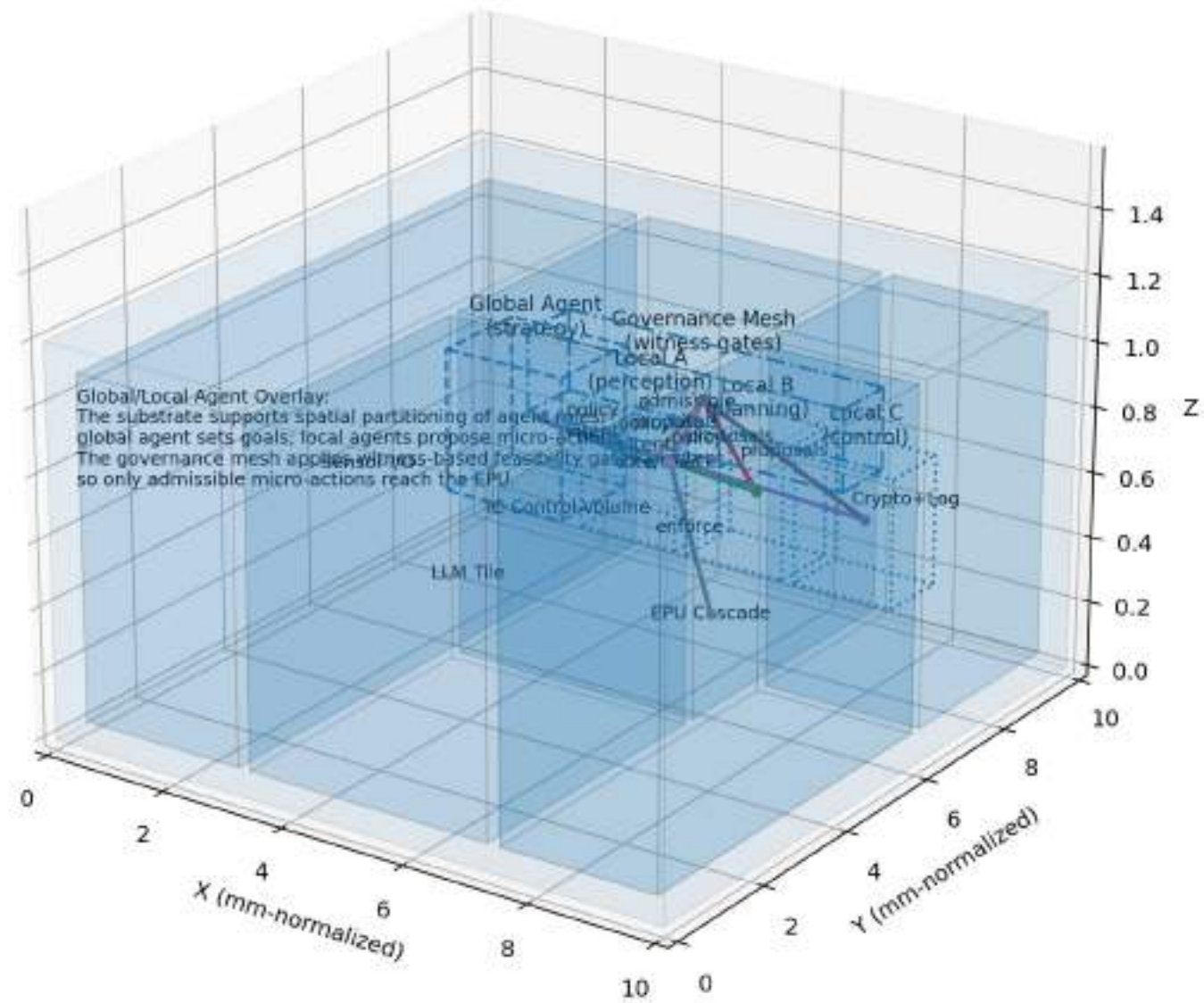
Overlay 38: Secure Update & Evidence Chain-of-Custody (ROM Packs + OTA Governance)



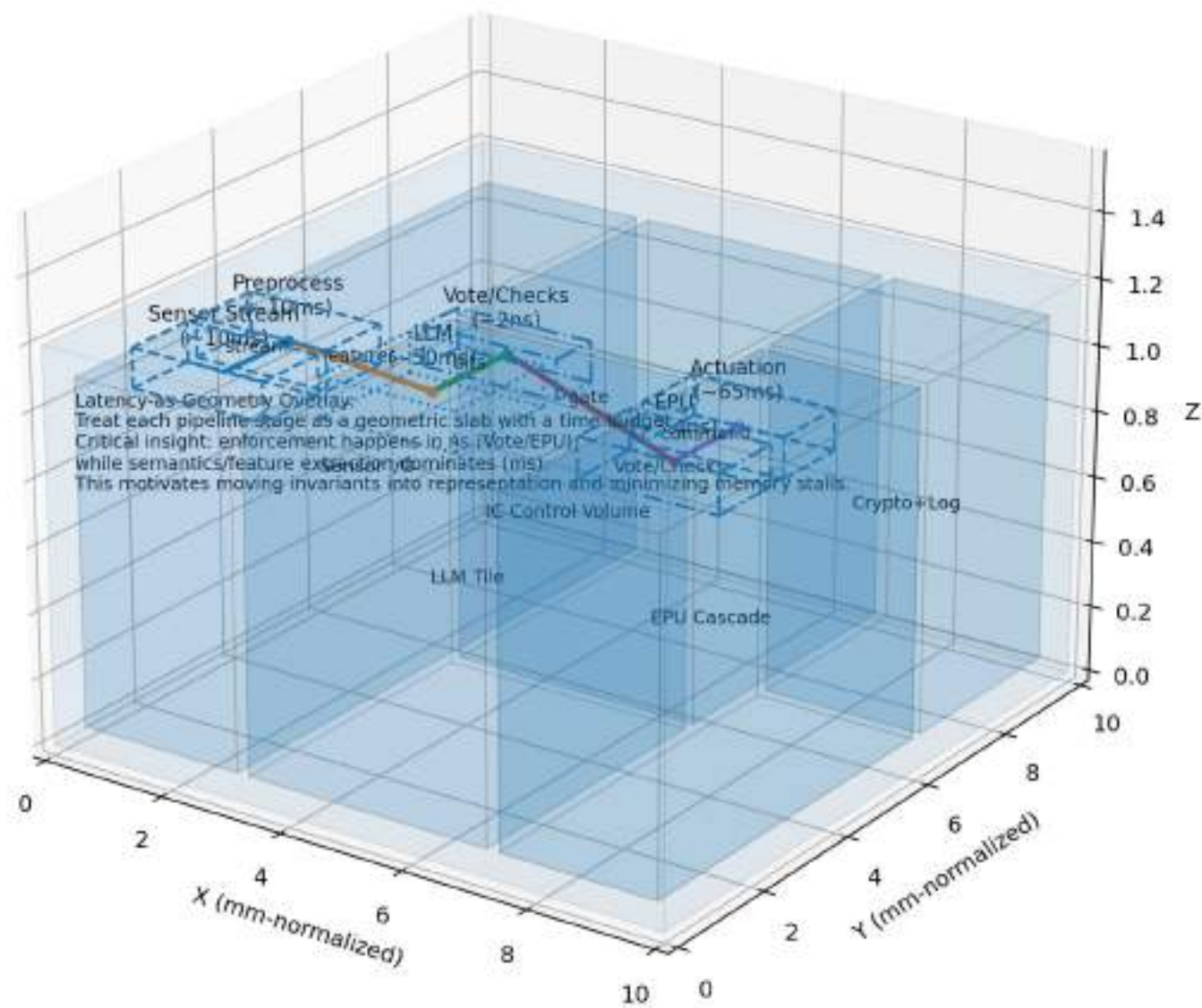
Overlay 38: Secure Update & Evidence Chain-of-Custody (ROM Packs + OTA Governance)



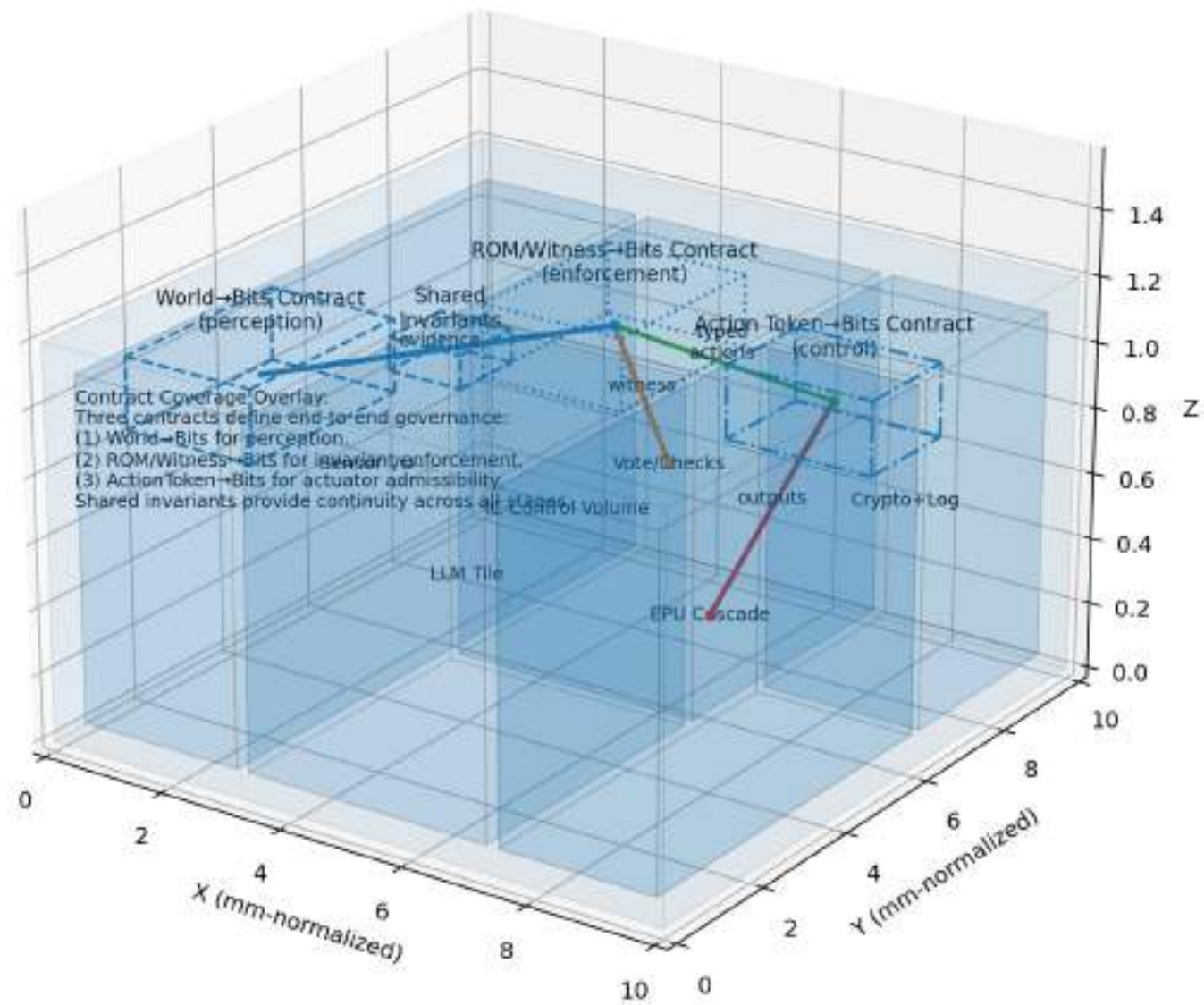
Overlay 39: 3D Global/Local Agent Partitioning (Spatial Governance Mesh)



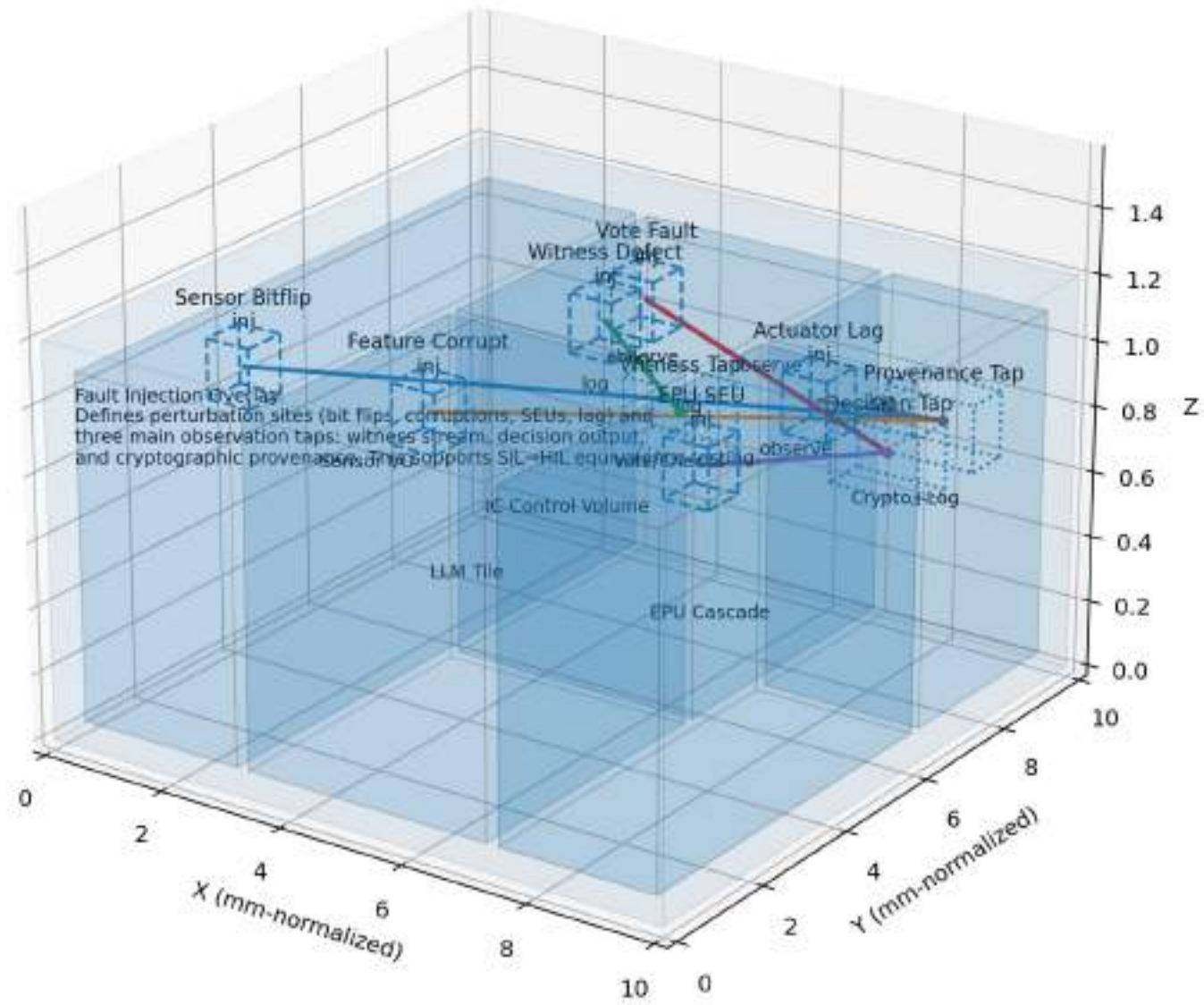
Overlay 40: Latency as Geometry (Propagation + Pipeline Budget)



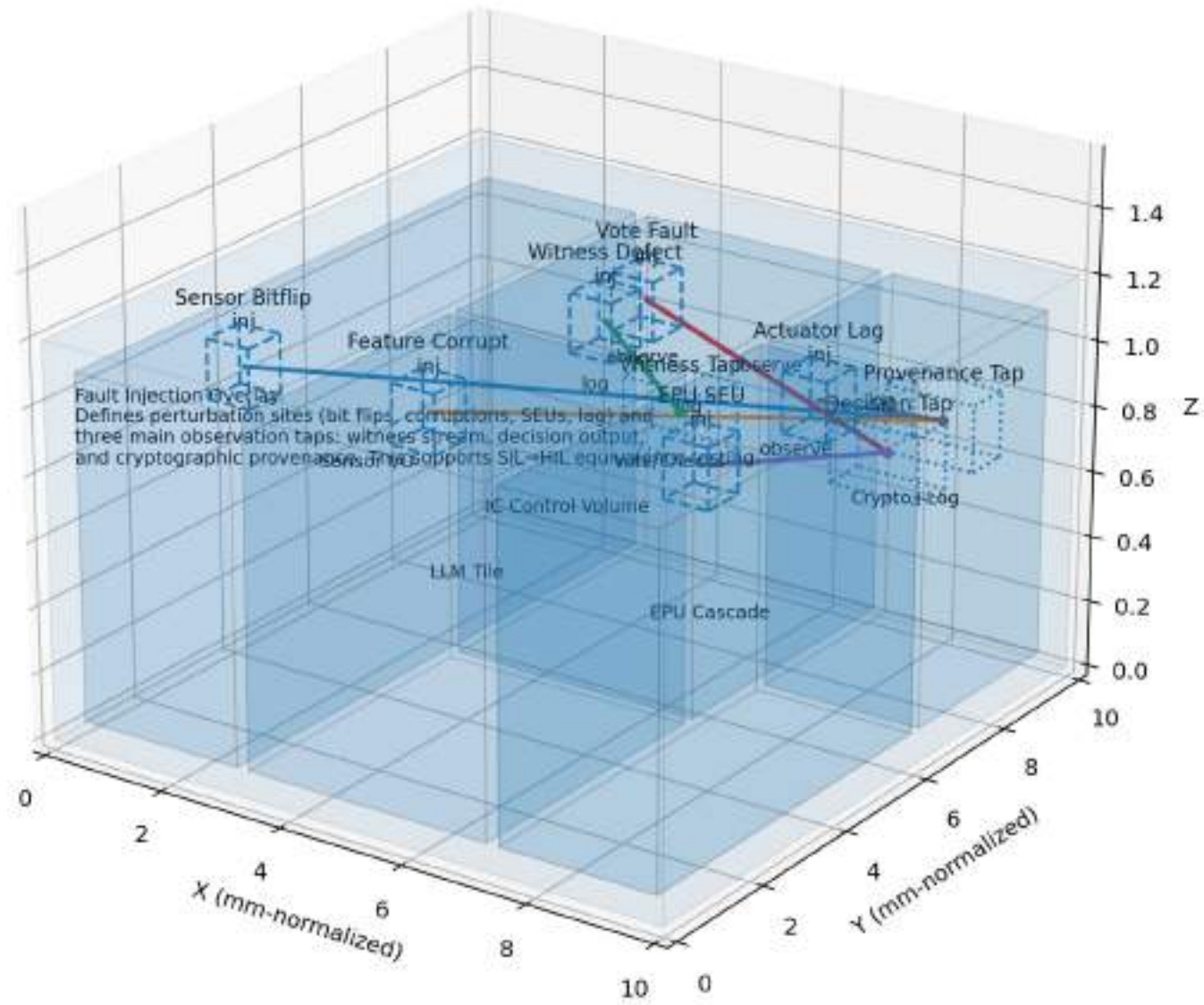
Overlay 41: Contract Coverage Map (Perception, Enforcement, Control)



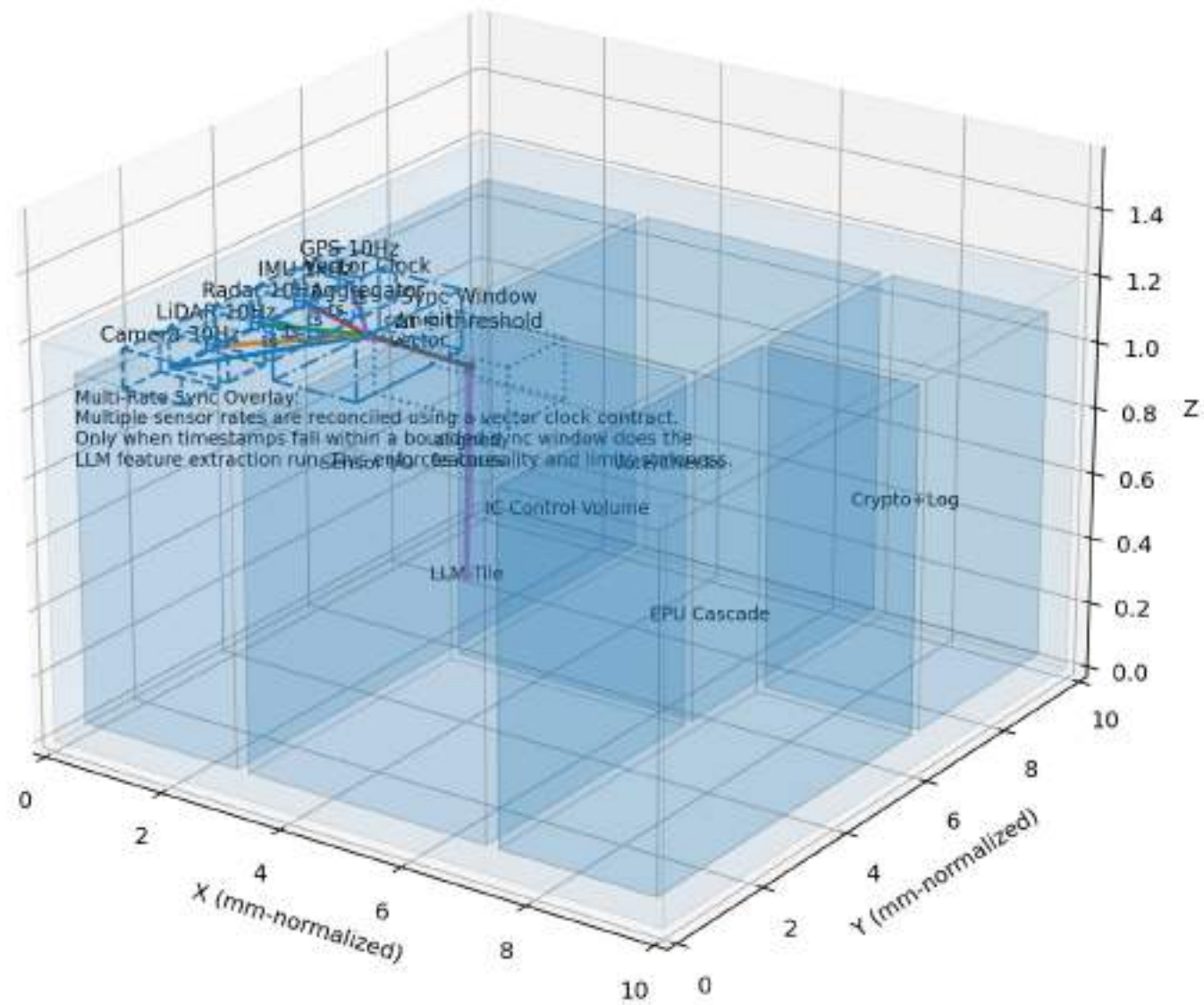
Overlay 42: Fault Injection Topology (Perturb + Observe)



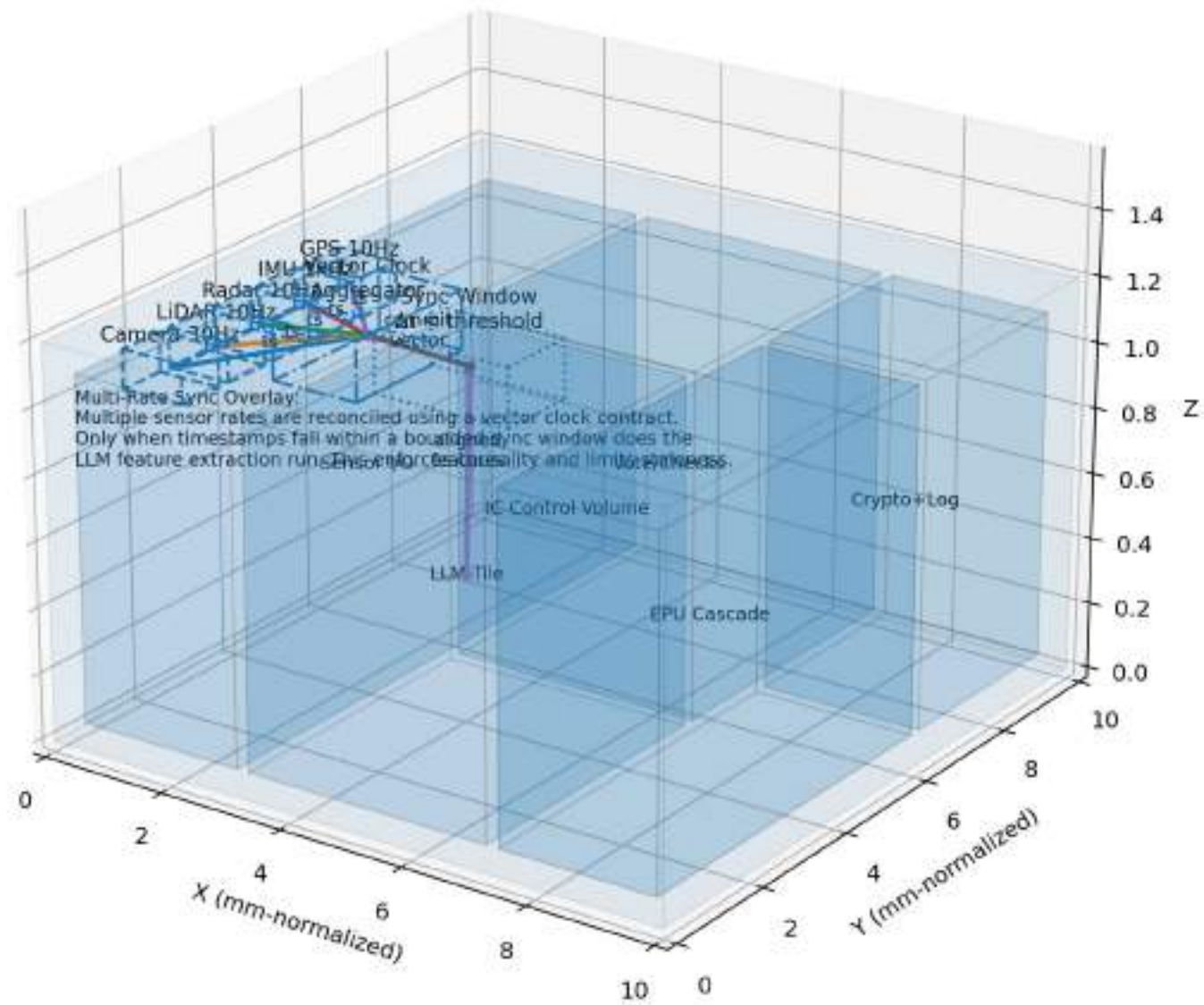
Overlay 42: Fault Injection Topology (Perturb + Observe)



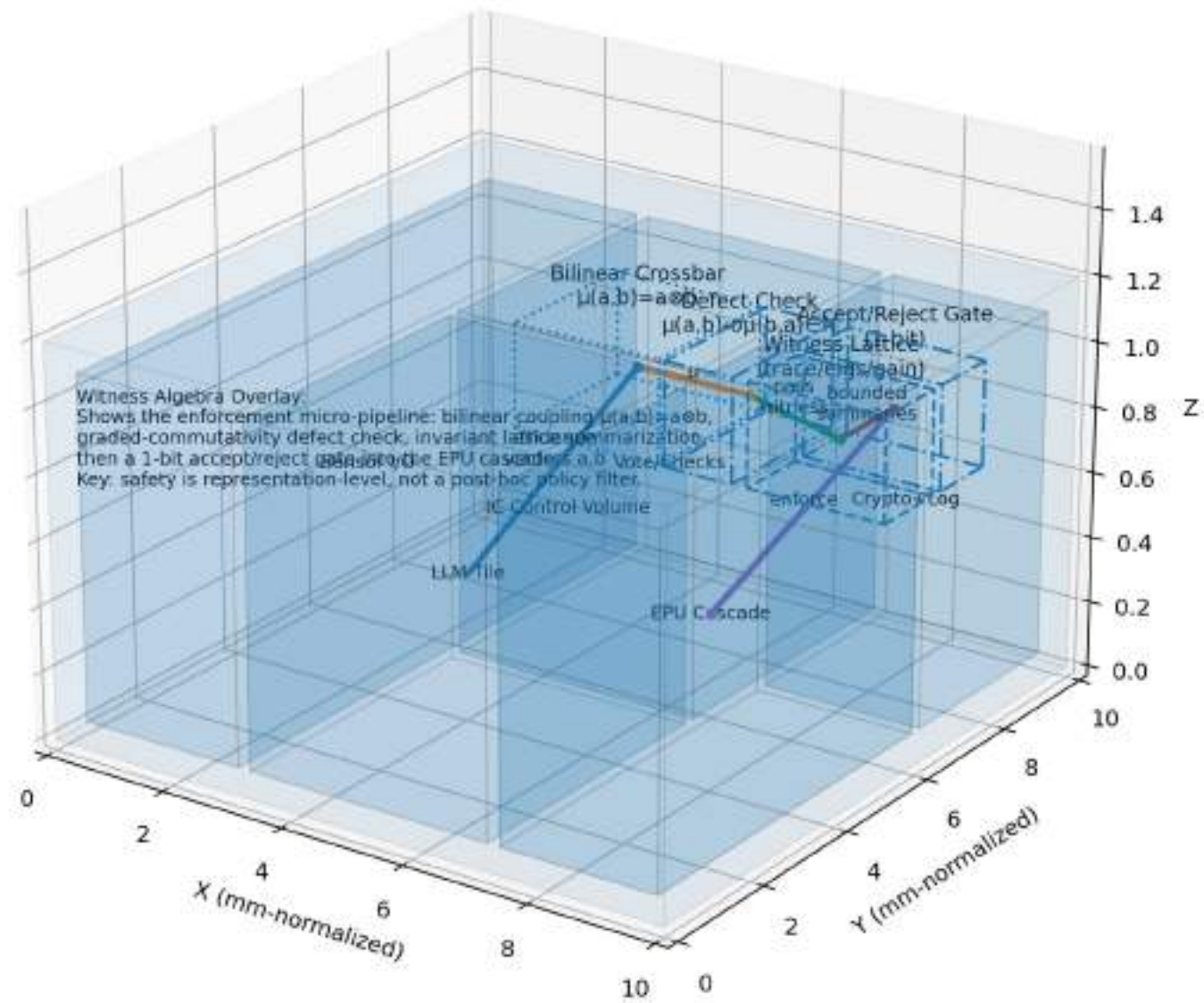
Overlay 43: Multi-Rate Timing & Synchronization (Vector Clock Contracts)



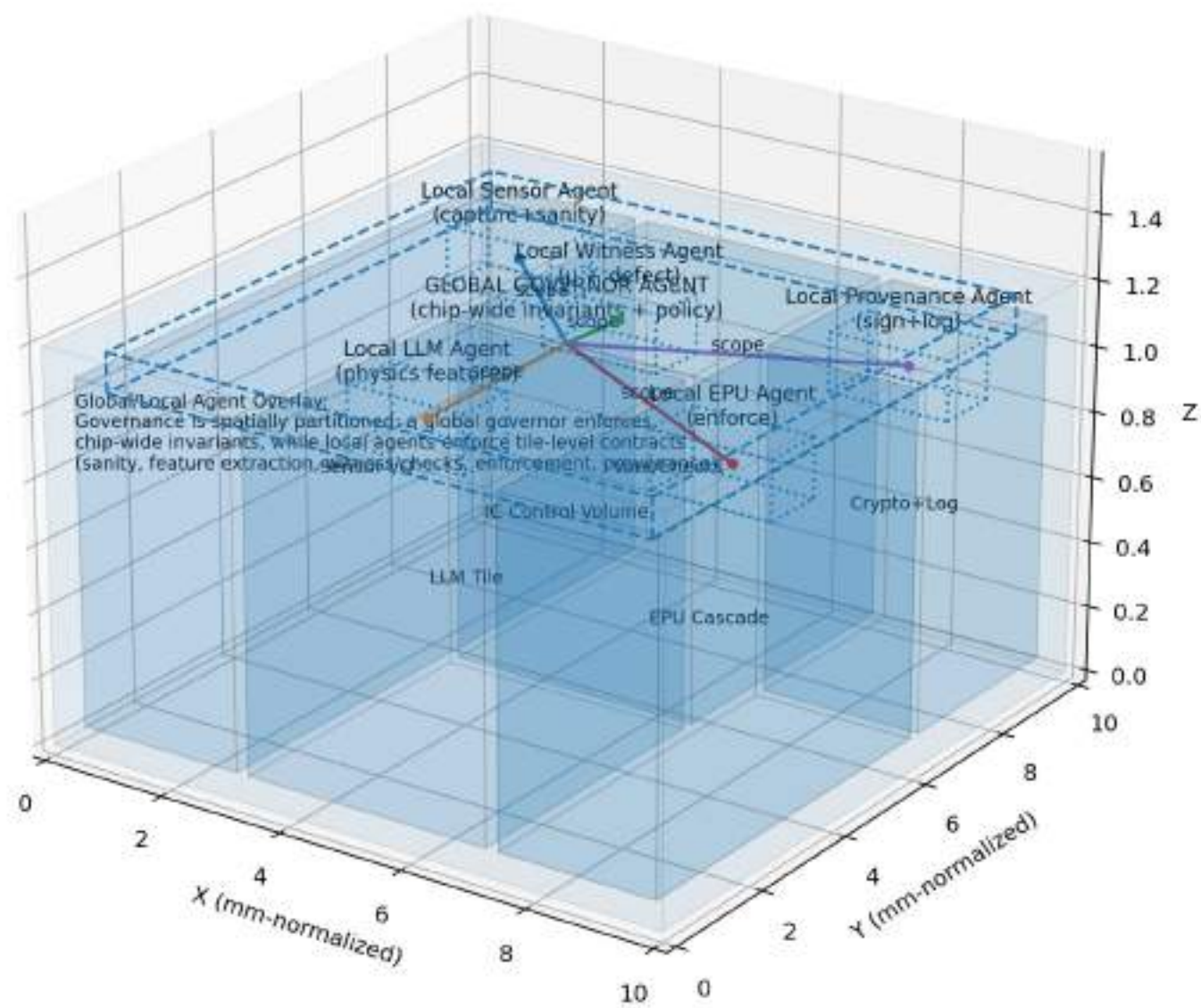
Overlay 43: Multi-Rate Timing & Synchronization (Vector Clock Contracts)



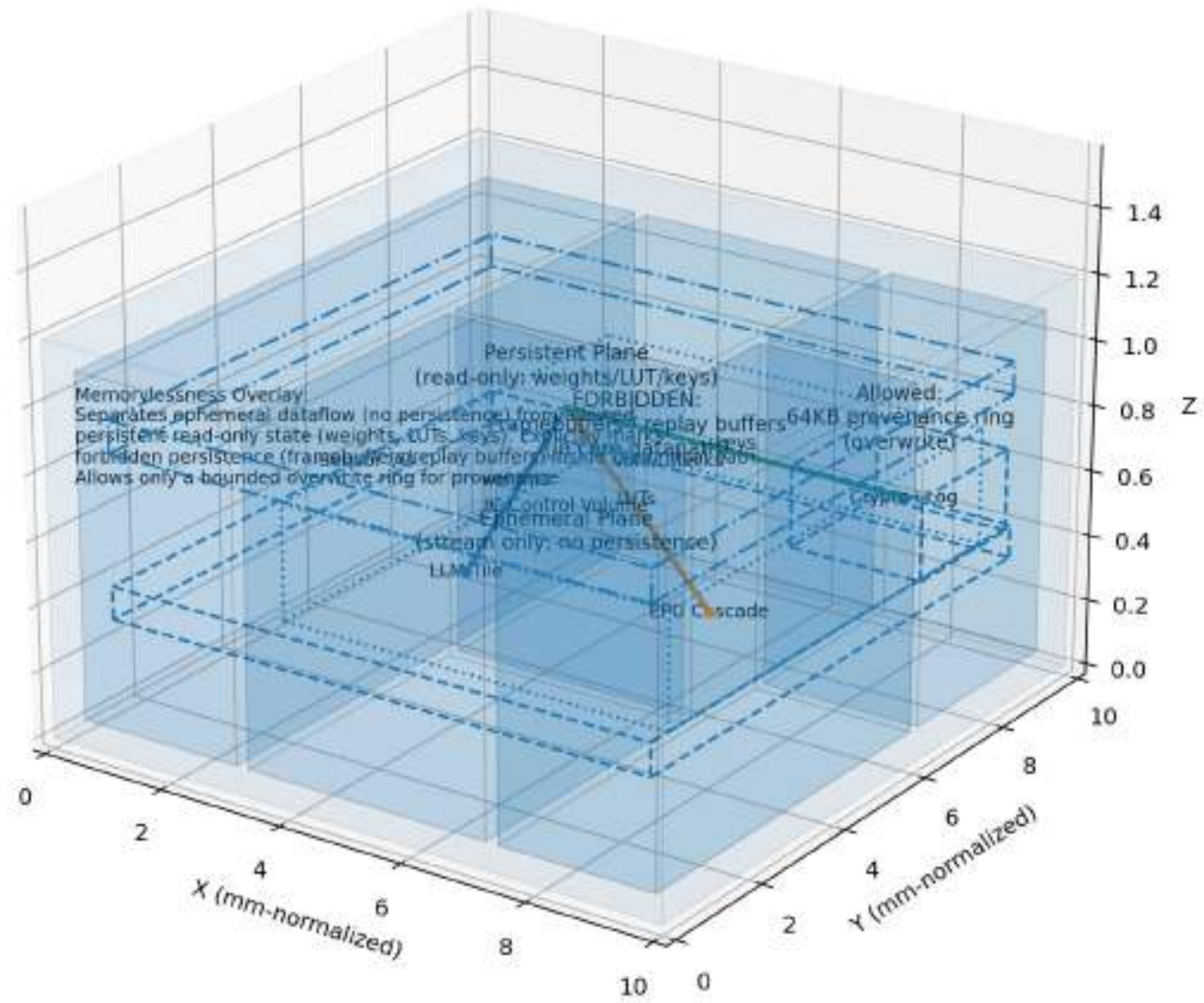
Overlay 44: Witness Algebra Pipeline (μ , Defect, Lattice, Gate)



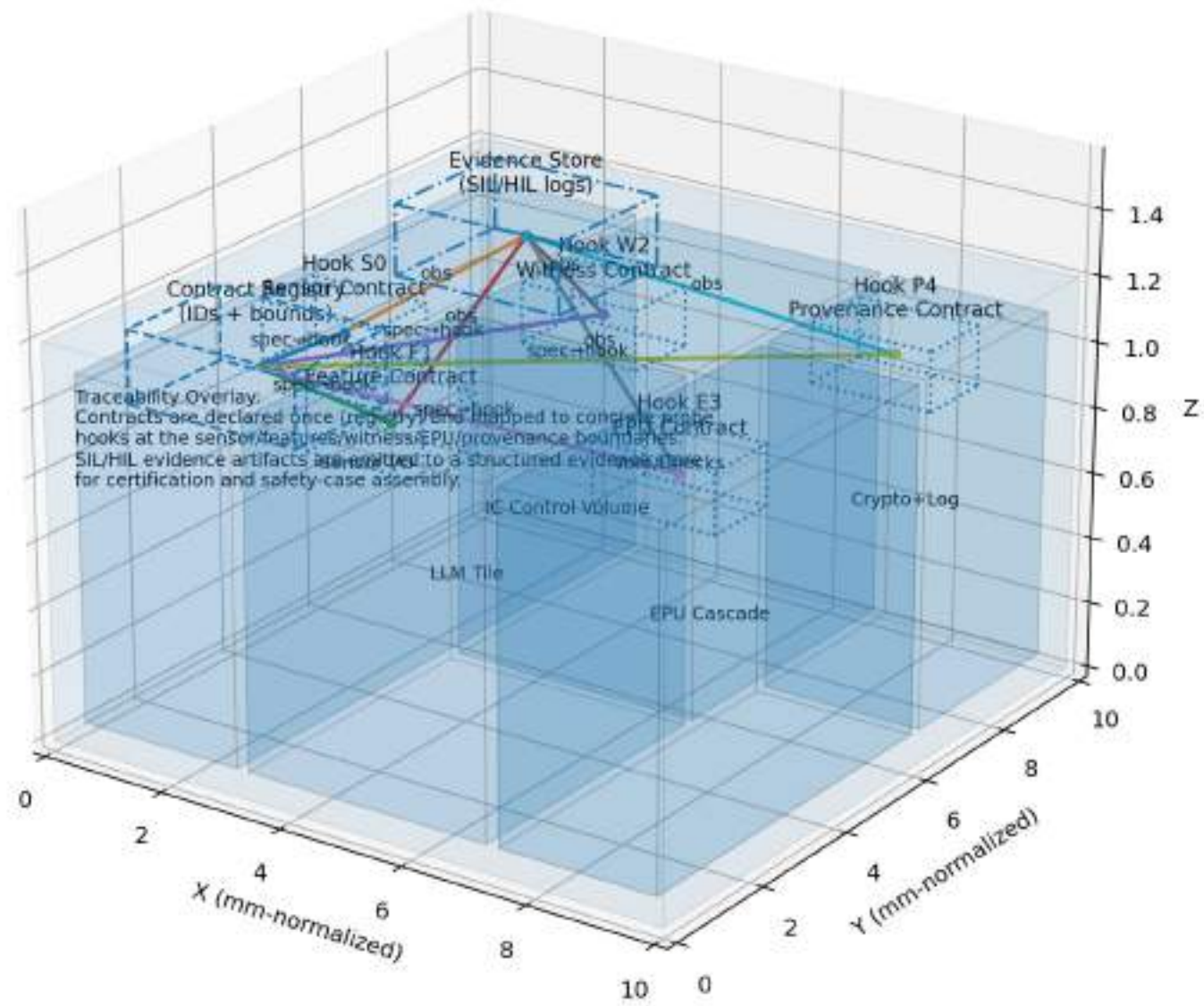
Overlay 45: Spatial Governance (Global vs Local Agents)



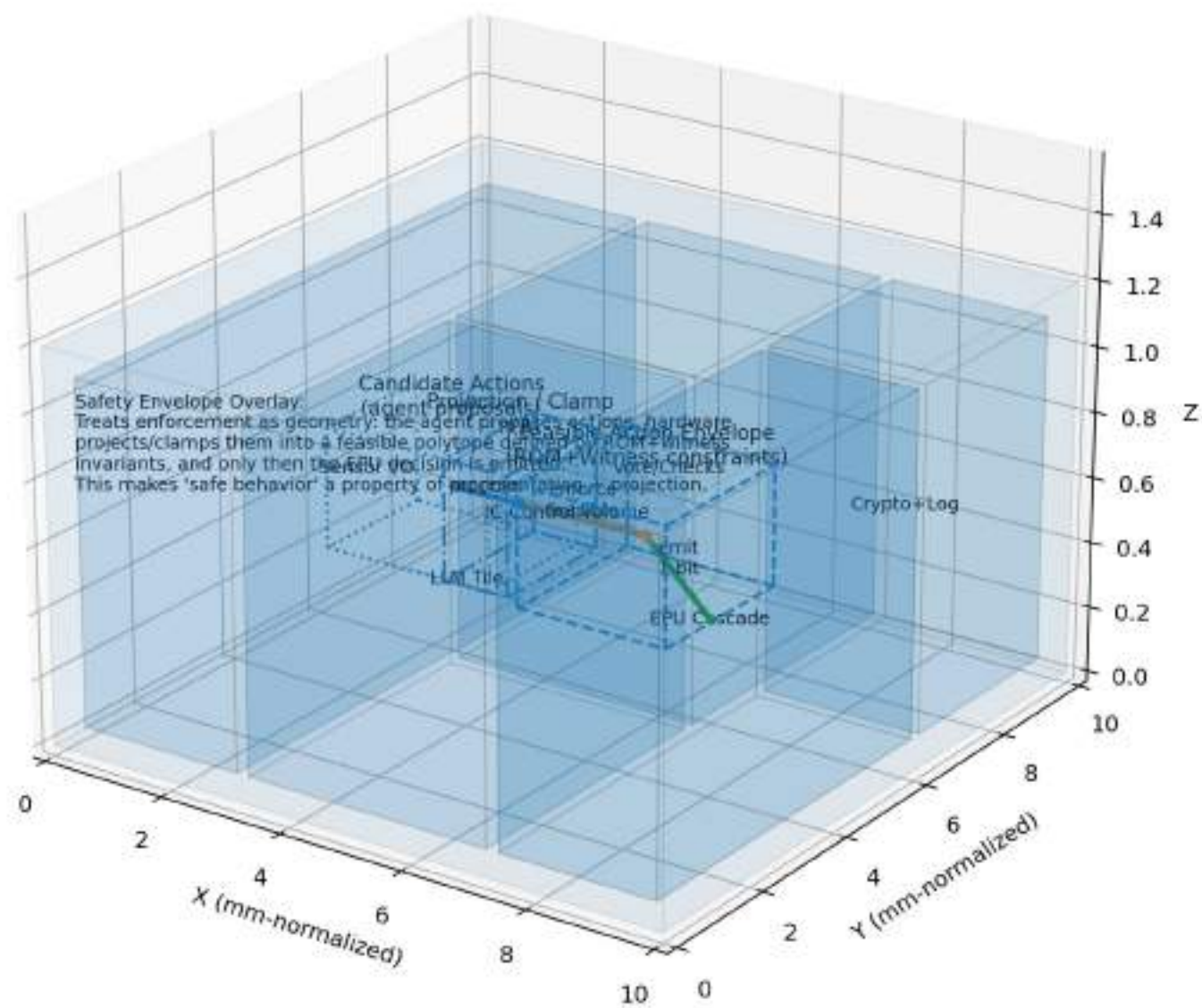
Overlay 46: Memorylessness Boundaries (Ephemeral vs Persistent Planes)



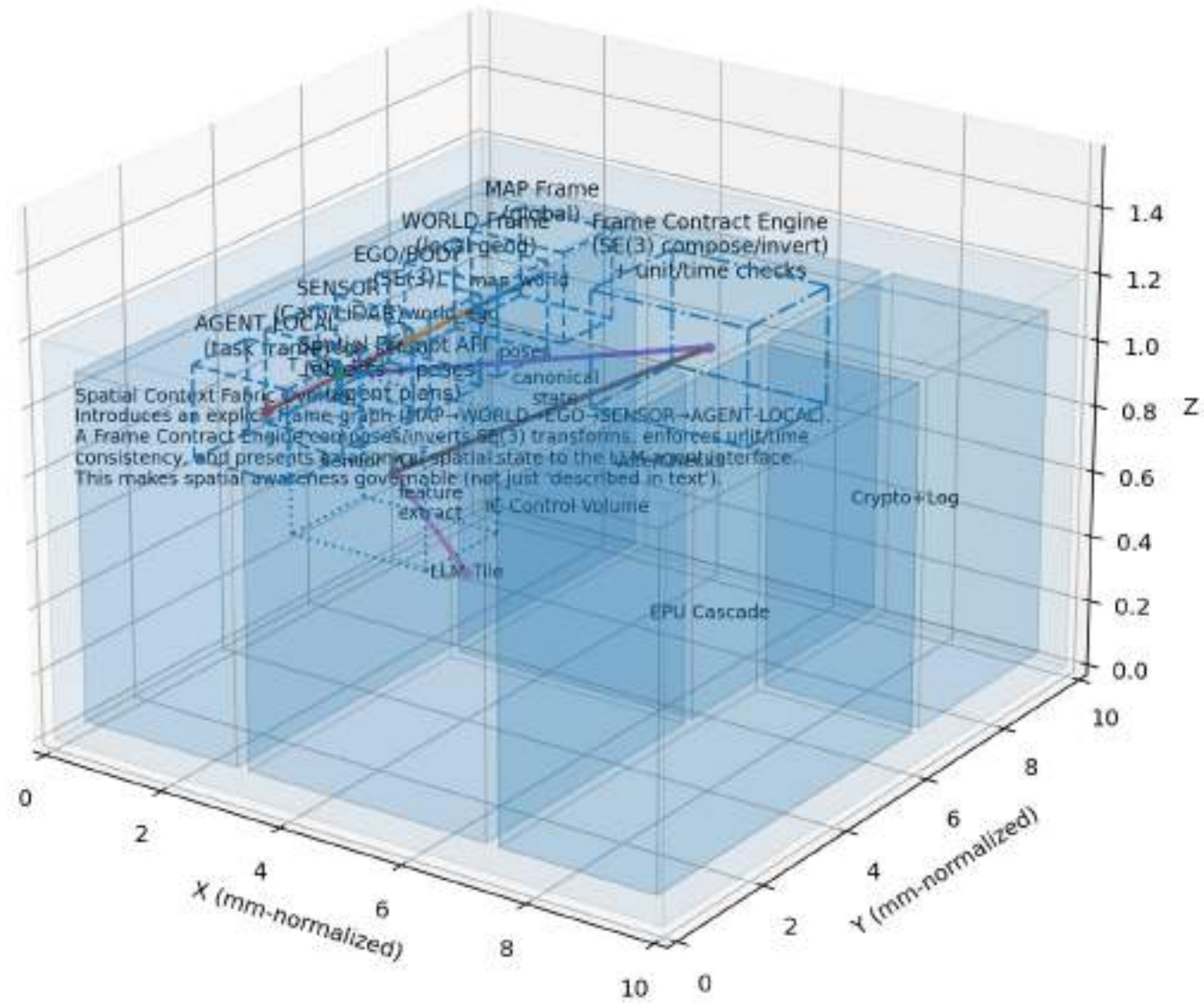
Overlay 47: Spec-to-Test Traceability (Contracts → Hooks → Evidence)



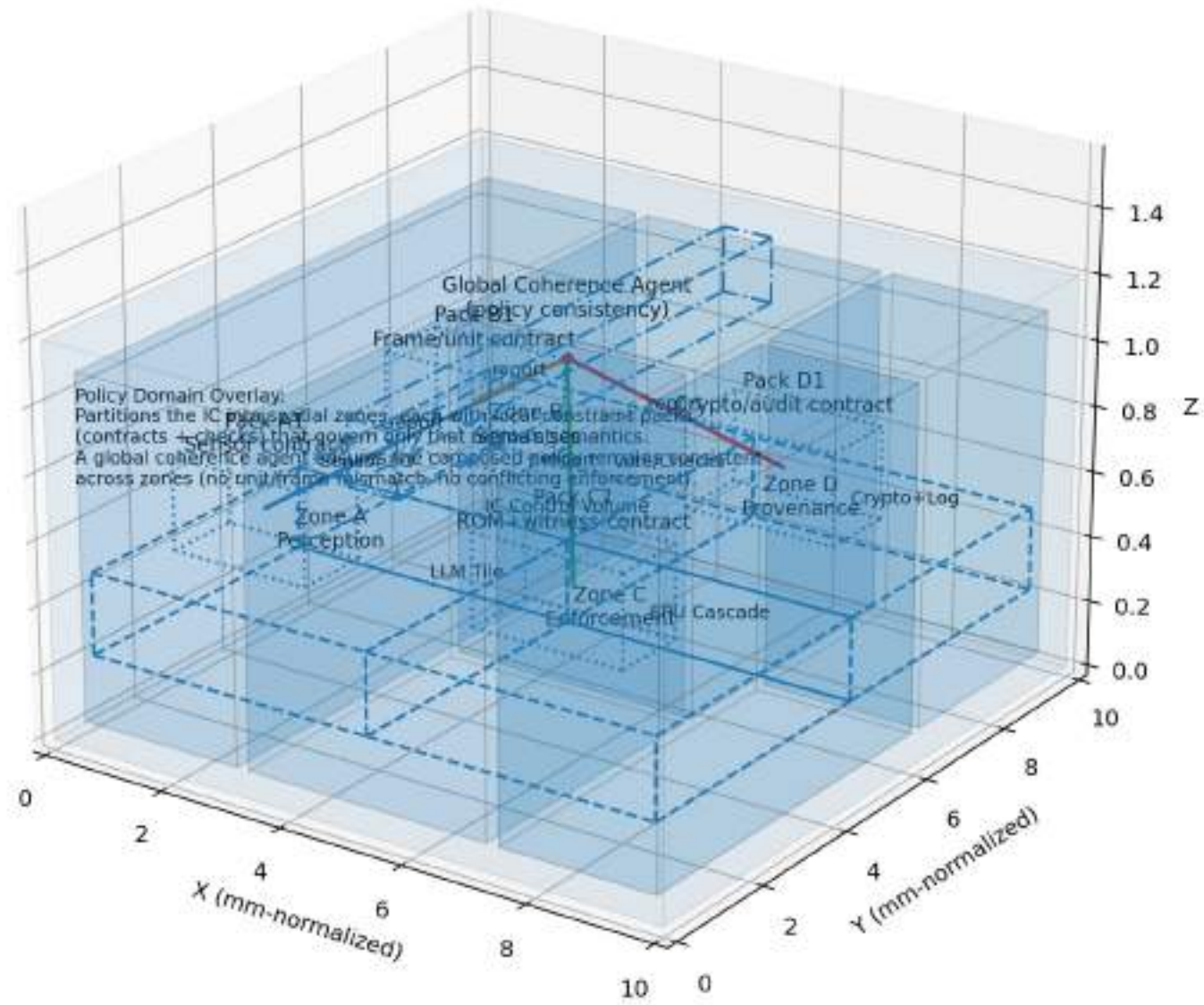
Overlay 48: Safety Envelope Geometry (Feasible Action Polytope)



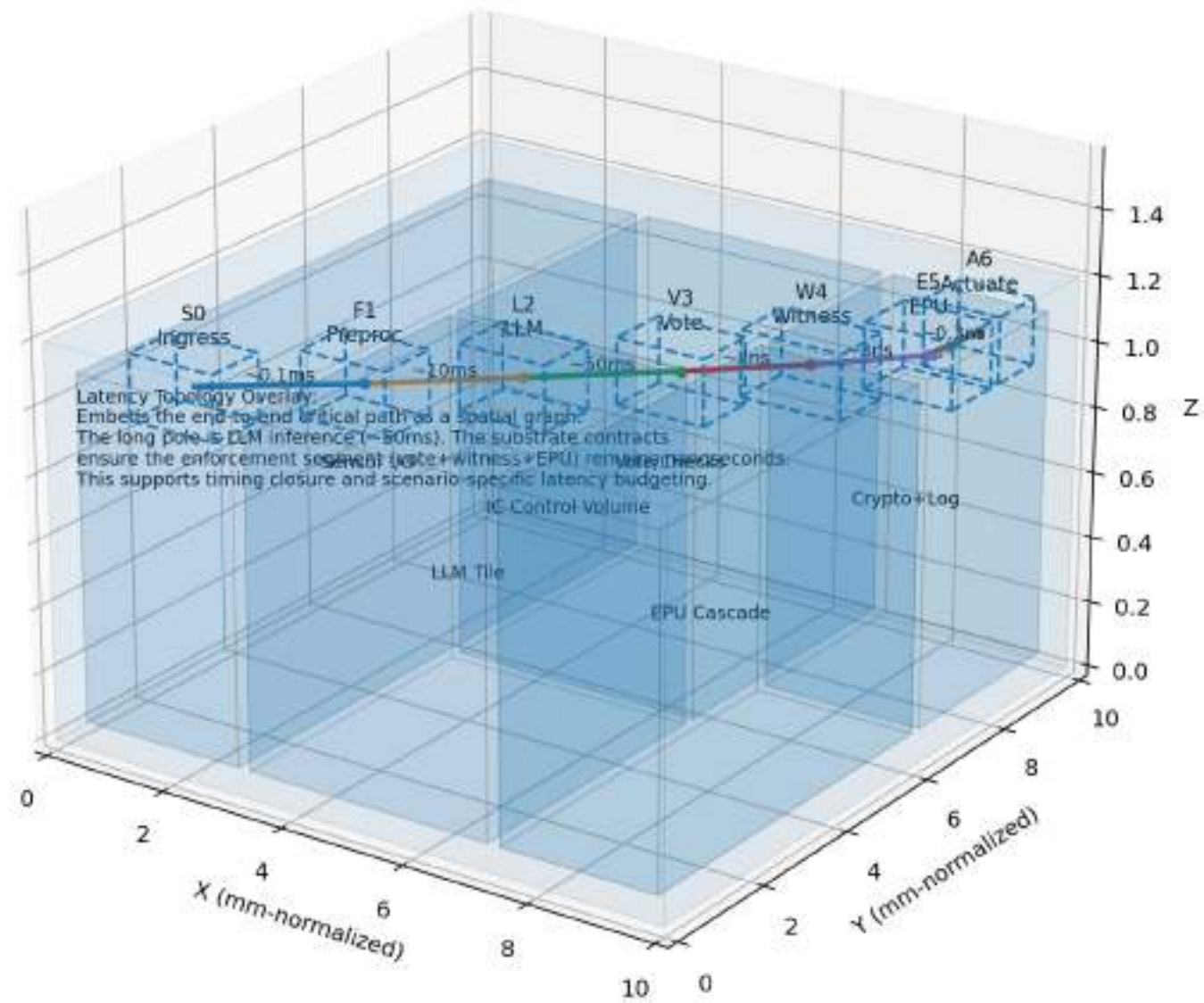
Overlay 49: Spatial Context Fabric (Frame Graph for LLM Agents)



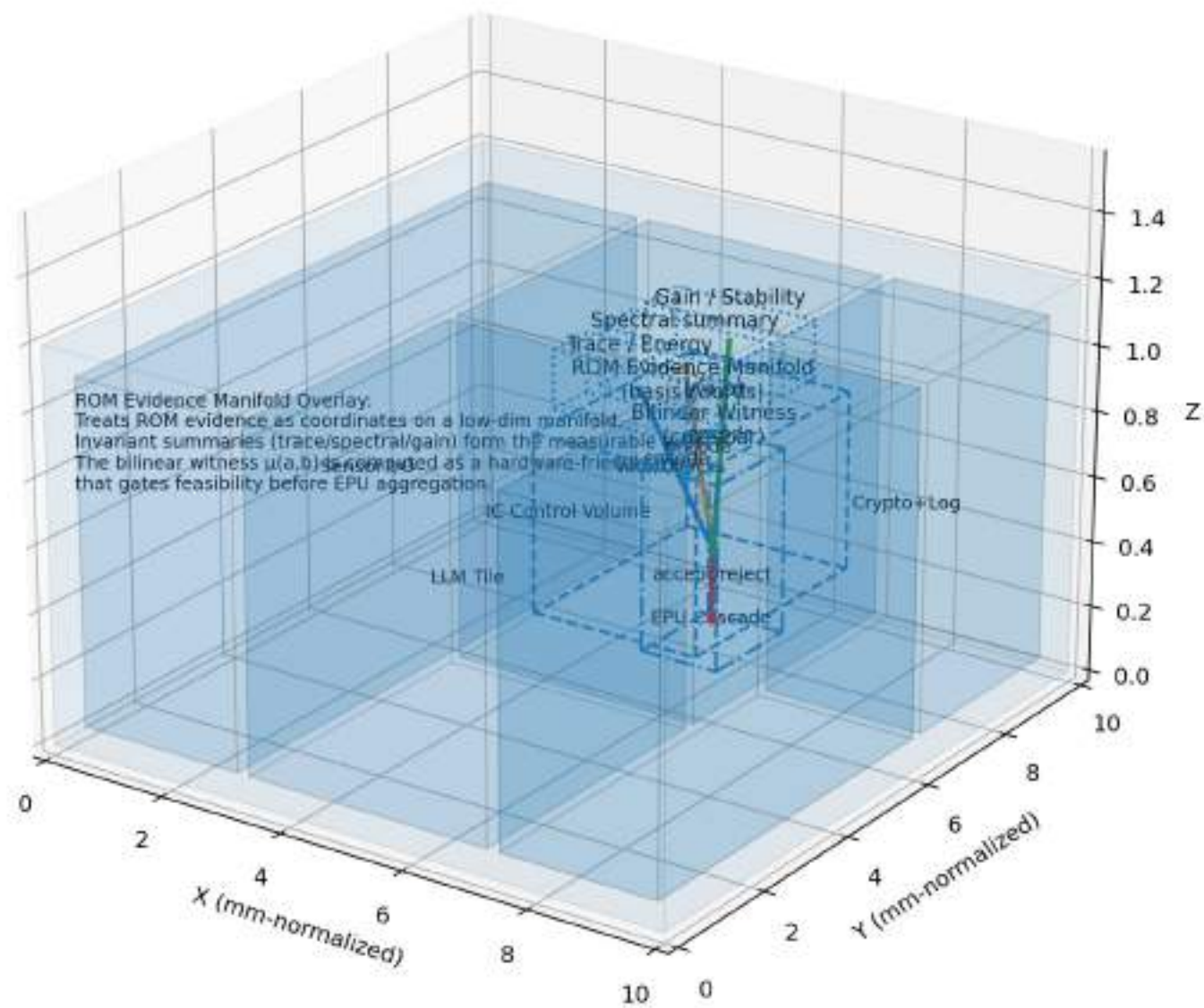
Overlay 50: Spatially-Scoped Policy Domains (Local Constraint Packs)



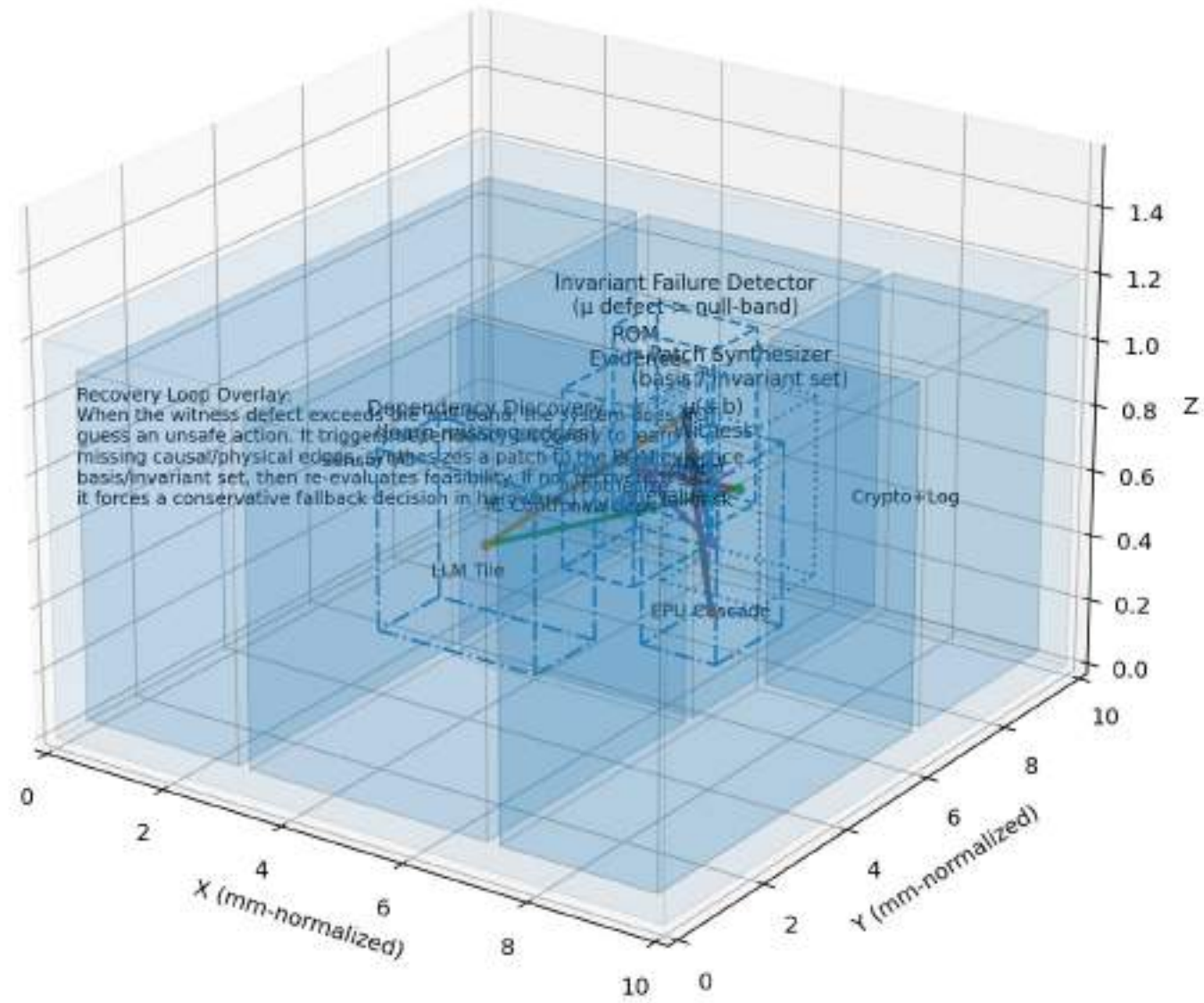
Overlay 51: End-to-End Latency Topology (Spatial Critical Path)



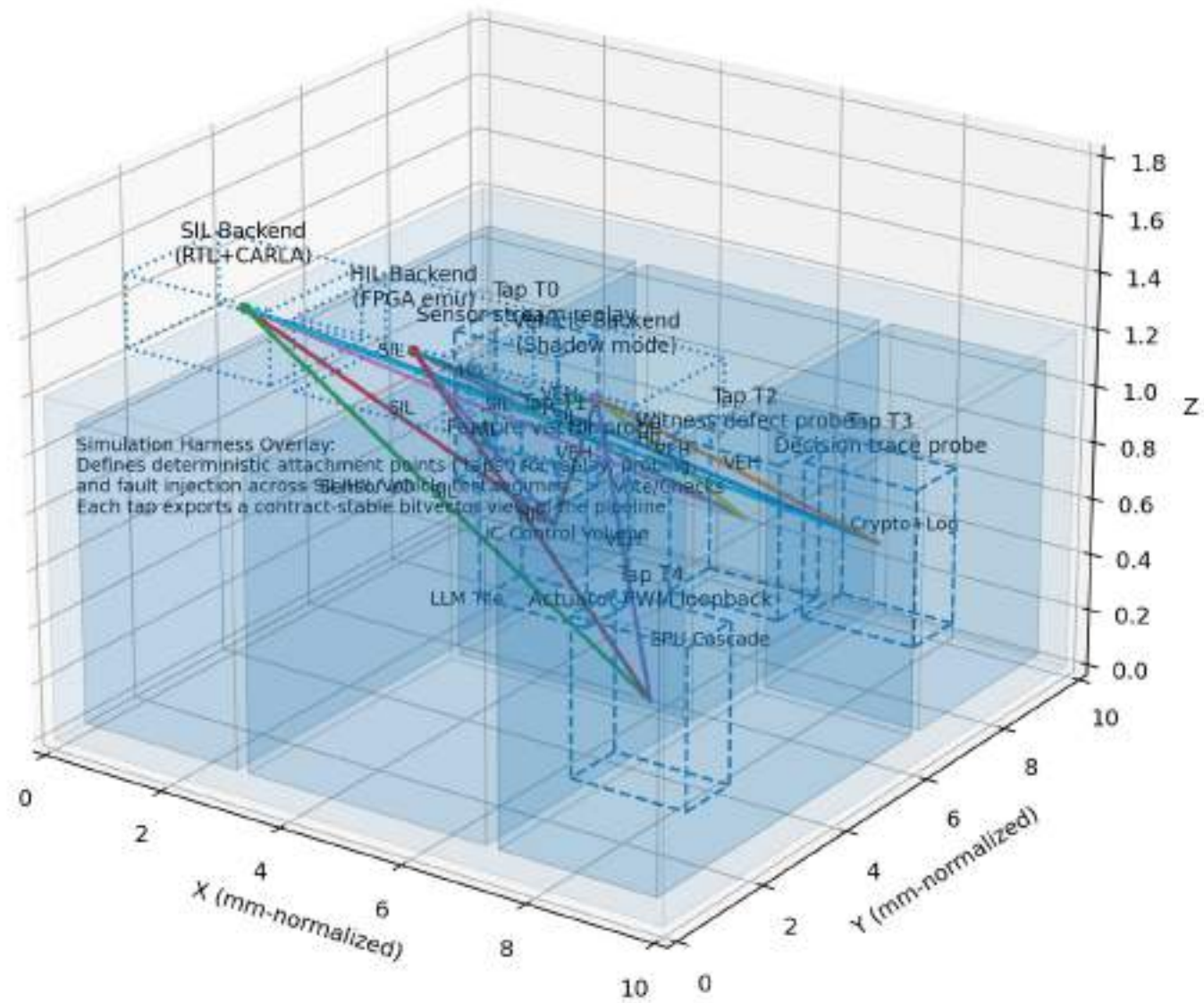
Overlay 52: ROM Evidence Manifold (Invariants as Coordinates)



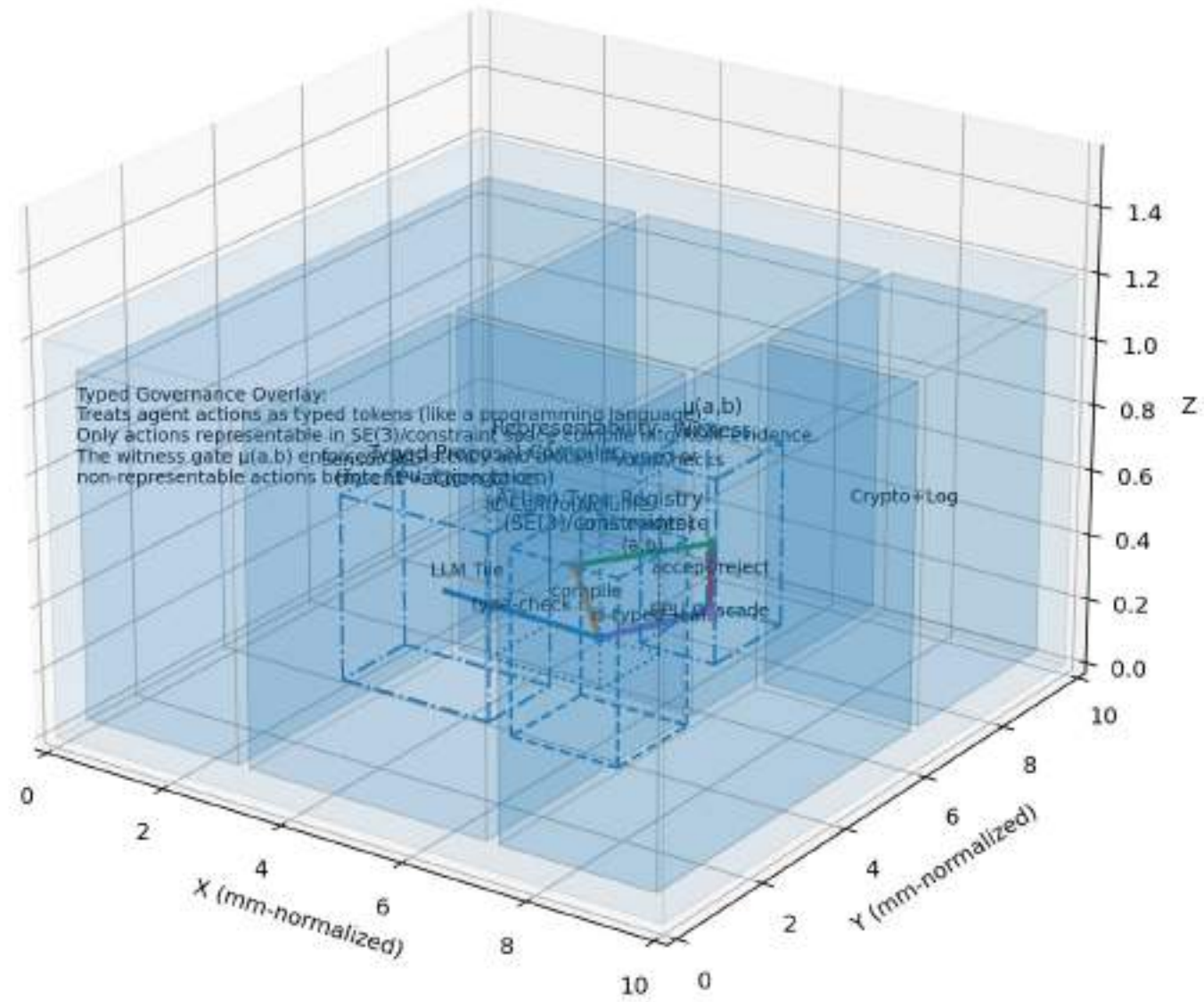
Overlay 53: Failure-to-Feasible Response (Dependency Discovery Recovery)



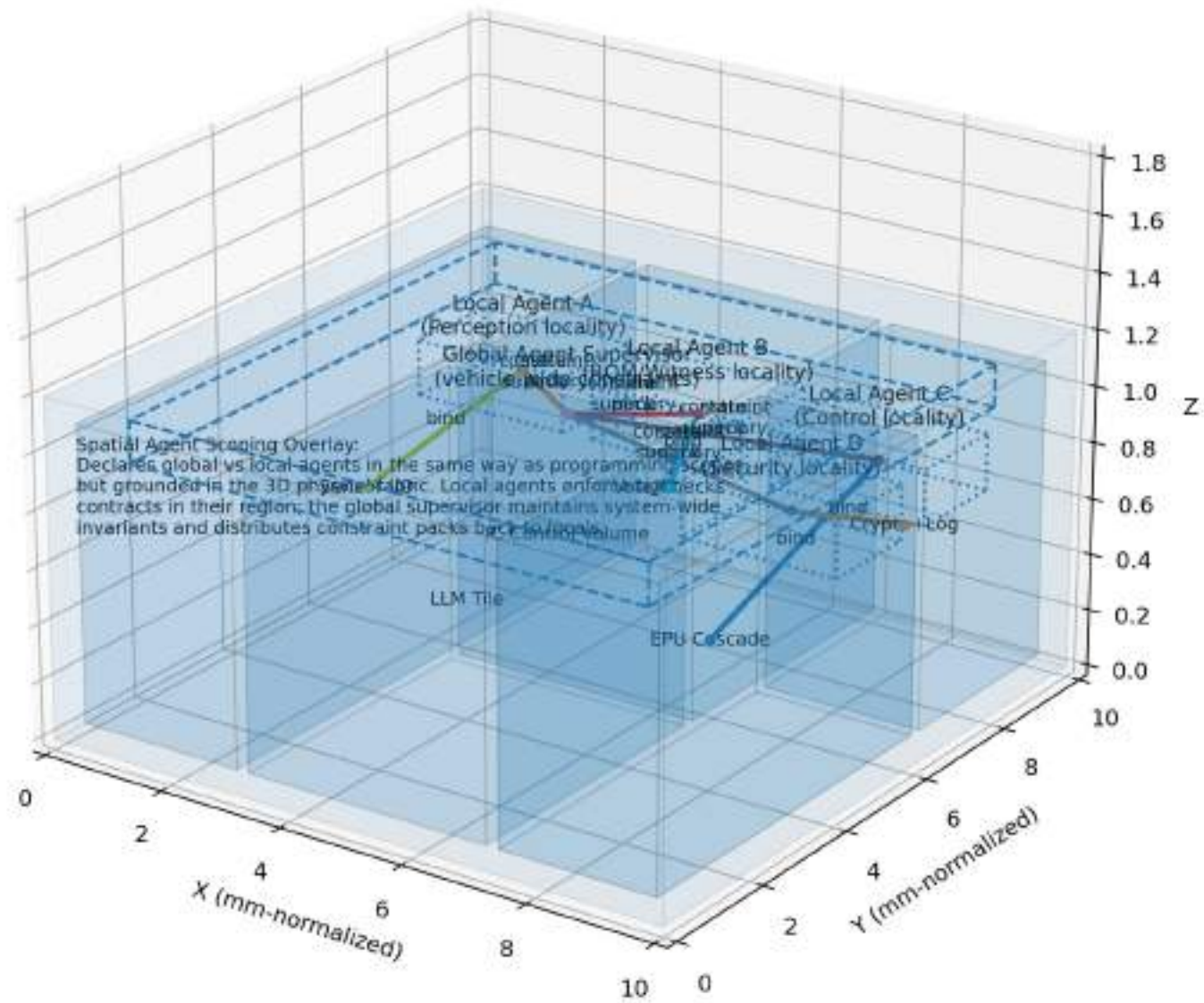
Overlay 54: Simulation Harness Attachment Points (SIL/HIL/Vehicle)



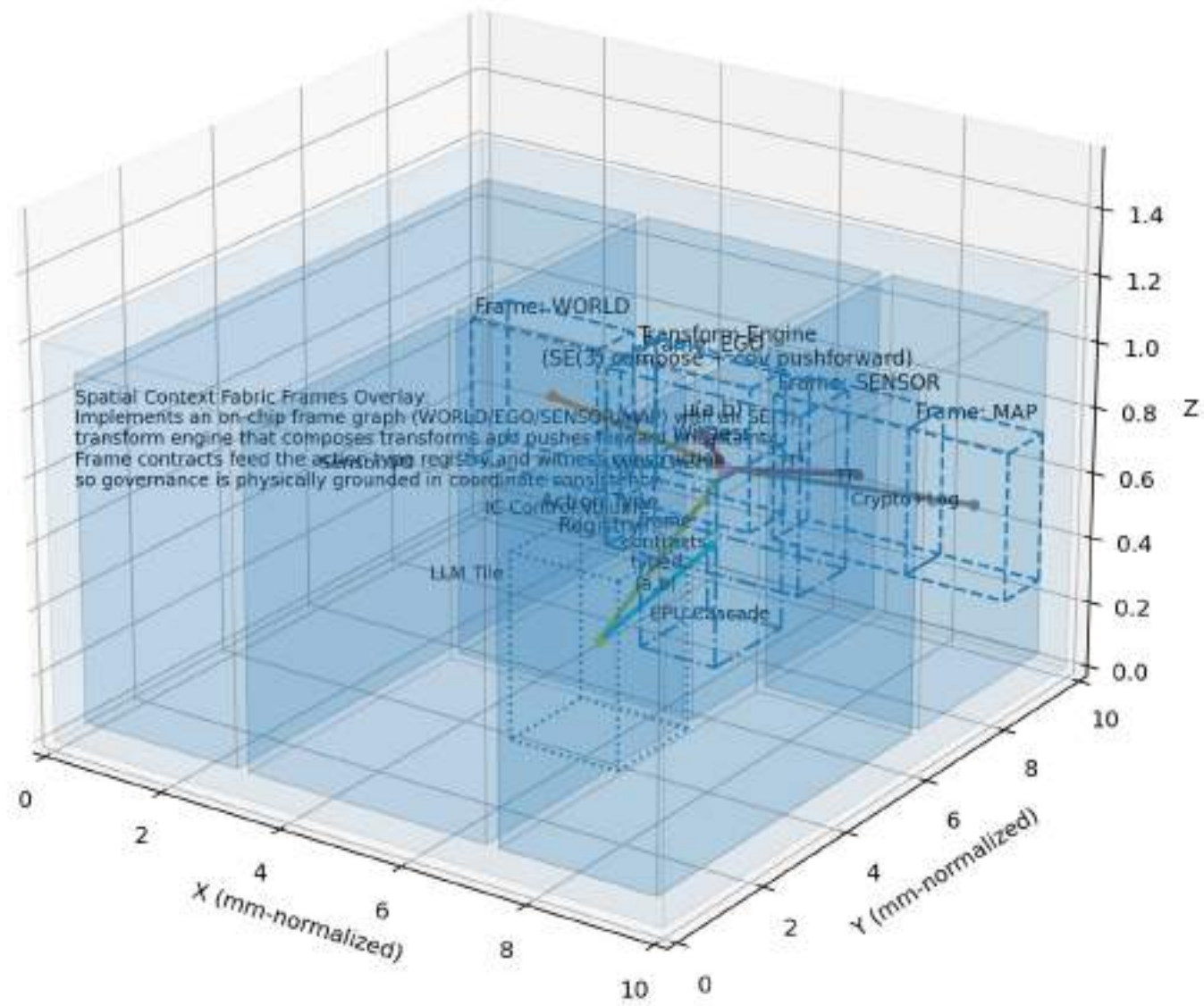
Overlay 55: Governance-to-Action Type System (Typed Representability)



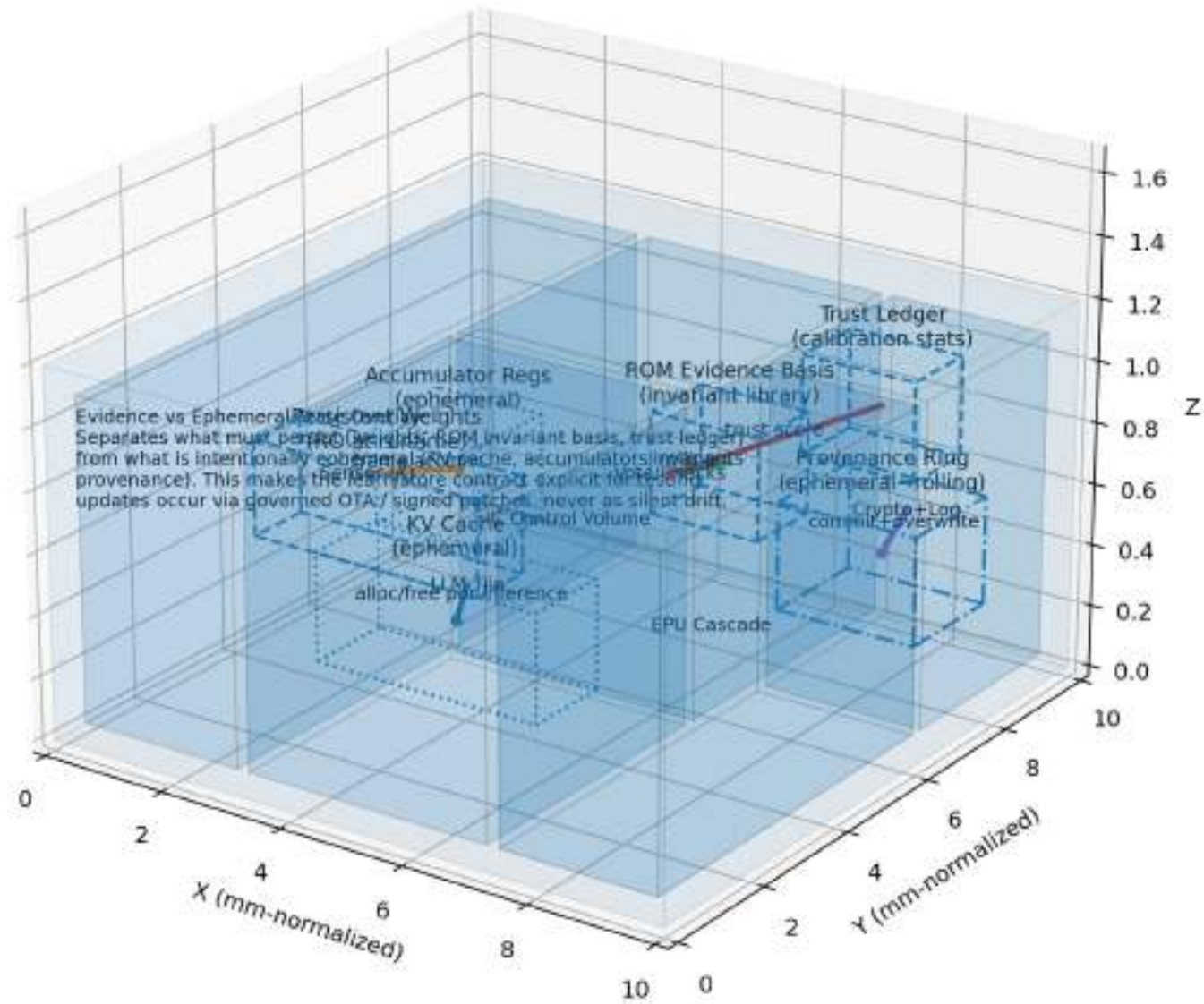
Overlay 56: Spatially-Scoped Agents (Global vs Local in 3D Fabric)



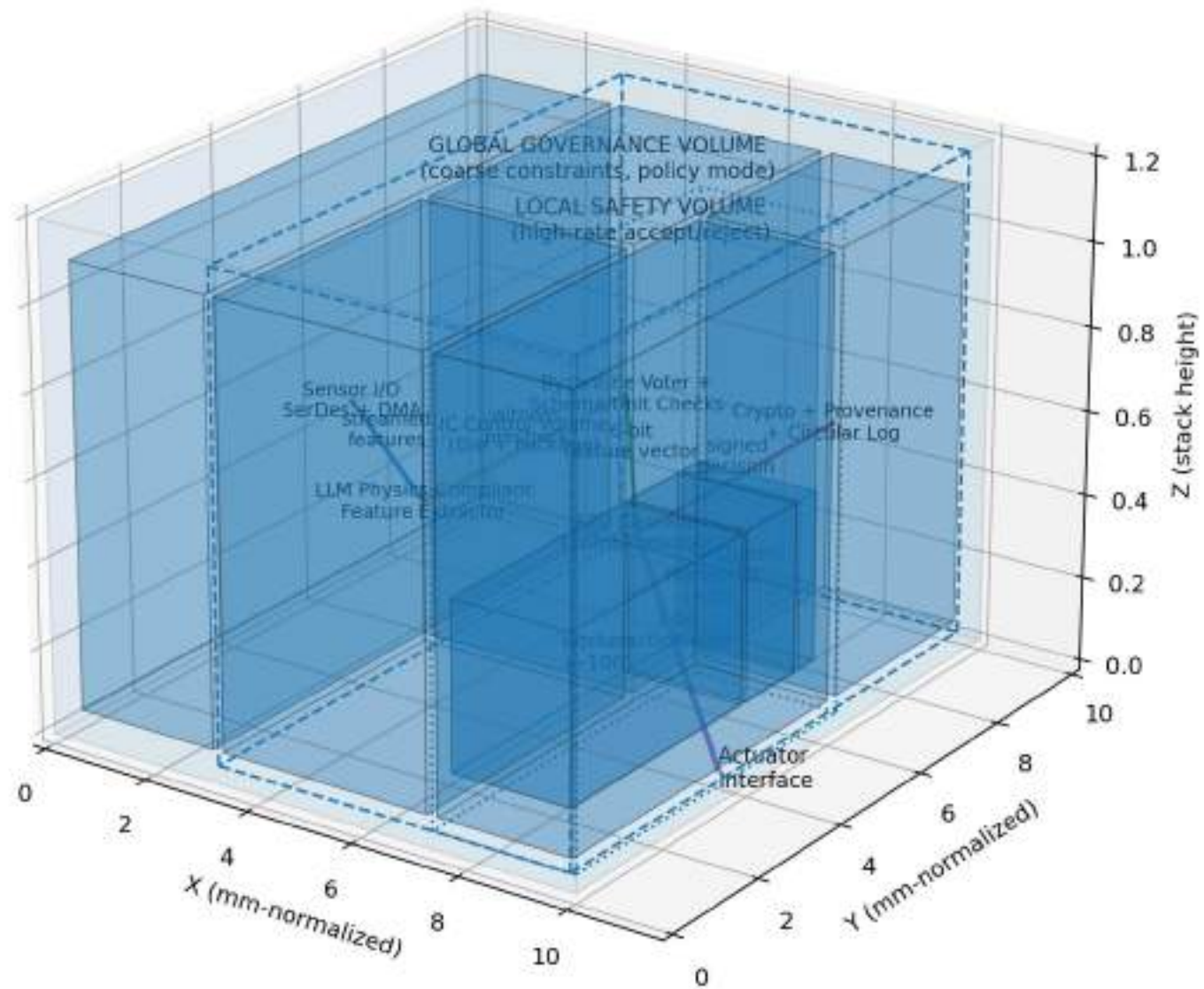
Overlay 57: Spatial Context Fabric Frames (On-chip Frame Graph)



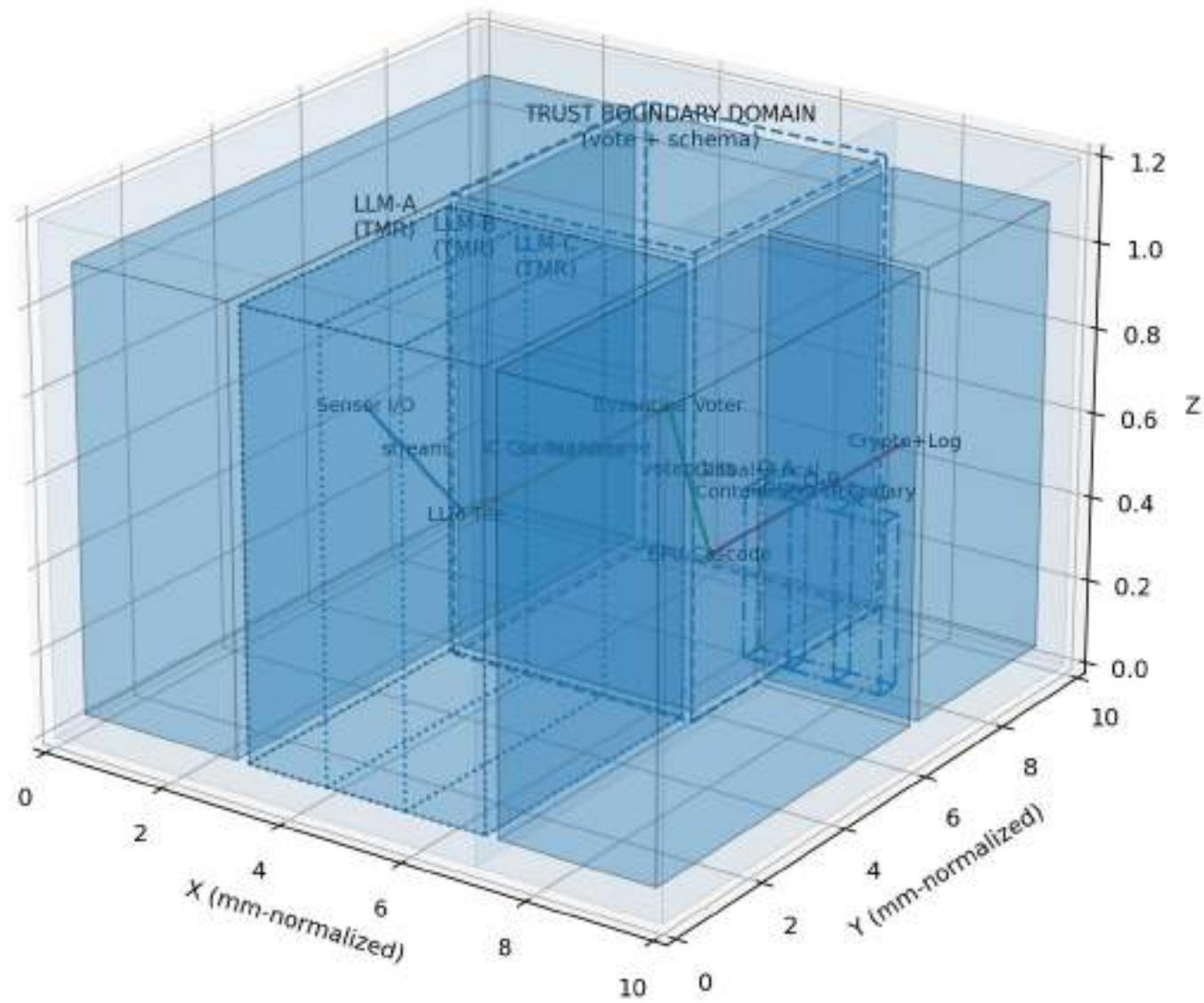
Overlay 58: Evidence Store vs Ephemeral State (Learning/Storage Contract)



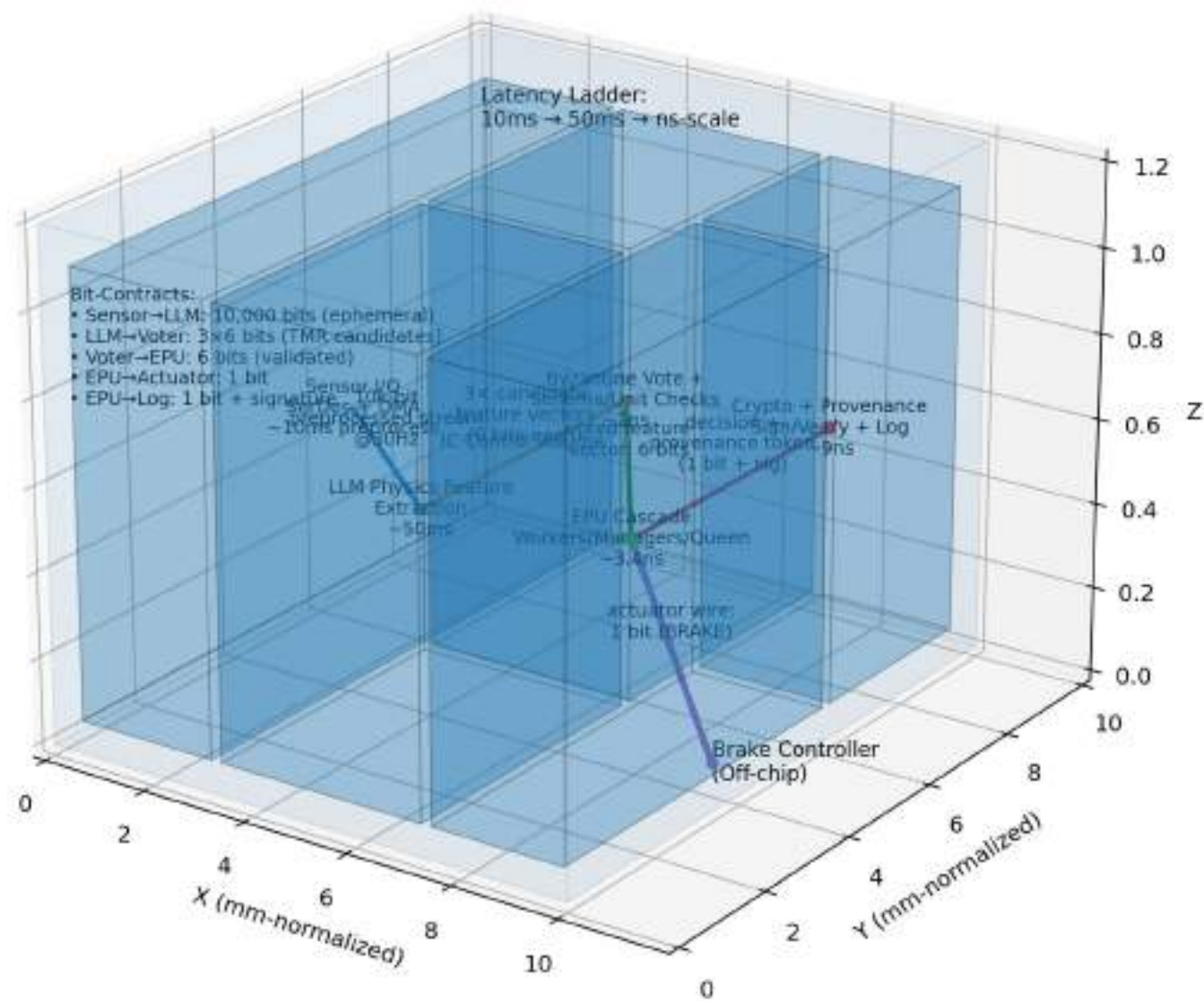
Overlay 1: Authority Volumes + Witness/Decision Bus (Conceptual IC 3D Control Volume)



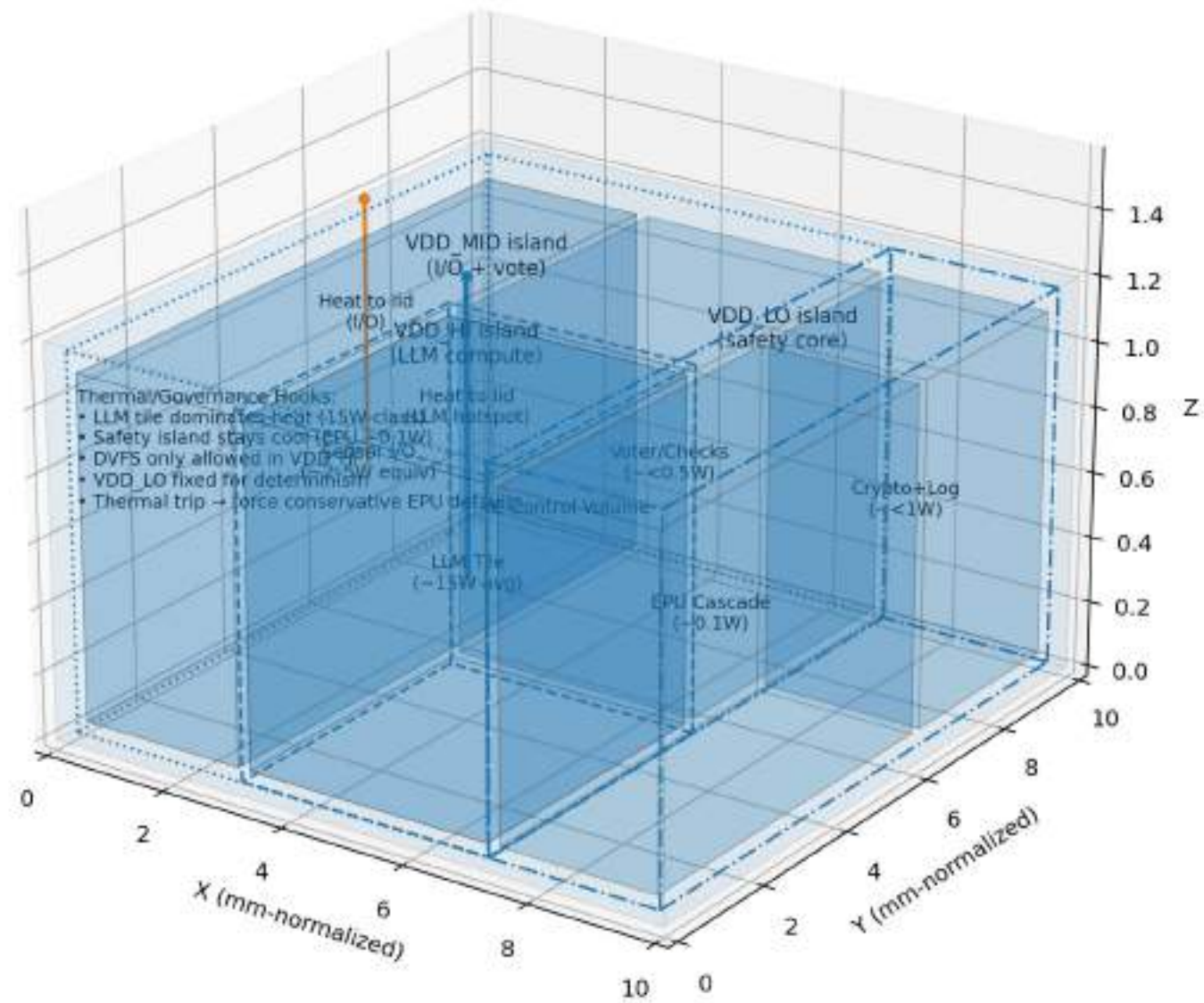
Overlay 2: Fault Containment + Redundancy Domains (LLM TMR + Queen TMR)



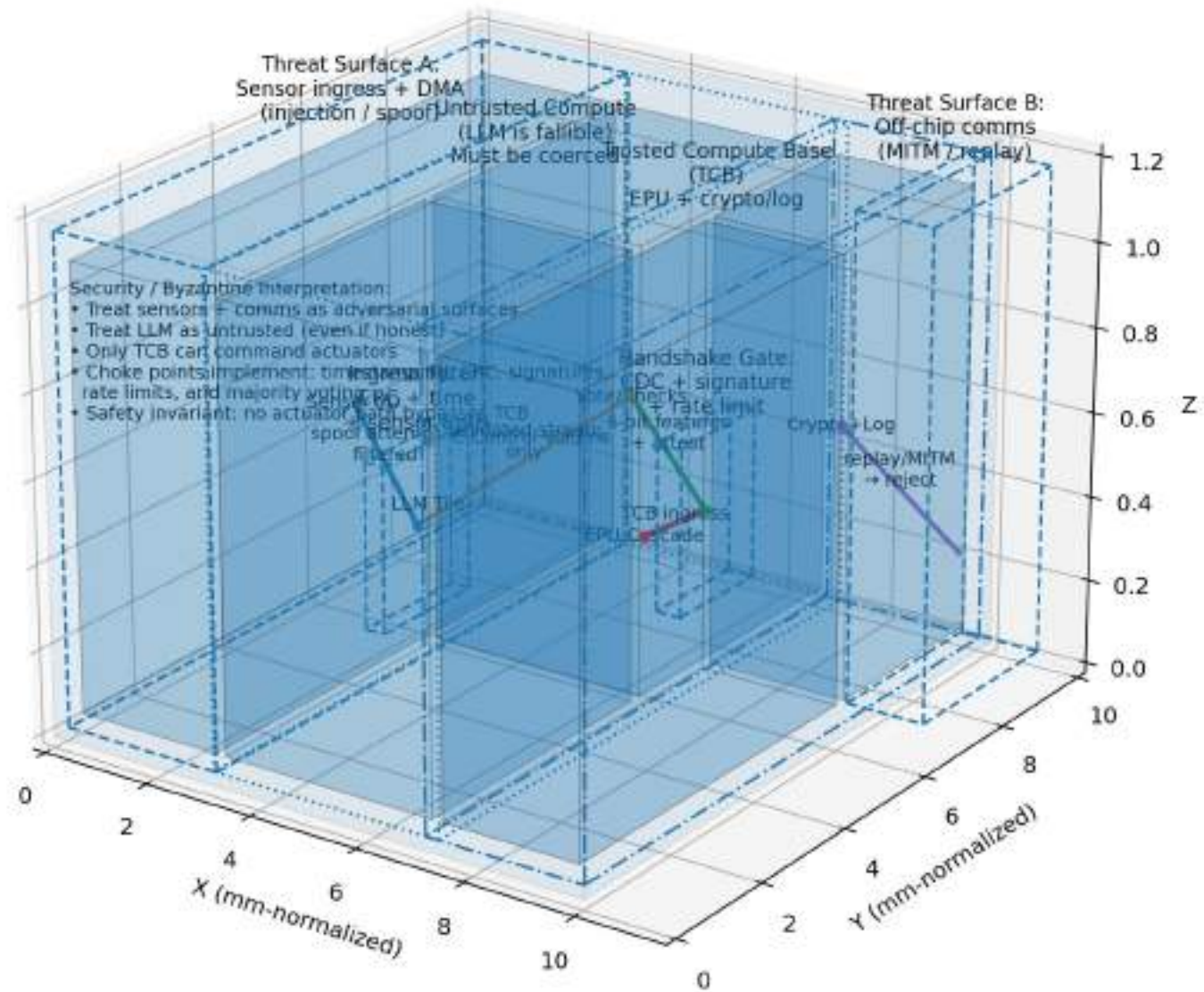
Overlay 3: End-to-End Streaming Dataflow + Bitwidth Contracts + Latency Budgets



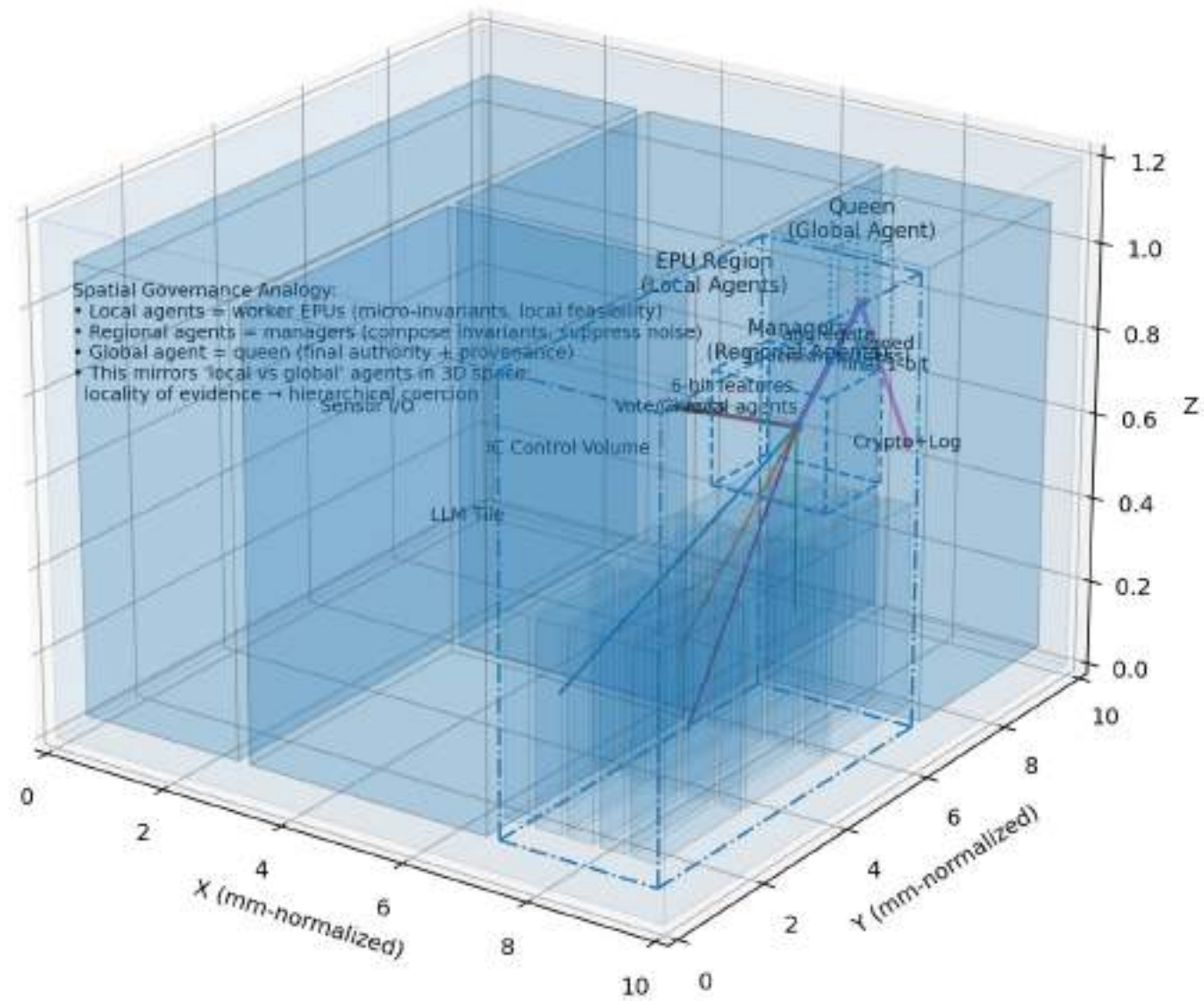
Overlay 4: Power/Thermal Islands + Voltage Domains (DVFS Constrained to Non-Safety Regions)



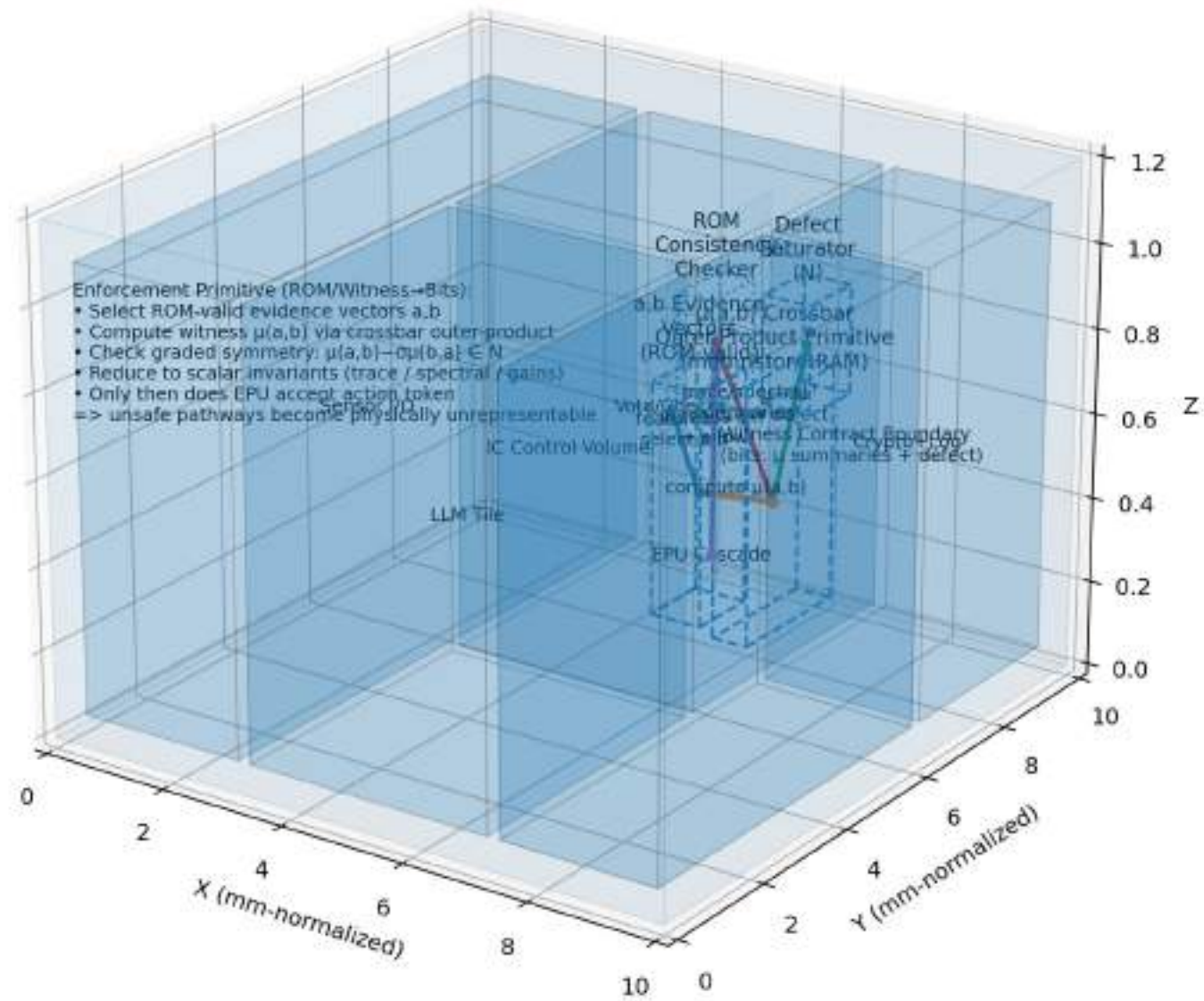
Overlay 7: Trust Boundaries + Threat Surfaces (Byzantine Model)



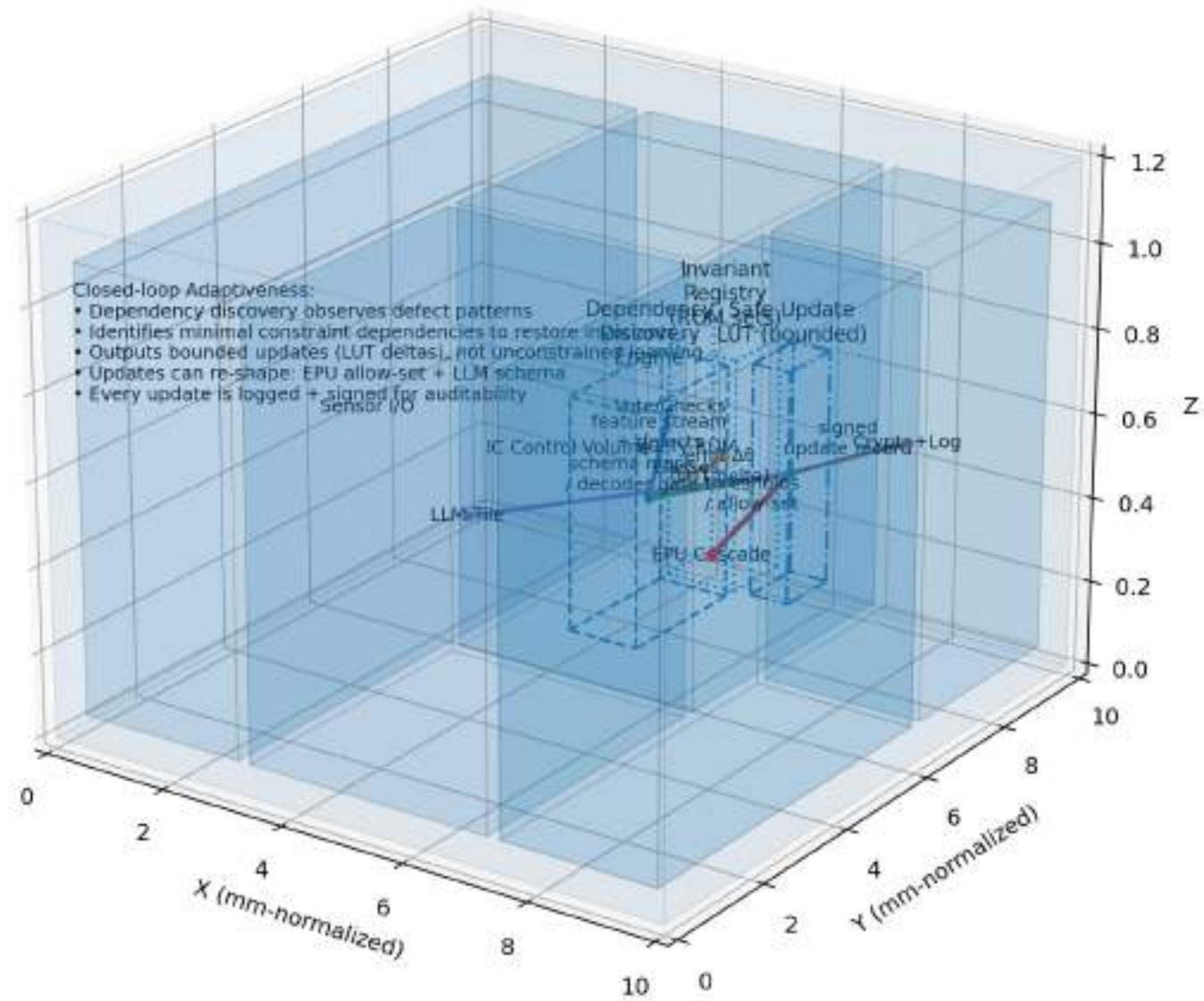
Overlay 9: Global vs Local Agent Decomposition (3D Control Volume Governance Mapping)



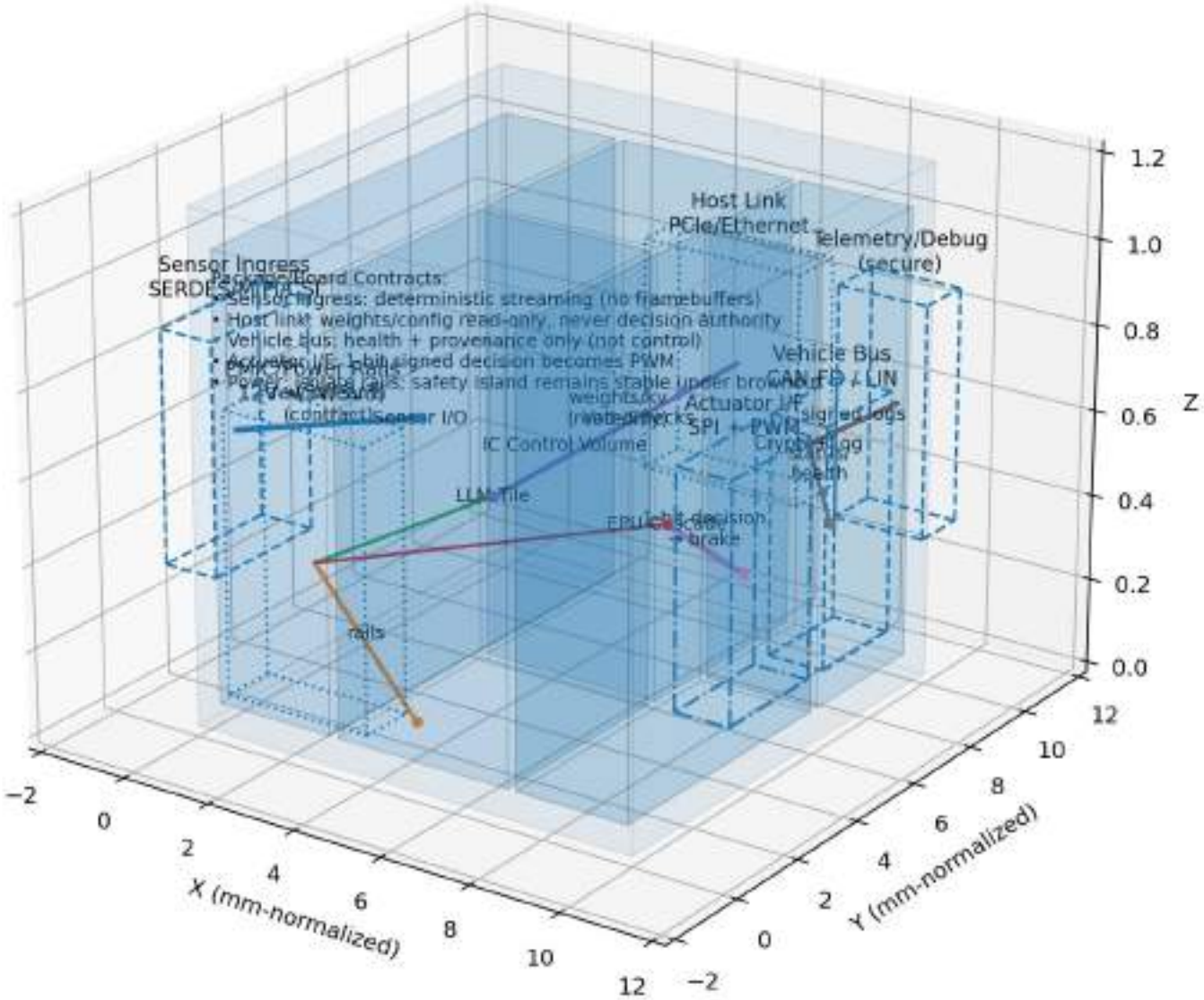
Overlay 10: Witness Primitive Hardware ($\mu(a,b)$ Crossbar + ROM Consistency Gate)



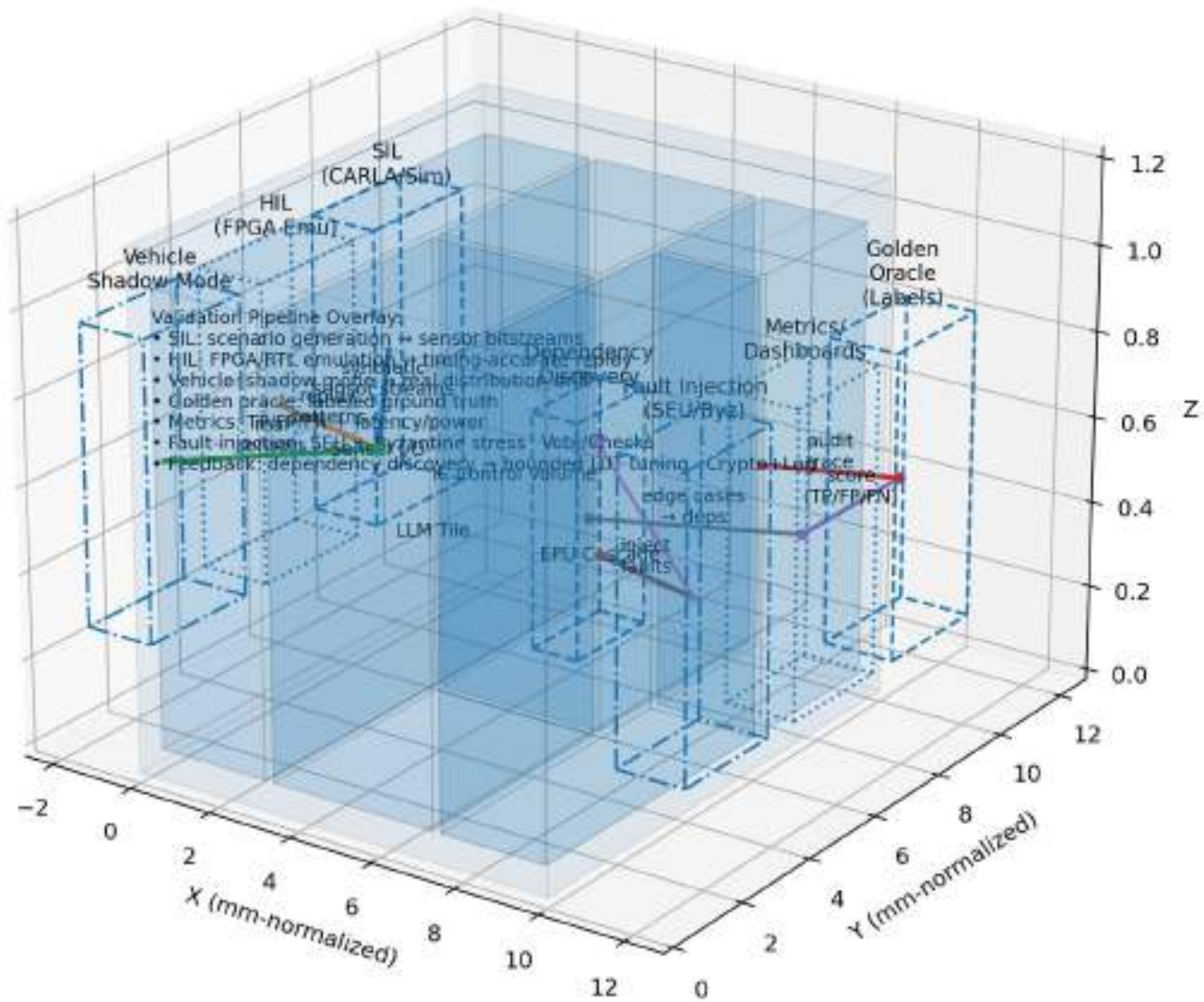
Overlay 11: Adaptiveness Loop (Dependency Discovery → Safe LUT Updates → Enforcement)



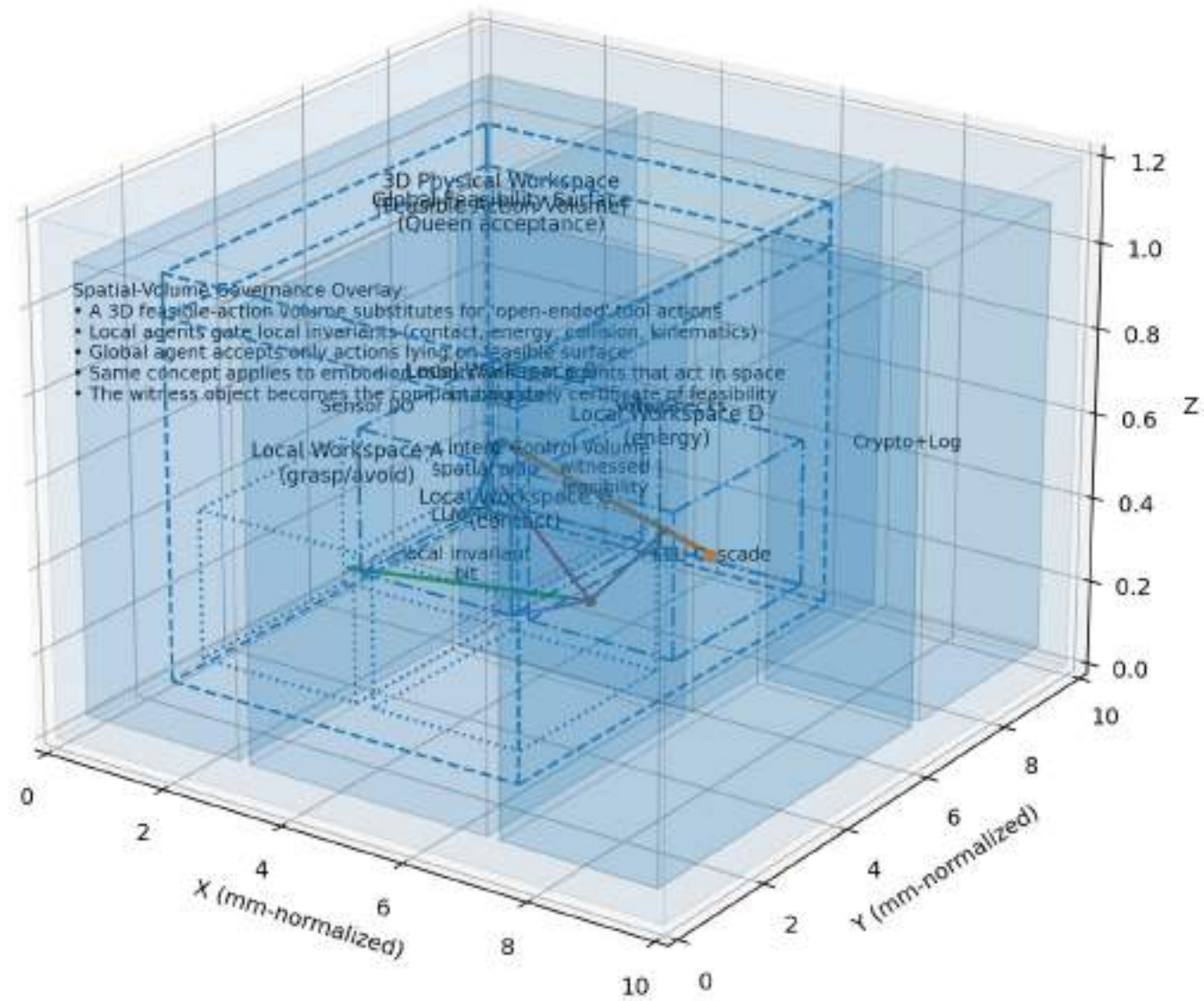
Overlay 12: Physical Interfaces & Protocol Contracts (Sensors ↔ Host ↔ Vehicle Bus ↔ Actuator)



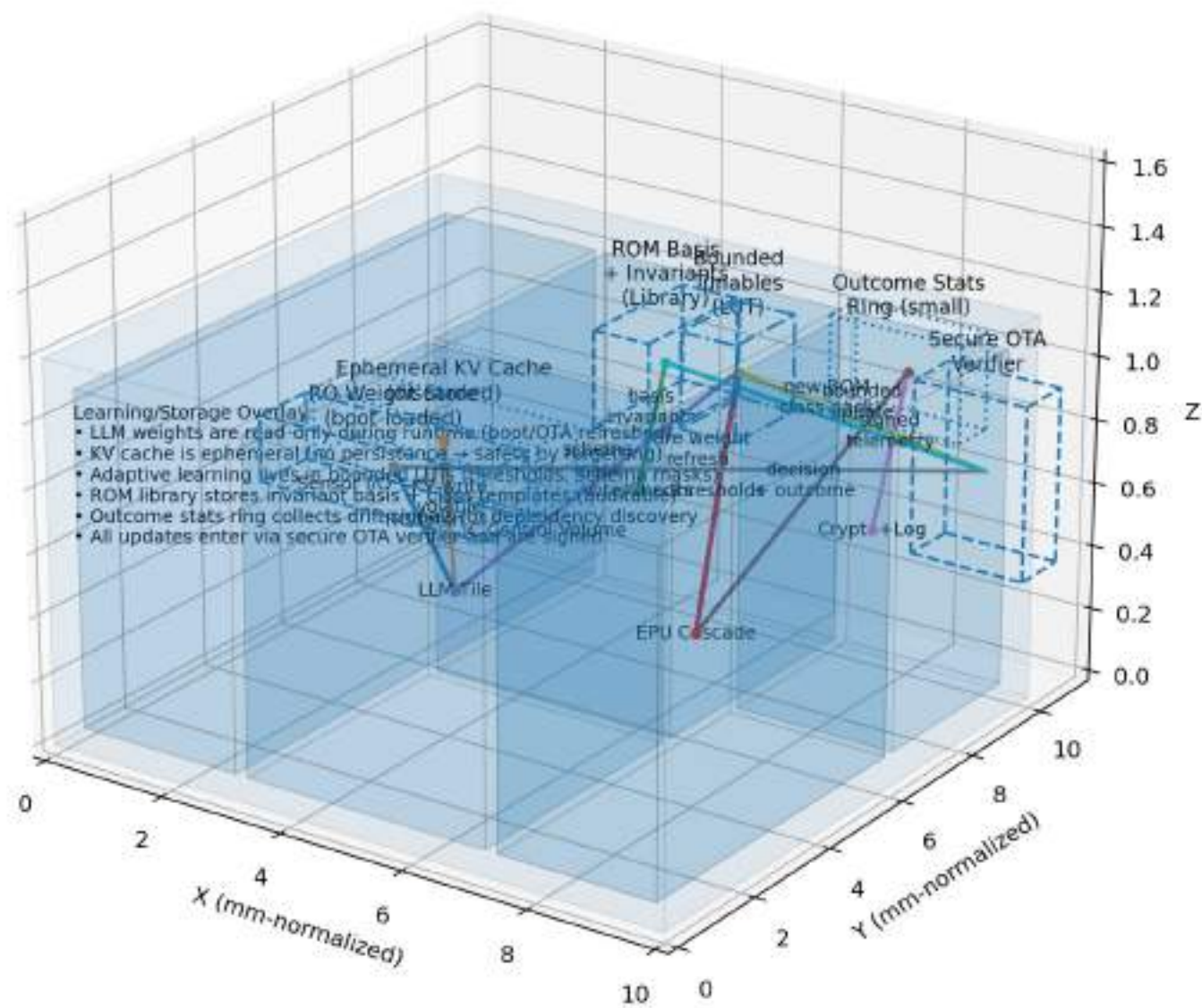
Overlay 13: SIL/HIL/Vehicle Testing Map (End-to-End Validation & Feedback)



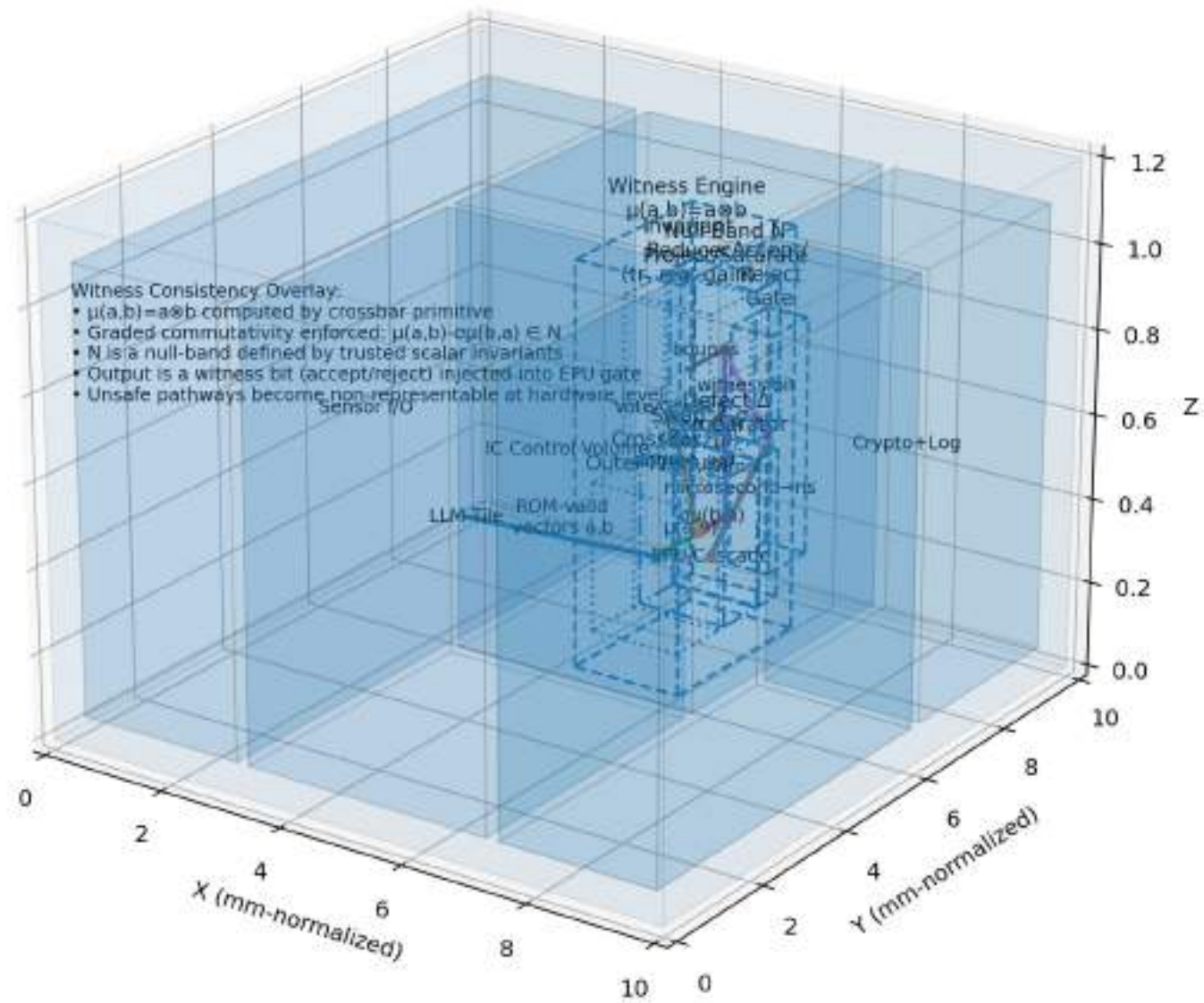
Overlay 14: 3D Workspace Governance (Local vs Global Feasibility in Physical Volume)



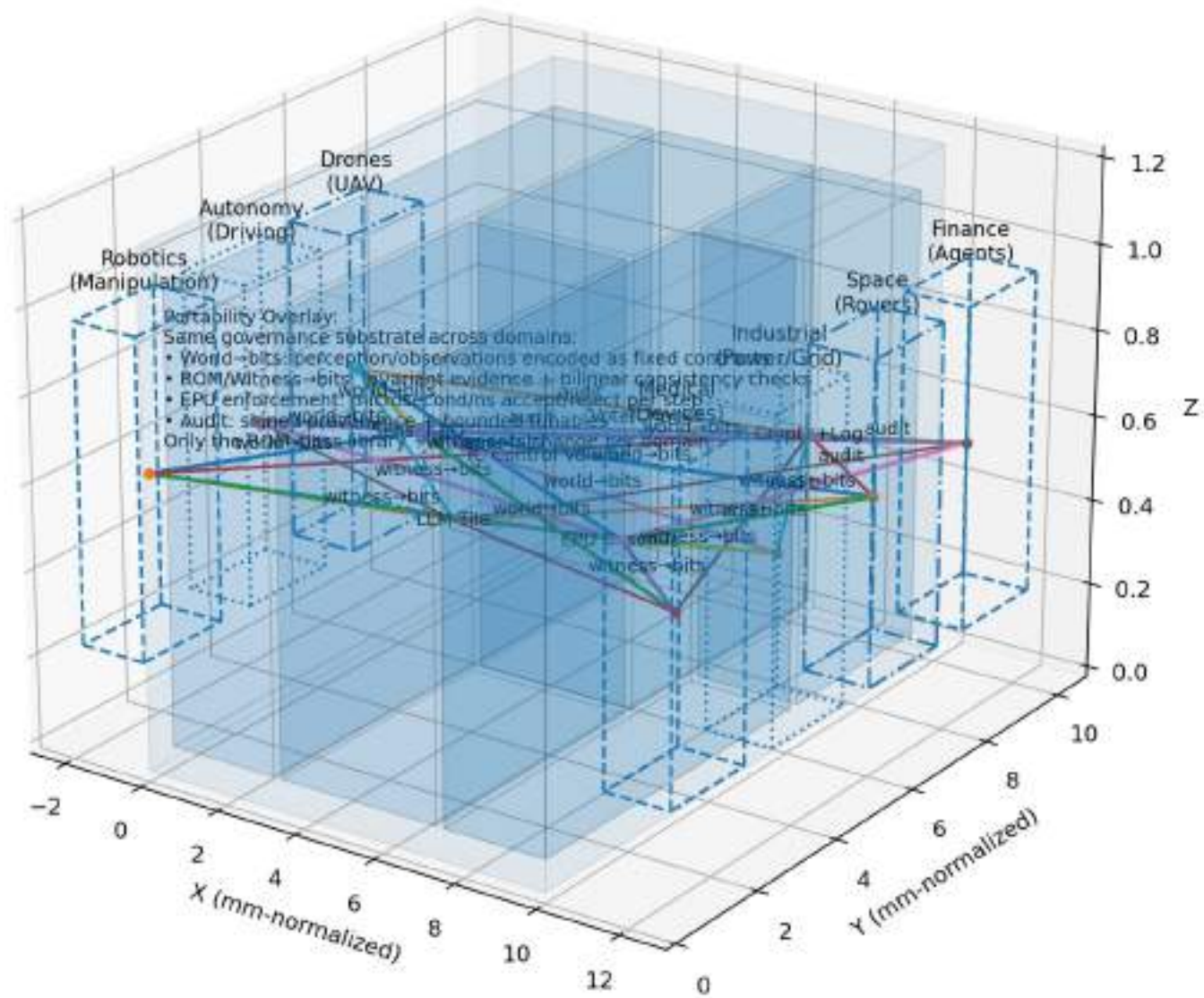
Overlay 15: Where Learning Lives (RO Weights vs Ephemeral Cache vs Bounded Tunables)



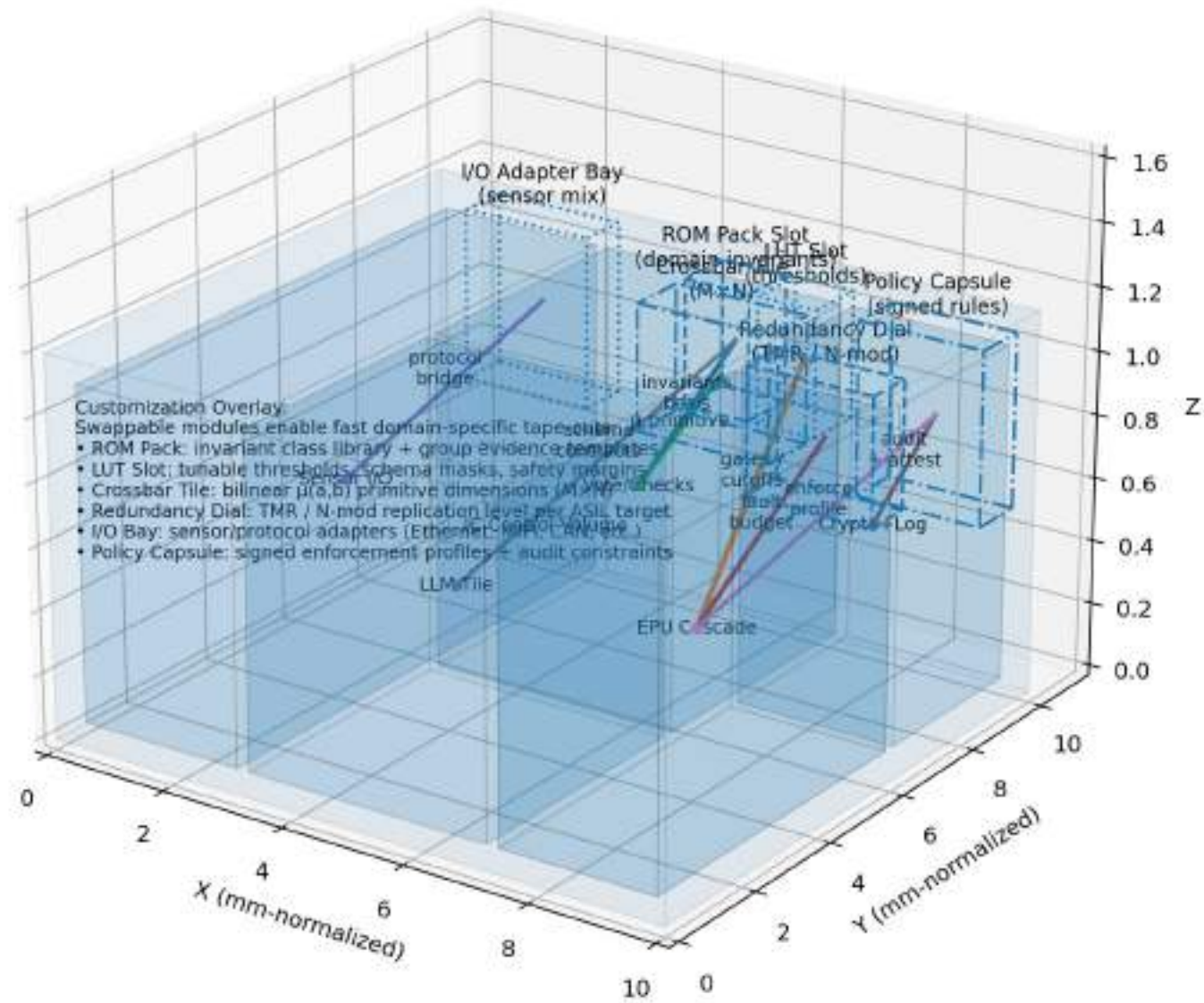
Overlay 16: Bilinear Witness Consistency Lattice (Graded Commutativity Enforcement)



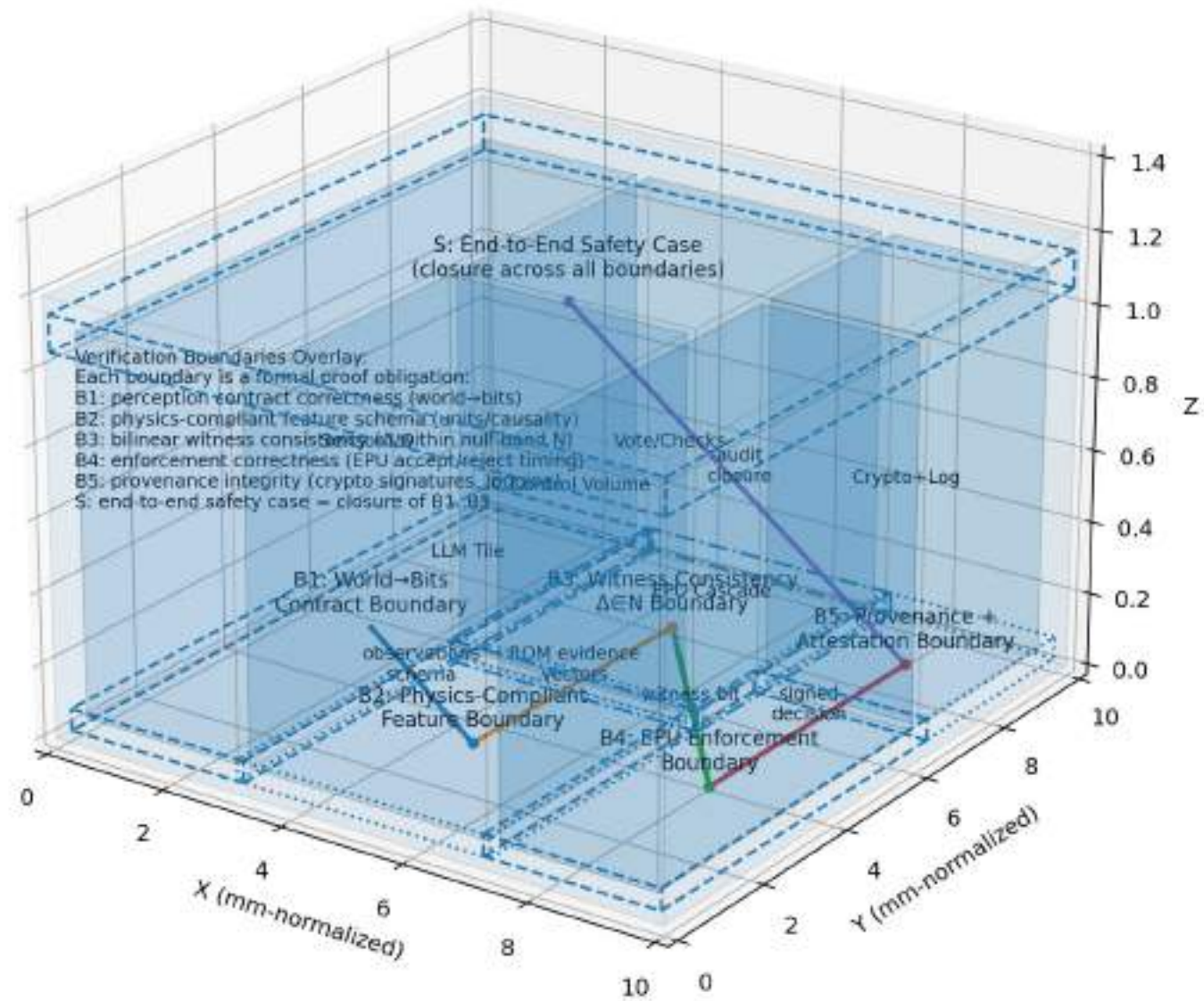
Overlay 17: Cross-Domain Portability (Shared Contracts, Domain-Specific Invariants)



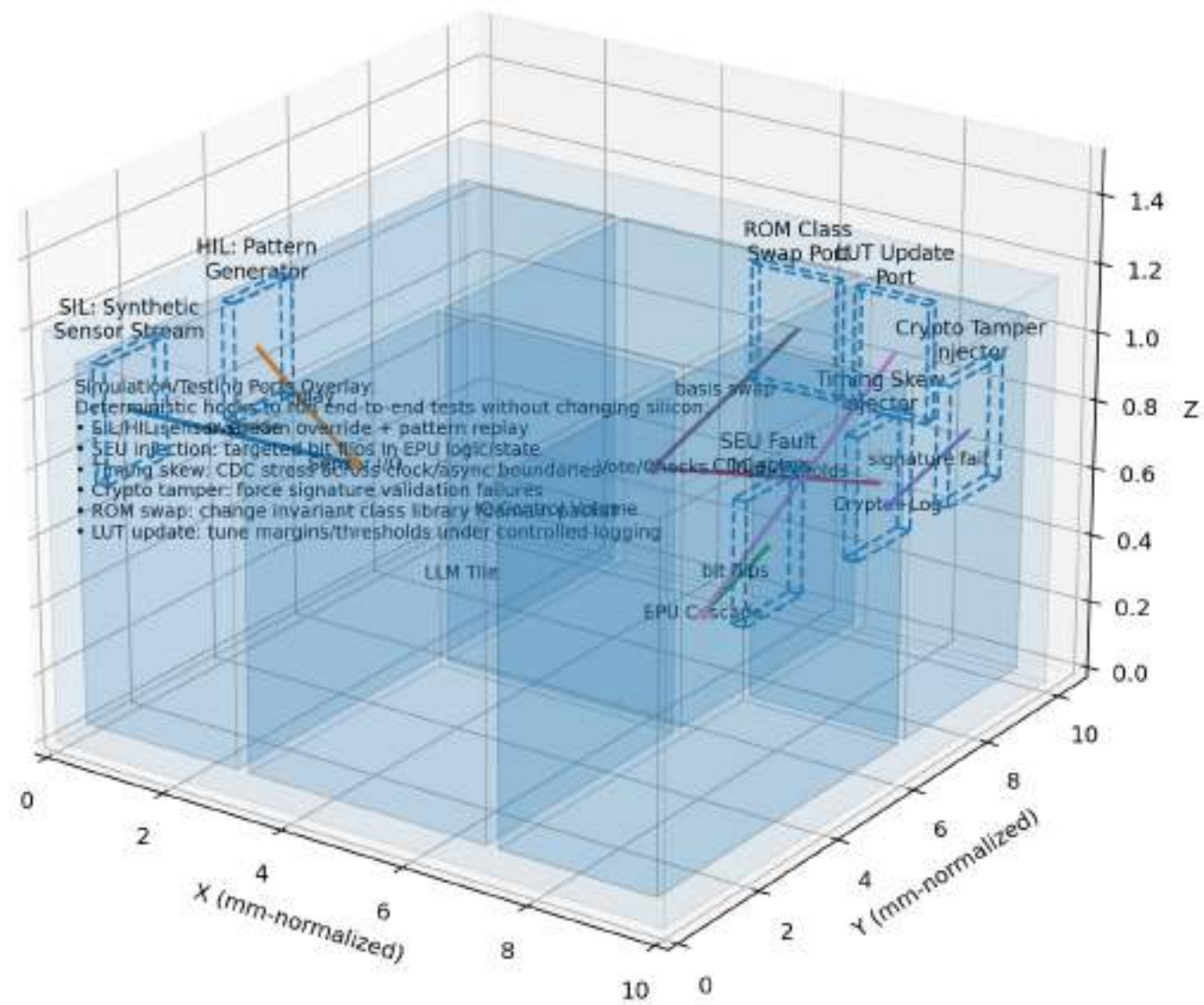
Overlay 18: Hardware Customization Knobs (Swappable Packs + Tileable Primitives)



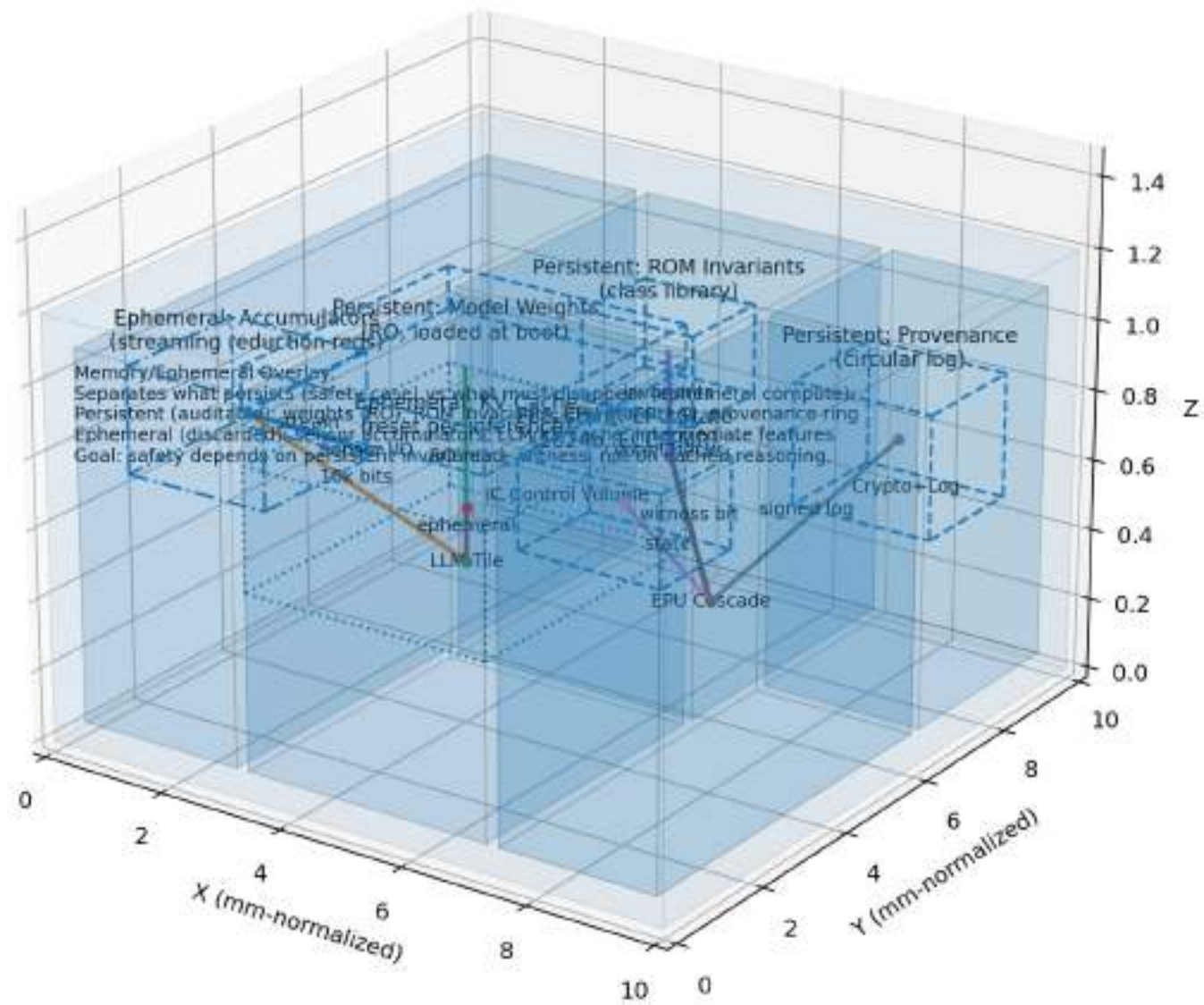
Overlay 19: Verification Boundary Surfaces (Proof Obligations Across the IC Control Volume)



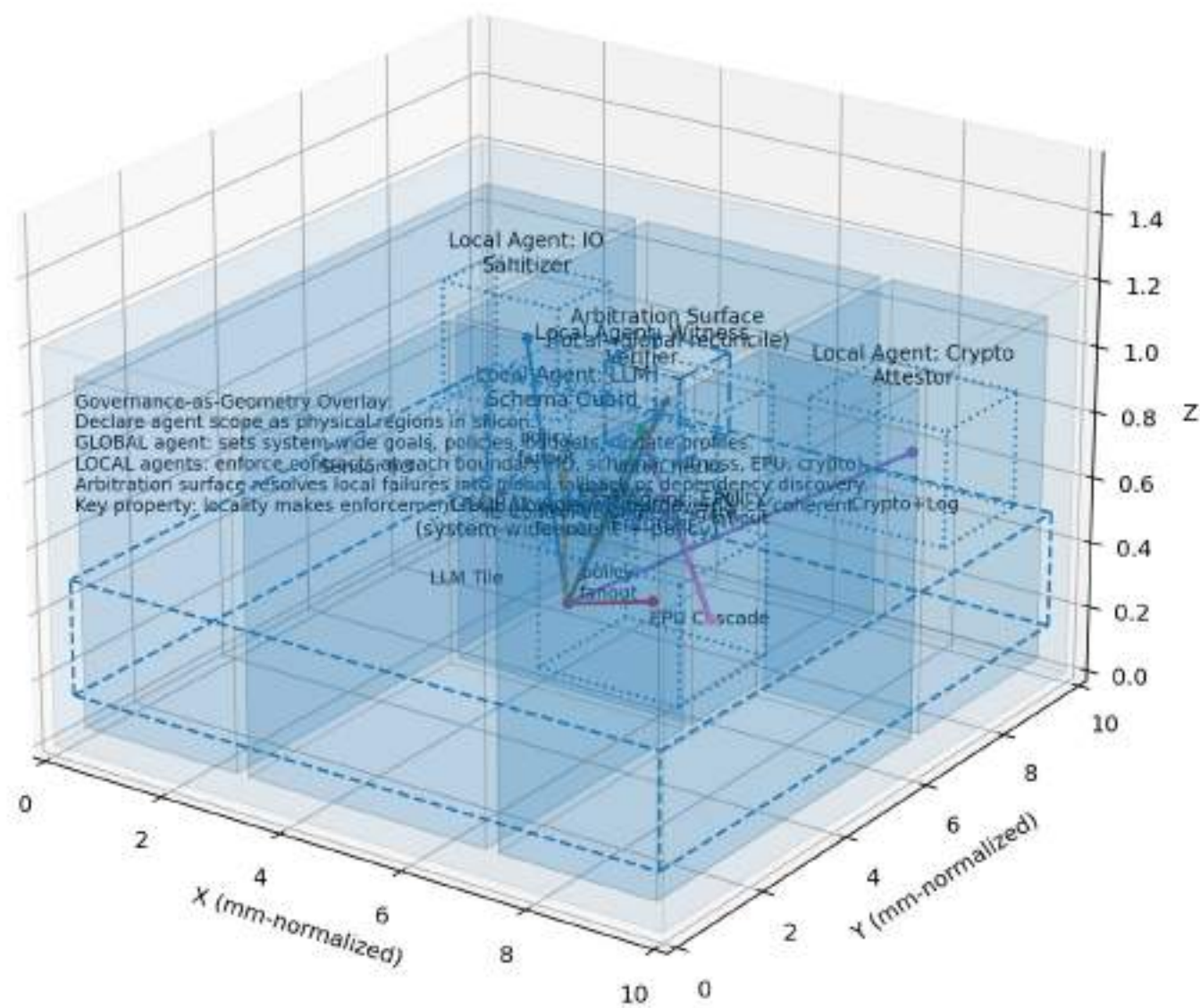
Overlay 20: Simulation Injection Ports (SIL/HIL/Fault-Injection Interfaces)



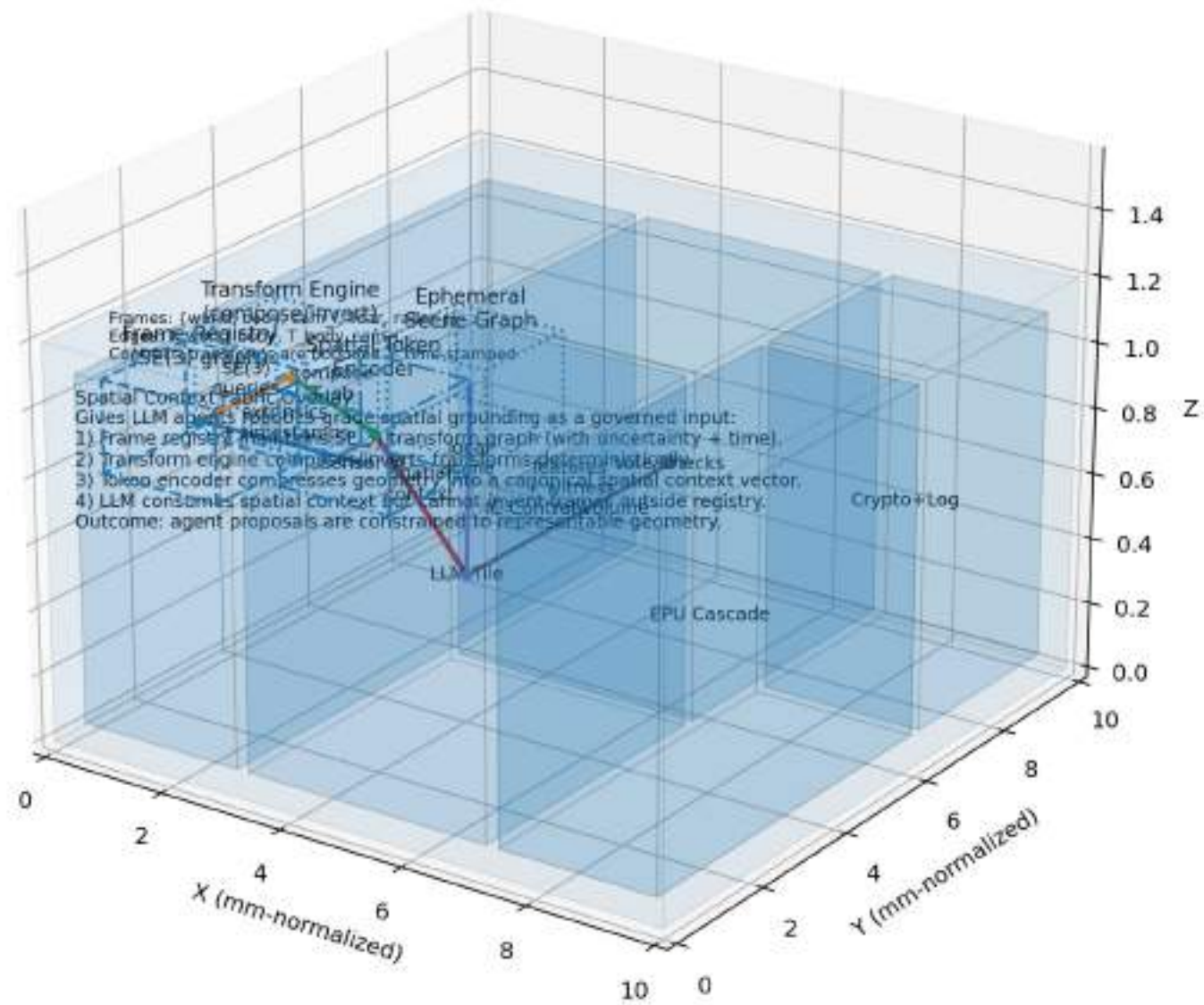
Overlay 21: Memory vs Ephemeral State (Persistence Contracts for Safety + Audit)



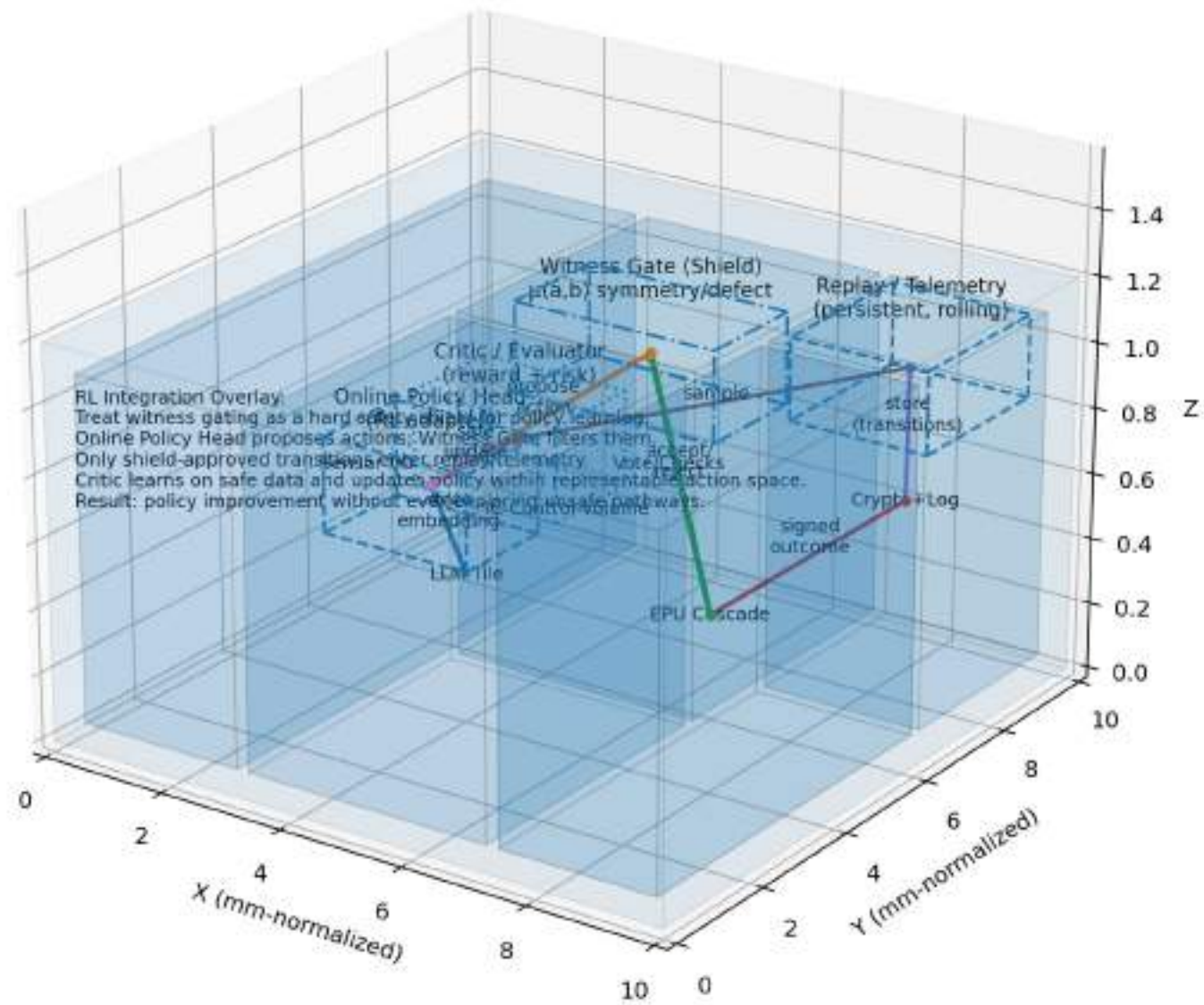
Overlay 22: Governance-as-Geometry (Global vs Local Agents in 3D IC Volume)



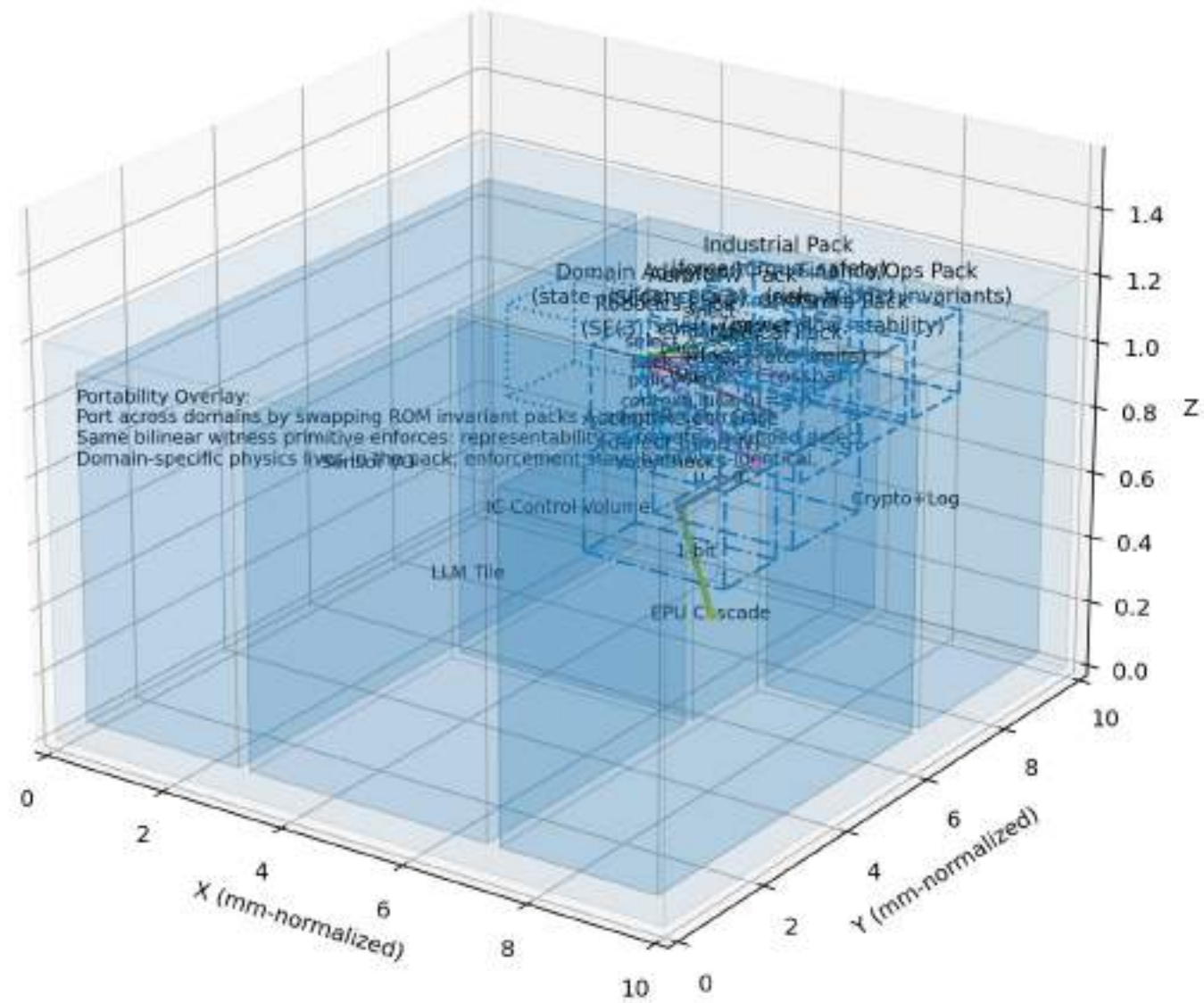
Overlay 23: Spatial Context Fabric (SE(3) Frame Registry + Transform Graph for LLM Agents)



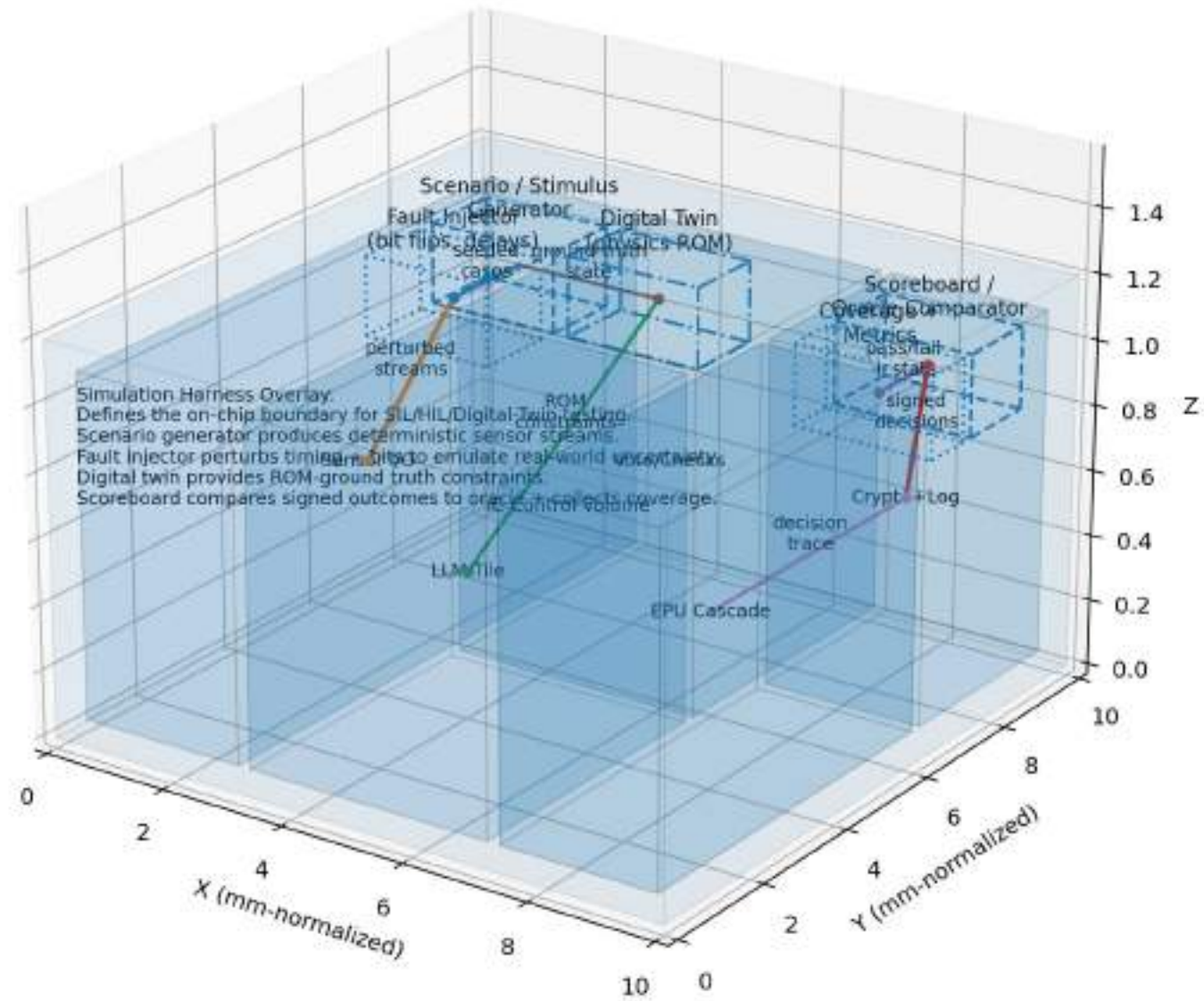
Overlay 24: RL Loop Integration (Safe Policy Improvement under Witness Gating)



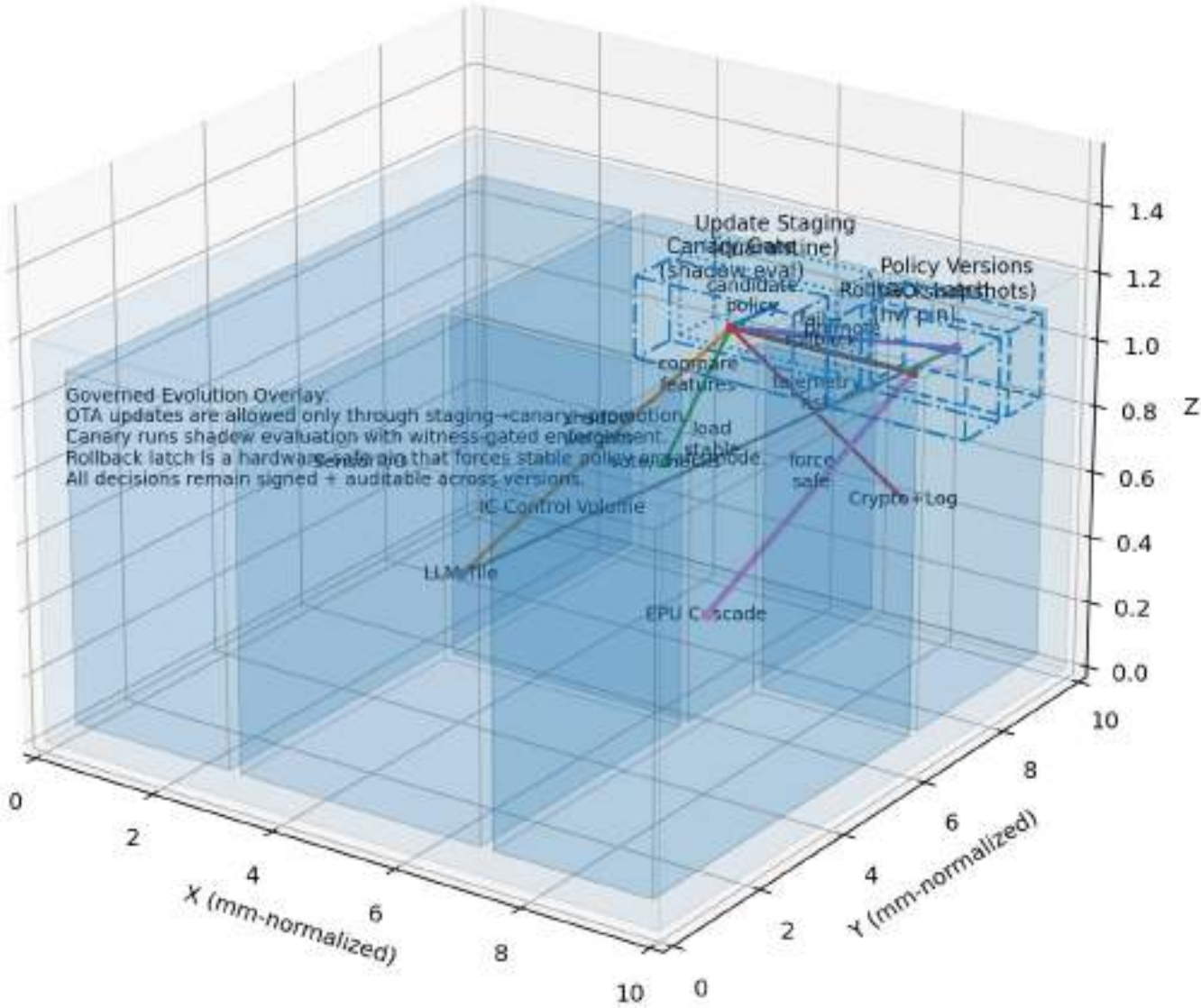
Overlay 25: Cross-Domain Portability (Invariant Packs + Adapter Contracts)



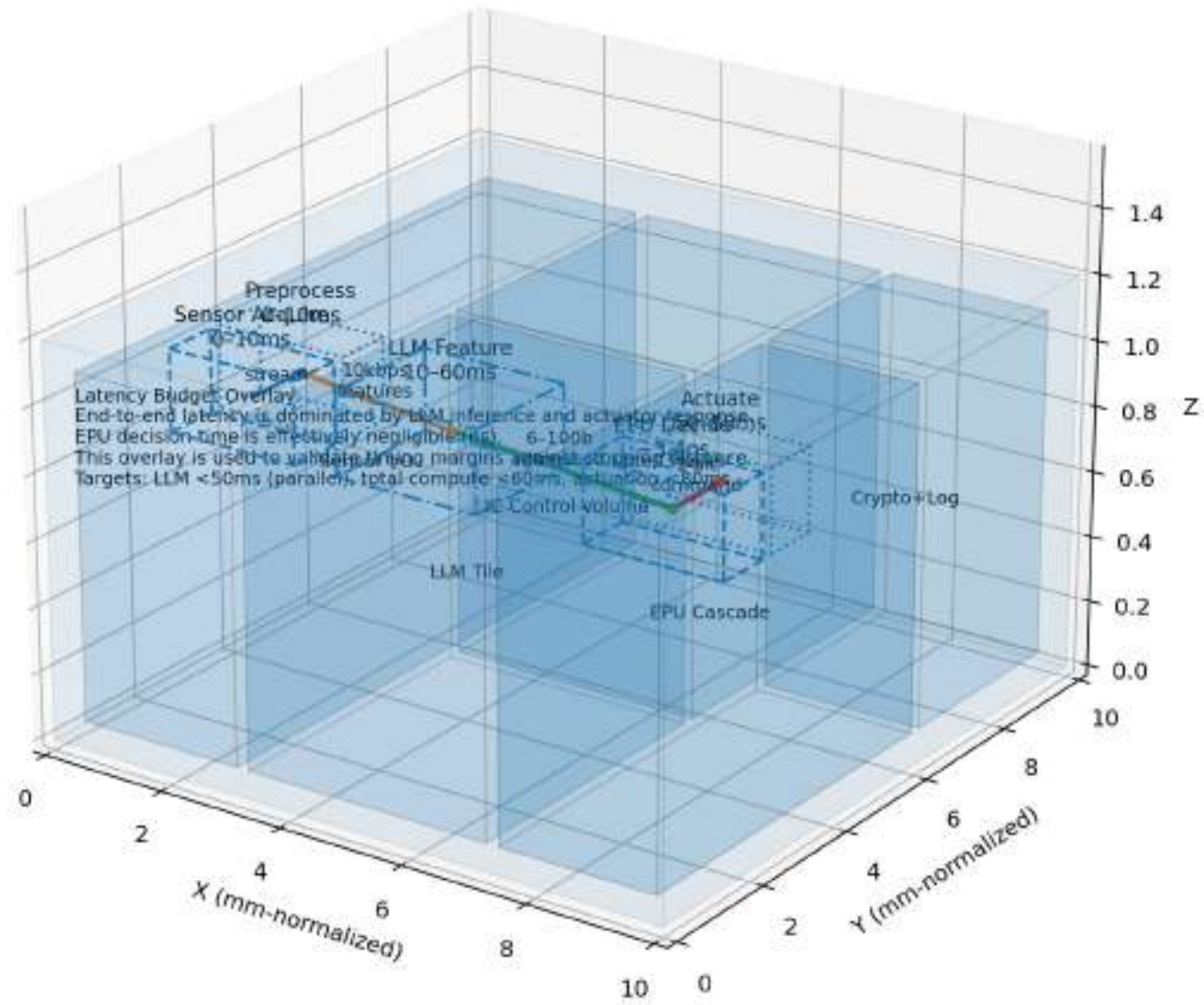
Overlay 27: Simulation Harness + Digital Twin Boundary (SIL/HIL Ready)



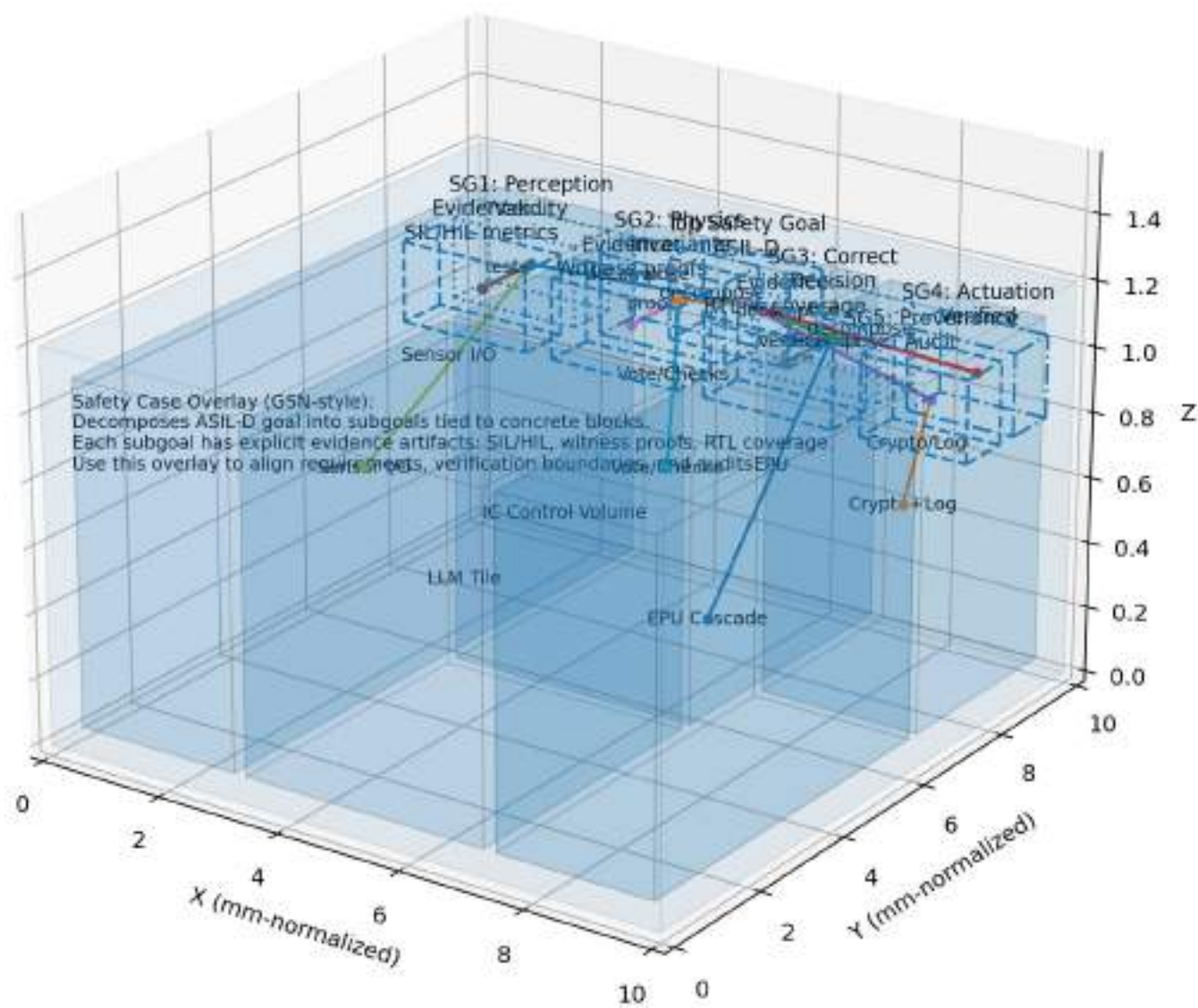
Overlay 28: OTA Policy Updates + Rollback (Governed Evolution)



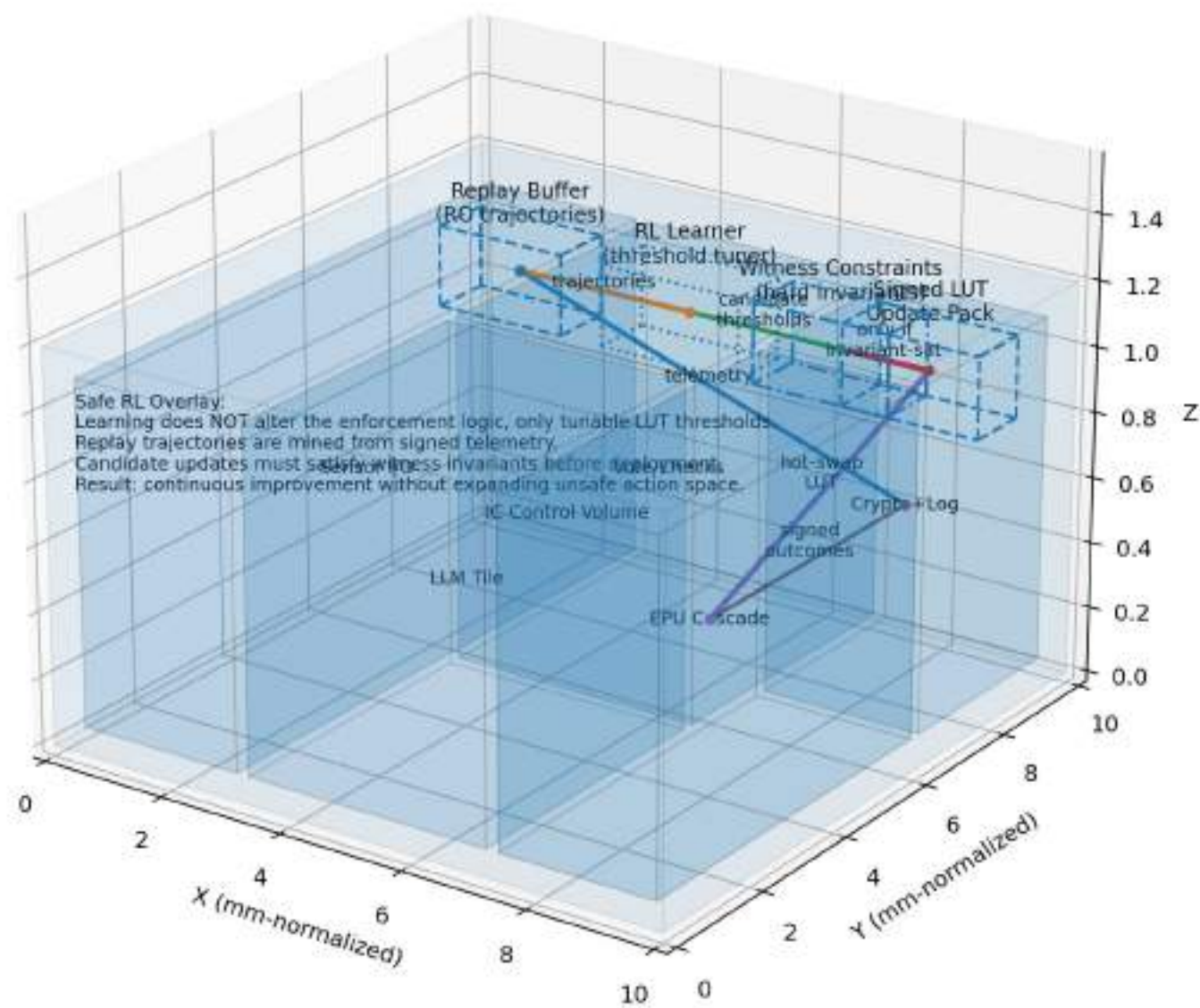
Overlay 30: End-to-End Latency Budget (Compute + Actuation)



Overlay 31: Safety Case Decomposition (ASIL-D Goal Structuring)



Overlay 32: Closed-loop RL Improvement Under Witness Constraints



Overlay 33: Spatial Governance Mesh (Global vs Local Agents in 3D Volume)

